Amalgamation, Congruence-Extension, and Interpolation Properties in Algebras

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To my brother Leo

Introduction

It is always exciting for a mathematician when close connections are discovered between seemingly distant notions or results from two different branches of mathematics, and this is especially true when the notions or results involved are important ones and the focus of considerable research in their respective areas. Thus, for instance, some recent developments have brought to light close and unexpected connections between two groups of results – metalogical interpolation theorems of which the first and best known is Craig's interpolation theorem for first-order logic, and algebraic theorems to the effect that certain classes of algebras have the amalgamation property. The main purpose of this paper is to study in detail mutual implications between these two kinds of results and using these implications to obtain new results of either kind.

It is known that the metalogical study of predicate logic (and various related logical systems) can be adequately replaced by an algebraic discussion of certain structures specifically designed for this purpose. Such a replacement turns out to be particularly desirable for the task at hand, and as the algebraic structures serving as the objects of our discussion we choose cylindric algebras.

Given any ordinal α, a *cylindric algebra of dimension* α is an algebra $\mathfrak{A}$ of the form $\langle A, +, \cdot, -, 0, 1, \mathsf{c}_\kappa, \mathsf{d}_{\kappa\lambda}\rangle_{\kappa,\lambda<\alpha}$ where $+, \cdot$ are binary operations on A, $-, \mathsf{c}_\kappa$ (for $\kappa<\alpha$) are unary operations, and $0, 1, \mathsf{d}_{\kappa\lambda}(\kappa, \lambda<\alpha)$ are distinguished elements of A. The class CA_α of all α-dimensional cylindric algebras is defined by a system of postulates having the form of identities, where the postulates involving only the operations $+, \cdot, -, 0, 1$ may be any one of the familiar postulate systems for Boolean algebras. Cylindric algebras are defined for any ordinal α, but in this introduction we shall be concerned only with infinite α.

An important kind of CA_α (we use 'CA_α' as an abbreviation for 'cylindric algebra

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[1]) This paper is a revised version of the author's doctoral dissertation submitted to the University of California. It is the result of an immensely enjoyable journey in the land between algebra and logic. The author wishes to express his gratitude to his guide on this journey, Professor Alfred Tarski, for his wit, his wisdom, and especially his patience. He also wishes to thank Professors Leon Henkin and Donald Monk for their advice and encouragement.

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of dimension α') arises naturally by means of a certain metalogical construction. We start with any first-order language Λ with equality which does not have any operation symbols; we suppose that the variables of Λ are arranged in a sequence of length α. With Λ we can associate in a natural way an algebraic structure $\mathfrak{F}_\Lambda$. The universe of $\mathfrak{F}_\Lambda$ is the set of formulas of Λ, and its fundamental operations are the usual sentential operations together with the operations of existential quantification with respect to each of the α variables. By a theory of the language Λ we mean of course any set of formulas of Λ containing all of its logical consequences. Given an arbitrary theory Θ we can construct by means of Θ a certain algebra of formulas. The fundamental operations of this algebra, like those of $\mathfrak{F}_\Lambda$, correspond to sentential operations and quantifications; however, in this case the elements of the algebra are sets of formulas rather than the formulas themselves. The algebra constructed is actually the quotient algebra obtained from $\mathfrak{F}_\Lambda$ by identifying two formulas just in case they are equivalent relative to Θ. By distinguishing certain elements of this algebra we get a CA_α, and the algebras constructed in this manner are called *cylindric algebras of formulas.*

It turns out that the class of all CA_α's of formulas, or rather the class of all isomorphic images of such algebras, has a simple intrinsic characterization. It is simply the class Lf_α of all locally finite CA_α's: a CA_α is said to be *locally finite* if for each of its elements x the set of all ordinals $\kappa < \alpha$ such that $\mathsf{c}_\kappa x \neq x$ is finite – this set is called the *dimension set* of x. Of special interest to us is the CA_α of formulas which is obtained from the smallest theory of Λ, i.e., the theory of all logical validities. Let us denote this algebra by $\mathfrak{F}_{\bar{\Lambda}}$. The reason we are especially interested in this algebra becomes clear when we recall what Craig's theorem says. Let Λ be any first-order language and let Θ_0 be the theory of logically valid formulas. If φ and ψ are formulas of Λ and the conditional $\varphi \to \psi$ is contained in Θ_0, then by Craig's theorem there is a formula θ such that only those non-logical constants which occur in both φ and ψ may occur in θ, and such that both the conditionals $\varphi \to \theta$ and $\theta \to \psi$ are in Θ_0. Craig's theorem does not in general hold when the theory Θ_0 of logical validities is replaced by an arbitrary theory Θ. Thus it is reasonable to expect that in its algebraic form the theorem will apply in general only to Lf_α's isomorphic to algebras of the form $\mathfrak{F}_{\bar{\Lambda}}$. Happily, it turns out that such algebras can be singled out from arbitrary CA_α's by means of a rather natural algebraic condition which is very closely related to the notion of a free algebra. There is no locally finite cylindric algebra which is at the same time a free CA_α, however, the algebras $\mathfrak{F}_{\bar{\Lambda}}$ prove to be just the locally finite algebras which come closest to being free in a sense we now make precise.

Consider any CA_α $\mathfrak{A}$ and any subset X of the universe of $\mathfrak{A}$ which generates $\mathfrak{A}$. We say that X *freely generates* $\mathfrak{A}$ *with dimension-restricting conditions* if a set Δx of ordinals $< \alpha$ can be associated with each element x of X such that the following condition holds: a function f from X into the universe of an arbitrary CA_α $\mathfrak{B}$ can be extended to a homomorphism from $\mathfrak{A}$ into $\mathfrak{B}$ provided only that the dimension set of

fx is included in Δx for every $x \in X$. Free dimension-restricted CA_α's are special cases of the general notion of a free algebra with defining relations. The Lf_α's $\mathfrak{F}_{\bar{\Lambda}}$ are, up to isomorphism, just the dimension-restricted free CA_α's which are locally finite. In view of this characterization the algebraic translation of Craig's theorem is obvious: for any given Lf_α $\mathfrak{A}$ and any set X freely generating $\mathfrak{A}$ with dimension-restricting conditions, the following condition is satisfied.

(I) If Y and Z are subsets of X, if y and z are any elements of the subalgebras of $\mathfrak{A}$ generated by Y and Z, respectively, and if $y \leq z$, then there is an element w of the subalgebra generated by $Y \cap Z$ such that $y \leq w \leq z$.

The condition described in (I) can of course be applied to any CA_α $\mathfrak{A}$ and set X generating $\mathfrak{A}$, and not just to dimension-restricted free Lf_α's. When (I) does hold for an algebra $\mathfrak{A}$ and generating set X we say that the algebra $\mathfrak{A}$ has the *strong interpolation property relative to* X, or simply that $\mathfrak{A}$ has the *strong interpolation property* (since it is usually clear from context which set X we have in mind). We say that the algebra $\mathfrak{A}$ has the (ordinary) *interpolation property* (*relative to* X) if, whenever the hypothesis of (I) is satisfied, there exists an element w which satisfies the conditions of the conclusion of (I) except that, instead of the inclusion $w \leq z$, we require only that the weaker inclusion

$$w \leq \mathsf{c}_{\kappa_0}\mathsf{c}_{\kappa_1} \cdots \mathsf{c}_{\kappa_{\lambda-1}} z$$

hold for some $\kappa_0, \kappa_1, \ldots, \kappa_{\lambda-1} < \alpha$. It is not difficult to show that in its algebraic translation Craig's theorem restricted to sentences says that every dimension-restricted free Lf_α has the interpolation property. (A distinction is not usually made between Craig's theorem in its restriction to sentences and its general form for arbitrary formulas. This is so because the two versions are easily interderivable. However, when restricted to a single fixed language Λ the two versions of the theorem are no longer interderivable, and this observation is reflected in the fact that the two versions of the interpolation property, the ordinary and strong properties, are not equivalent – indeed, it is easy to give an example of a CA_α which has the interpolation property but does not have the property in the strong sense.)

The ordinary interpolation property is also interesting for another reason. It is proved in Theorem 2.1.11 that a necessary and sufficient condition for the CA_α $\mathfrak{A}$ to have the interpolation property relative to the generating set X is the following:

(II) If Y and Z are any subsets of X whose union is all of X, if R and S are any congruence relations on the subalgebras of $\mathfrak{A}$ generated by Y and Z, respectively, and if the congruence relation on the subalgebra generated by $Y \cap Z$ which is induced by R is identical with the one induced by S, then there is a congruence relation T on the

whole algebra $\mathfrak{A}$ which induces respectively R and S on the subalgebras generated by Y and Z.

Condition (II), as opposed to condition (I), is formulated in terms of the general theory of algebras, and hence can be referred to arbitrary algebraic structures $\mathfrak{A} = \langle A, Q_i \rangle_{i \in I}$ where A, the universe of $\mathfrak{A}$, is an arbitrary non-empty set and $\langle Q_i \rangle_{i \in I}$ is an arbitrary system of finitary operations on A. We say that an arbitrary algebra $\mathfrak{A}$ has the *congruence-extension property relative to* a generating set X if $\mathfrak{A}$ and X satisfy condition (II).

We have seen how consideration of the theorem of Craig leads naturally to a property applicable to algebras of the most general character – the congruence-extension property, and we have seen, moreover, that the content of Craig's theorem is that this property actually applies to every locally finite dimension-restricted free cylindric algebra. We can now explain the second basic notion of our discussion, that of *amalgamation property*. It is a property applicable, not to individual algebras, but to arbitrary classes of algebras. A class K is said to have this property if it satisfies the following condition:

(III) If $\mathfrak{A}, \mathfrak{B}, \mathfrak{C} \in \mathsf{K}$ and f and g are any isomorphisms from $\mathfrak{C}$ into $\mathfrak{A}$ and $\mathfrak{B}$, respectively, then there is a $\mathfrak{D} \in \mathsf{K}$, an isomorphism h from $\mathfrak{A}$ into $\mathfrak{D}$, and an isomorphism k from $\mathfrak{B}$ into $\mathfrak{D}$, such that $h \circ f = k \circ g$.

Thus the main task of this paper can now be described as the study of relations between the amalgamation and congruence-extension properties – or more precisely between the amalgamation property of a class K of algebras and the congruence-extension property of members of this class.

In the main result of Chapter 1, Theorem 1.2.8, the exact relation between the amalgamation property and the congruence-extension property is stated; it takes the form of certain mutual implications which turn out to be very easy to formulate when applied exclusively to classes K which are *algebraically closed* in the sense that they are closed under the formation of homomorphic images, subalgebras, and direct products of arbitrary systems of algebras of K. As a direct corollary of 1.2.8 we get that an algebraically closed class K has the amalgamation property if and only if every free algebra $\mathfrak{A}$ of K has the congruence-extension property (relative to any set X freely generating $\mathfrak{A}$). As it is actually stated 1.2.8 applies to a much wider variety of classes of algebras, but its formulation is correspondingly more complicated. Our main interest in the general form of 1.2.8 lies in the fact that in this form the result can be applied to the class Lf_α of locally finite CA_α's, while the corollary previously mentioned cannot be so applied (since Lf_α is not algebraically closed). In fact, by applying 1.2.8 to Lf_α we conclude that Lf_α has the amalgamation property if and only if all dimension-

restricted free Lf_α's have the interpolation property (which for CA_α's is, as we have previously observed, equivalent to the congruence-extension property). By comparing this conclusion with our earlier observations we see that as a consequence of 1.2.8 Craig's theorem (for any standard system of predicate logic) is equivalent to the statement that Lf_α has the amalgamation property. Thus a purely algebraic proof of the latter statement yields automatically a similar proof of Craig's theorem. Daigneault in [7] and [8] has actually provided such a proof, and, without knowing or using Theorem 1.2.8 which as far as we know appears here for the first time, he has also shown how Craig's theorem follows from the amalgamation property of Lf_α; some of the methods of proof applied in [8] have strongly influenced techniques of the present paper. (Strictly speaking, Daigneault deals with locally finite polyadic algebras with equality, but it is well known that these algebras are essentially the same as locally finite cylindric algebras.)

We now turn to the question whether the amalgamation property applies to various important classes of infinite-dimensional CA_α's different from Lf_α. No results in this direction had been previously known. Now, however, using 1.2.8 we are able to answer this question for a number of such classes.

A CA_α is called *dimension-complemented* if for every element x of the algebra there is an ordinal $\kappa < \alpha$ such that $\mathsf{c}_\kappa x = x$, i.e., the dimension set of x does not contain every ordinal $< \alpha$; the class of all such algebras is denoted by Dc_α. These algebras are interesting because, though considerably more general than Lf_α's, they turn out to have many important properties of Lf_α's. In Theorem 2.2.6 and Remark 2.2.9 we show that every dimension-restricted free CA_α which is a Dc_α has the strong interpolation property, while the class Dc_α itself fails to have the amalgamation property. On the other hand, the class Di_α of all so-called *diagonal* CA_α's (which includes Dc_α) has the amalgamation property by 2.2.20; the same applies to the class of semisimple CA_α's by 2.2.24. Turning now to the class RCA_α of all representable CA_α's, which is more comprehensive than all the preceding classes, we see from 2.3.4 that no free RCA_α with four or more free generators has the interpolation property; since RCA_α is algebraically closed, we conclude directly from this that RCA_α fails to have the amalgamation property. By 2.3.8 this result applies also to every class K of CA_α's which contains all RCA_α's; this latter result extends to infinite α what Comer has shown in [4] to hold for every finite $\alpha \geq 2$. Thus in the increasing sequence

$$\mathsf{Lf}_\alpha \subset \mathsf{Dc}_\alpha \subset \mathsf{Di}_\alpha \subset \mathsf{RCA}_\alpha$$

the first and third classes have the amalgamation property while the second and fourth classes fail to have it.

In addition to the amalgamation property, a weaker property, the so-called *embedding property*, has been sometimes discussed in the literature. A class K is said to have this property if any two algebras $\mathfrak{A}$, $\mathfrak{B}$ in K can both be isomorphically embedded

as subalgebras in the same algebra $\mathfrak{C}$ of K. We shall use here the term embedding property in a still weaker sense; in fact, we assume that the algebras $\mathfrak{A}$ and $\mathfrak{B}$ have isomorphic minimal subalgebras. In [4] Comer shows that for every finite $\alpha \geq 2$ the class RCA_α fails to have, not only the amalgamation property, but also the embedding property. On the other hand, we see from 2.2.15 that for infinite α the situation changes: RCA_α does have in this case the embedding property (although, as was previously mentioned, it fails to have the amalgamation property).

A systematic summary of the results of this paper together with some related results of other authors can be found in the two tables of Section 2.4.

Chapter 1 is devoted exclusively to investigations of a general algebraic character. In Chapter 2 the results of the first chapter are applied to the theory of cylindric algebras. The source of the particular treatment of algebraic structures in general and cylindric algebras in particular that forms the basis of this paper is the book of Henkin, Monk, and Tarski which is listed in the bibliography and hereafterward referred to by the mnemonic $\mathscr{HMT}$. This book is also the source of practically all the notation and terminology we shall use here. Finally, we refer the reader to $\mathscr{HMT}$ for further historical and bibliographic information.

A bibliography appears at the end of the paper and is arranged alphabetically by author. With the exception of $\mathscr{HMT}$ the items in it are numbered consecutively and referred to by this number in the paper.

Chapter 1

AMALGAMATION AND CONGRUENCE-EXTENSION PROPERTIES IN ARBITRARY ALGEBRAS

The material of this chapter lies entirely within the general theory of algebras. We assume that the reader is familiar with the basic elements of this theory, and so they are just briefly reviewed in Section 1.1; some special notions important for our work are discussed in more detail. In Section 2.2 the connection between the amalgamation and congruence-extension properties is investigated.

1.1 Preliminaries

One can naturally divide the present section into two parts. In the first part we follow very closely (and sometimes literally) the development presented in the Preliminaries and Chapter 0 of $\mathscr{HMT}$. After a quick review of set-theoretical and metalogical notation we will be mainly concerned with recalling the basic notions of the general theory of algebras and with introducing the special notion and terminology of this theory that we shall use. The discussion of free algebras is however less perfunctory partly because of their particular importance for our work and partly because their treatment in $\mathscr{HMT}$ has a few novel features. The second part of the section is taken up in a fairly detailed discussion of the notion of an algebra with a distinguished system of generators, a notion which plays an important role in this work. This notion does not occur in $\mathscr{HMT}$ but its fundamental properties are easy consequences of the general theory.

The set-theoretical notation and terminology employed in this paper will be essentially identical to that used in $\mathscr{HMT}$ and explained in detail in the Preliminaries of that work. Most of this notation is either standard or self-explanatory and in these preliminary remarks we shall only mention a few things which might lead to confusion if encountered in the text without explanation. For all unexplained notation and terminology (of a metalogical and algebraic as well as set-theoretical character) the reader is referred to $\mathscr{HMT}$.

The empty set is denoted by 0. The complement of two sets A and B is usually denoted by $A \sim B$, but in a context where A is fixed and we want to emphasize the unary character of this operation we often write ${}_A\sim B$ or simply $\sim B$ if the set A is understood from context. The union over all members of a family F of sets is usually denoted by $\bigcup F$; similarly for intersection. $\subset$ is used for the relation of strict inclusion between sets and $\subseteq$ is used for ordinary inclusion. The set of all subsets of a set A is denoted by $Sb\ A$. If R is any binary relation, then $Do\ R$, $Rg\ R$, and R^{-1} shall denote the domain, range, and converse (or inverse) of R, respectively. If A is

any set,

$$R^*A = \{x: aRx \quad \text{for some} \quad a \in A\}$$

and

$$A \upharpoonright R = \{\langle a, x\rangle : aRx, \ a \in A\};$$

here aRx means $\langle a, x\rangle \in R$. If R is an equivalence relation on A and $x \in A$, then x/R is the equivalence class $R^*\{x\}$ of x over R, and $X/R = \{x/R : x \in X\}$ whenever $X \subseteq A$.

All notation applying to binary relations applies in particular to functions. Id denotes the identity relation on the universal class so that, for any set A,

$$A \upharpoonright Id = \{\langle x, x\rangle : x \in A\}.$$

If f is a function and $x \in Do\, f$, then the value of f at x may be denoted in any of the following ways: $f(x), fx, f_x, f^{(x)}$. ${}^AB$ represents the set of all functions from A into B. If τ is any term such that $\tau(x)$ is a well-determined element for each $x \in A$, we let $\langle \tau(x) : x \in A\rangle$ denote the function with domain A which takes $\tau(x)$ as its value at each $x \in A$. If $F = \langle F_i : i \in I\rangle$ is a system of sets indexed by I, then $\mathsf{P}F$ or $\mathsf{P}_{i \in I} F_i$ denotes the Cartesian product of the system F, that is, the set of all $f \in {}^I(\bigcup_{i \in I} F_i)$ such that $fi \in Fi$ for each $i \in I$. For any element i we define pj_i, the *i'th projection*, to be the function whose domain is the class of all functions with $i \in Do\, f$ and such that $\mathsf{pj}_i f = fi$ for each such f.

An ordinal number is equal to the set of all ordinals less than it. Thus zero, the first ordinal, is the empty set 0, and the following conditions are all equivalent: $\alpha < \beta$, $\alpha \in \beta$, $\alpha \subset \beta$. Cardinal numbers are identified with minimal ordinals but $\alpha + \beta$ always denotes the ordinal sum. The cardinality of A is denoted by $|A|$. By a *sequence* we mean any system indexed by an ordinal. An ordered pair $\langle x, y\rangle$ is identified with the sequence $\{\langle 0, x\rangle, \langle 1, y\rangle\}$ so that $A \times B = \mathsf{P}\{\langle 0, A\rangle, \langle 1, B\rangle\}$ and in particular $A \times A = {}^2A$. A finite sequence a of length κ is often written as $\langle a_0, a_1, \cdots, a_{\kappa-1}\rangle$. Finally, $\widehat{a\,b}$ shall denote the concatenation of the two sequences a and b.

The first-order predicate languages with which we deal are quite standard in their construction except for one detail: because we want to use them to construct cylindric algebras of formulas of all possible dimensions, we must consider languages with arbitrary numbers of individual variables. It is also convenient to have the same non-logical constants and variables for all languages considered, so we shall assume that a fixed variable v_ξ is assigned to each ordinal ξ, and that the variables of any given language Λ are always the first α variables in this ordering where α is determined by Λ and can be any finite or infinite ordinal. If we want to indicate explicitly what this ordinal is we write Λ_α for Λ.

Terms and formulas of a language Λ are conceived of as finite sequences of symbols from the vocabulary of Λ, and hence are constructed from their respective subterms and subformulas by the operation of concatenation. For example, if C is the symbol

for conjunction, then the conjunction of the two formulas φ and ψ is the sequence $\langle C\rangle\widehat{\ }\widehat{\varphi}\,\psi$ and this is denoted by $\varphi \wedge \psi$. An *equation* is any atomic formula of the form $\tau = \sigma$ where τ and σ are terms; an *identity* is a sentence of the form $\forall_{v_{\kappa_0}} \cdots \forall_{v_{\kappa_{\lambda-1}}} \varphi$ where φ is an equation.

Given an algebra $\mathfrak{A}$ (or more generally an arbitrary relational structure $\mathfrak{A}$) and a discourse language Λ_α for $\mathfrak{A}$, we assume that it is known what it means for a sequence $x \in {}^\alpha A$ to satisfy a formula φ of Λ_α. φ is said to *hold* or be *identically satisfied* in $\mathfrak{A}$, or $\mathfrak{A}$ is said to be a *model* of φ, if x satisfies φ for every $x \in {}^\alpha A$. The notions of logical consequence, logical validity, and logical equivalence are defined in terms of satisfaction in the usual way. A class K of algebras is said to be *equational* if it is exactly the class of models of some set of identities. By a well-known theorem of Garrett Birkhoff K is equational iff it is algebraically closed (see the Introduction).

The phrase 'if and only if' is usually abbreviated iff.

We turn now to the general theory of algebras.

DEFINITION 1.1.1. *If $\kappa = \langle \kappa_i : i \in I\rangle$ is any system of finite (possibly zero) ordinal numbers, then by an* ***algebra*** *(or an* ***algebraic structure****) of type κ we mean a pair $\mathfrak{A} = \langle A, Q\rangle$ where A is a non-empty set and Q is a function with domain I which correlates with each $i \in I$ a finitary operation Q_i of rank κ_i on and to elements of A; we usually use the notation $\mathfrak{A} = \langle A, Q_i\rangle_{i \in I}$.*

A is called the ***universe*** *of $\mathfrak{A}$, and the Q_i are called the* ***fundamental operations*** *of $\mathfrak{A}$. A fundamental operation of rank 0 is identified with the unique element in its range and is called a* ***distinguished element*** *of $\mathfrak{A}$.*

Algebras and systems of algebras are represented by German capital letters, $\mathfrak{A}, \mathfrak{B}, \ldots$; if an algebra, or a system of algebras, is denoted by a given German letter, say $\mathfrak{A}$, it will be automatically assumed that the corresponding Roman letter, A, denotes the universe of $\mathfrak{A}$, or the system of universes of terms of $\mathfrak{A}$.

Two algebras $\mathfrak{A}$ and $\mathfrak{B}$ are called *similar* if they have the same similarity type. All the considerations of this chapter apply to arbitrary similarity types, but never involve relations between algebras of different similarity type. Thus in any given context we automatically assume that all the algebras are of a fixed type. The letters K and L are intended to represent arbitrary classes of similar algebras, and they are reserved exclusively for this purpose.

We assume it is known what is meant by saying that $\mathfrak{B}$ is a *subalgebra* of $\mathfrak{A}$; symbolically this is written $\mathfrak{B} \subseteq \mathfrak{A}$. An algebra is said to be *minimal* if it has no proper subalgebra. The class of all subalgebras of algebras in K is denoted by SK. A *subuniverse* B of an algebra $\mathfrak{A}$ is any subset of A which is closed under all fundamental operations; notice that a subuniverse always contains all distinguished elements of $\mathfrak{A}$. If $X \subseteq A$, the *subuniverse generated by* X, i.e., the intersection of all subuniverses of

$\mathfrak{A}$ which include X, is denoted by $Sg\ X$. $Sg\ X = 0$ iff $X = 0$ and no distinguished elements are included among the fundamental operations of $\mathfrak{A}$; if $Sg\ X \neq 0$, then the subalgebra of $\mathfrak{A}$ with universe $Sg\ X$ is denoted by $\mathfrak{Sg}\ X$. $Sg\ X$ and $\mathfrak{Sg}\ X$ (if it exists) depend of course on the algebra $\mathfrak{A}$; when we want to make this explicit we write $Sg^{(\mathfrak{A})}X$ and $\mathfrak{Sg}^{(\mathfrak{A})}X$. A similar remark applies to most of the other terms introduced in this section.

The set of all *homomorphisms* from $\mathfrak{A}$ onto $\mathfrak{B}$ is denoted by $Ho(\mathfrak{A}, \mathfrak{B})$; the set of homomorphisms into $\mathfrak{B}$, that is, onto some subalgebra of $\mathfrak{B}$, is denoted by $Hom(\mathfrak{A}, \mathfrak{B})$. The sets of *isomorphisms* onto and into are denoted respectively by $Is(\mathfrak{A}, \mathfrak{B})$ and $Ism(\mathfrak{A}, \mathfrak{B})$. We write $\mathfrak{A} \geqslant \mathfrak{B}$ or $\mathfrak{B} \leqslant \mathfrak{A}$ if $Ho(\mathfrak{A}, \mathfrak{B}) \neq 0$, and $\mathfrak{A} \cong \mathfrak{B}$ if $Is(\mathfrak{A}, \mathfrak{B}) \neq 0$; we denote the classes of all homomorphic and isomorphic images of members of K by $\boldsymbol{H}\mathsf{K}$ and $\boldsymbol{I}\mathsf{K}$. If $h \in Ho(\mathfrak{A}, \mathfrak{B})$ and $\mathfrak{A}' \subseteq \mathfrak{A}$, then h^*A' is a non-empty subuniverse of $\mathfrak{B}$; the algebra whose universe is h^*A' is denoted by $h^*\mathfrak{A}'$. If $h \in Hom(\mathfrak{A}, \mathfrak{B})$, then $h \in Ho(\mathfrak{A}, h^*\mathfrak{A})$ and $h^*Sg^{(\mathfrak{A})}X = Sg^{(\mathfrak{B})}h^*X$; if $h \in Ho(\mathfrak{A}, \mathfrak{B})$, then $h^*\mathfrak{Sg}^{(\mathfrak{A})}X = \mathfrak{Sg}^{(\mathfrak{B})}h^*X$ provided $Sg^{(\mathfrak{A})}X \neq 0$.

The set of all *congruence relations* on $\mathfrak{A}$ is denoted by $Co\ \mathfrak{A}$, and, for any $X \subseteq A \times A$, $Cg\ X$ is the *congruence relation generated* by X. If $R \in Co\ \mathfrak{A}$, then $\langle x/R : x \in A\rangle \in Ho(\mathfrak{A}, \mathfrak{A}/R)$ where $\mathfrak{A}/R$ denotes the *quotient algebra* of $\mathfrak{A}$ over R. On the other hand, if $h \in Ho(\mathfrak{A}, \mathfrak{B})$, then $R = \{\langle x, y\rangle : x, y \in A,\ hx = hy\} \in Co\ \mathfrak{A}$, and $\{\langle x/R, hx\rangle : x \in A\rangle \in Is(\mathfrak{A}/R, \mathfrak{B})$. An algebra is *simple* if it has exactly two congruence relations. If $R \in Co\ \mathfrak{A}$, then $\mathfrak{A}/R$ is simple iff R is a maximal proper congruence relation, i.e., R is a maximal element of $Co\ \mathfrak{A} \sim \{A \times A\}$ under the set-theoretical inclusion relation.

Let $\mathfrak{A} = \langle \mathfrak{A}_i : i \in I\rangle$ be any system of algebras. Then the *direct product* of $\mathfrak{A}$ is written $\mathsf{P}\mathfrak{A}$ or $\mathsf{P}_{i \in I}\mathfrak{A}_i$. The class of all isomorphic images of direct products of arbitrary systems of algebras in K is denoted by $\boldsymbol{P}\mathsf{K}$. We have $\mathsf{pj}_i \in Ho(\mathsf{P}\mathfrak{A}, \mathfrak{A}_i)$ for each $i \in I$, and, if $h_i \in Ho(\mathfrak{B}, \mathfrak{A}_i)$ for every $i \in I$, then $\mathfrak{B}$ is homomorphic with a subalgebra of $\mathsf{P}\mathfrak{A}$; in fact, $\langle\langle h_i x : i \in I\rangle : x \in B\rangle \in Hom(\mathfrak{B}, \mathsf{P}\mathfrak{A})$. A *subdirect product* of the system $\mathfrak{A}$ is any $\mathfrak{B} \subseteq \mathsf{P}\mathfrak{A}$ such that $\mathsf{pj}_i{}^*\mathfrak{B} = \mathfrak{A}_i$ for each $i \in I$. An algebra is called *semisimple* if it is isomorphic to a subdirect product of some system of simple algebras.

A class K of similar algebras is called *algebraically closed* if $\boldsymbol{H}\mathsf{K} = \boldsymbol{S}\mathsf{K} = \boldsymbol{P}\mathsf{K} = \mathsf{K}$. We always have $\boldsymbol{SH}\mathsf{K} \subseteq \boldsymbol{HS}\mathsf{K}$, $\boldsymbol{PH}\mathsf{K} \subseteq \boldsymbol{HP}\mathsf{K}$, and $\boldsymbol{PS}\mathsf{K} \subseteq \boldsymbol{SP}\mathsf{K}$ so that K is algebraically closed iff $\boldsymbol{HSP}\mathsf{K} = \mathsf{K}$; in the general case $\boldsymbol{HSP}\mathsf{K}$ is the smallest algebraically closed class including K. As was mentioned above K is algebraically closed iff it is an equational class.

We shall have occasion in Chapter 2 to use the notion of a reduced product. Let $\mathfrak{A} = \langle \mathfrak{A}_i : i \in I\rangle$ be any system of algebras and let M be a filter on I, i.e., M is a subset of $Sb\ I$ such that $I \in M$, $X \cap Y \in M$ whenever $X, Y \in M$, and $Y \in M$ whenever $X \in M$ and $X \subseteq Y \subseteq I$. Set $\bar{M}^{(\mathfrak{A})} = \bar{M} = \{\langle x, y\rangle : x, y \in \mathsf{P}\mathfrak{A}, \{i : i \in I, x_i = y_i\} \in M\}$. Then $\bar{M} \in Co\ \mathsf{P}\mathfrak{A}$. The quotient algebra $\mathsf{P}\mathfrak{A}/\bar{M}$ is called a *reduced product* of the system $\mathfrak{A}$. If $Rg\ \mathfrak{A} \subseteq \mathsf{K}$ we always have $\mathsf{P}\mathfrak{A}/\bar{M} \in \boldsymbol{HP}\mathsf{K} \subseteq \boldsymbol{HSP}\mathsf{K}$.

We now turn to the discussion of free algebras; they play an especially important role in the subsequent development. The notion of a free algebra is of model-theoretical origin. In $\mathscr{HMT}$ a free algebra with α generators over a class K of similar algebras is characterized as an algebra (with the same similarity type) which is generated by a set G with cardinality α satisfying the following condition: for any equation (in any discourse language of K) with κ variables, $\kappa \leq \alpha$, and any assignment of distinct generators to these variables, the given equation is satisfied by this assignment iff it is identically satisfied in every algebra in K. This characterization implies that any two free algebras over the same class K and with the same number of generators are isomorphic. Actually, in $\mathscr{HMT}$ there is chosen, for each K and each α, a well-defined algebra of this kind with a fixed set of generators, and this algebra is referred to as the *free algebra over* K *with* α *generators*, in symbols, $\mathfrak{Fr}_\alpha\mathsf{K}$.

The model-theoretical characterization of free algebras leads to a simple metamathematical construction of $\mathfrak{Fr}_\alpha\mathsf{K}$; following $\mathscr{HMT}$ the definition of $\mathfrak{Fr}_\alpha\mathsf{K}$ which we shall give here can be viewed as an exact mathematical paraphrase of this metamathematical construction. Any construction of a free algebra will depend of course on the similarity type we have in mind. The following definition will be made for a fixed but representative type; it is clear how the definition is to be modified so as to apply to any similarity type.

DEFINITION 1.1.2. *Consider the similarity type* $3 \upharpoonright Id = \langle\langle 0, 0\rangle, \langle 1, 1\rangle, \langle 2, 2\rangle\rangle$, *i.e., the type of an algebra with one distinguished element, one unary, and one binary operation.*

(i) *For every non-zero cardinal* α *we let*

$$Fr_\alpha = \bigcap \{X : \alpha \cup \{\langle 0, 0\rangle\} \subseteq X, \{1\} \times X \subseteq X, \{2\} \times {}^2X \subseteq X\}$$

and

$$\mathfrak{Fr}_\alpha = \langle Fr_\alpha, \langle\langle 0, x\rangle : x \in {}^0X\rangle, \langle\langle 1, x\rangle : x \in X\rangle, \langle\langle 2, x\rangle : x \in {}^2X\rangle\rangle .$$

$\mathfrak{Fr}_\alpha$ *is called the* ***absolutely free algebra (with similarity type*** $3 \upharpoonright Id$***) with*** α ***generators.*** *The ordinals* $\xi < \alpha$ *are referred to as* ***letters*** *and the elements* $x \in Fr_\alpha$ *as* ***words*** *of the algebra* $\mathfrak{Fr}_\alpha$.

(ii) *For every class* K *of algebras (with similarity type* $3 \upharpoonright Id$*) and for every cardinal* $\alpha \neq 0$ *we let*

$$Cr_\alpha\mathsf{K} = \bigcap \{R : R \in Co\, \mathfrak{Fr}_\alpha \quad and \quad \mathfrak{Fr}_\alpha/R \in \mathbf{IS}\mathsf{K}\} ;$$

we also let

$$Fr_\alpha\mathsf{K} = Fr_\alpha/Cr_\alpha\mathsf{K} \quad and \quad \mathfrak{Fr}_\alpha\mathsf{K} = \mathfrak{Fr}_\alpha/Cr_\alpha\mathsf{K} .$$

$\mathfrak{Fr}_\alpha\mathsf{K}$ *is called the* ***free algebra over*** K ***with*** α ***generators.***

The same symbols $\mathfrak{Fr}_\alpha$ and Fr_α will be used for all similarity types since it will always be clear from the context what we have in mind.

We always have $\mathfrak{Fr}_\alpha \mathsf{K} \in \boldsymbol{SP}\mathsf{K}$ and for any L such that $\mathsf{K} \subseteq \mathsf{L} \subseteq \boldsymbol{HSP}\mathsf{K}$ we have $\mathfrak{Fr}_\alpha \mathsf{K} = \mathfrak{Fr}_\alpha \mathsf{L}$.

DEFINITION 1.1.3. *Given an algebra $\mathfrak{A}$ and a class K of algebras, we say that X K-**freely generates** $\mathfrak{A}$ if $\mathfrak{A} = \mathfrak{Sg}\, X$* and, for all $\mathfrak{B} \in \mathsf{K}$ *and $f \in {}^X B$, there is an $h \in Hom(\mathfrak{A}, \mathfrak{B})$ such that $h \supseteq f$.*

An extremely useful characterization (up to isomorphism) of the free algebras is given in

THEOREM 1.1.4. *Let K be a class of algebras, and let $\mathfrak{A}$ be any algebra and $X \subseteq A$. Then the following two conditions are equivalent:*

(i) $\mathfrak{A} \in \boldsymbol{SP}\mathsf{K}$ *and X K-freely generates $\mathfrak{A}$;*

(ii) *there is an $h \in Is(\mathfrak{A}, \mathfrak{Fr}_{|X|}\mathsf{K})$ such that $h^*X = |X|/Cr_{|X|}\mathsf{K}$.*

In $\mathscr{HMT}$ there is briefly considered a certain generalization of the notion of free algebra – the so-called *free algebra with defining relations.* Like that of free algebra itself, this notion is of model-theoretical origin. But again, as in the case of free algebras, the model-theoretical characterization leads to a metamathematical construction which in turn can be easily given a purely algebraic formulation.

DEFINITION 1.1.5. *Assume K is a class of algebras and S any set of ordered pairs of words of $\mathfrak{Fr}_\alpha$. Let*

$$Cr_\alpha^{(S)}\mathsf{K} = \bigcap \{R : R \in Co\, \mathfrak{Fr}_\alpha,\ \mathfrak{Fr}_\alpha/R \in \boldsymbol{IS}\mathsf{K},\ S \subseteq R\}$$

and let

$$\mathfrak{Fr}_\alpha^{(S)}\mathsf{K} = \mathfrak{Fr}_\alpha/Cr_\alpha^{(S)}\mathsf{K}.$$

*$\mathfrak{Fr}_\alpha^{(S)}\,\mathsf{K}$ is called the **free algebra over** K **with α generators subject to the defining relations** S.*

Actually, we shall be interested only in a very special case of the notion of free algebra with defining relations. We get this special case by considering only defining relations which do not involve more than one letter at a time. Thus we obtain algebras that are not necessarily free but are generated by elements which are in a sense independent of one another.

DEFINITION 1.1.6. *Let K be a class of algebras and α a cardinal $\neq 0$. $\mathfrak{A}$ is called an **independent algebra over** K **with α generators** if $\mathfrak{A} = \mathfrak{Fr}_\alpha^{(S)}\,\mathsf{K}$ for some $S \subseteq \bigcup_{\xi<\alpha} {}^2 Sg^{(\mathfrak{Fr}\alpha)}\{\xi\}$.*[2]

[2] The notion of independent algebra defined here is very closely related to that of *weak independence* introduced in Grätzer [11]. In fact, an algebra proves to be independent over a class K in the sense of 1.1.6 iff it is generated by a set of elements weakly independent in K.

DEFINITION 1.1.7. *Given an algebra* $\mathfrak{A}$ *and a class* K *of algebras, we say that* X ***K-independently generates*** $\mathfrak{A}$ *if* $\mathfrak{A} = \mathfrak{Sg}\, X$ *and, for all* $\mathfrak{B} \in \mathsf{K}$ *and* $f \in \mathsf{P}_{x \in X} Hom(\mathfrak{Sg}^{(\mathfrak{A})}\{x\}, \mathfrak{B})$, *there is an* $h \in Hom(\mathfrak{A}, \mathfrak{B})$ *such that* $h \supseteq \bigcup_{x \in X} f_x$.

Corresponding to 1.1.4 we have the following characterization of independent algebras.

THEOREM 1.1.8. *Let* K *be a class of algebras and let* $\mathfrak{A}$ *be any algebra and* $X \subseteq A$. *Then the following two conditions are equivalent:*

(i) $\mathfrak{A} \in \mathbf{SP}\mathsf{K}$ *and* X *K-independently generates* $\mathfrak{A}$;

(ii) *there is an independent algebra* $\mathfrak{F}\mathfrak{r}_\alpha^{(S)}\mathsf{K}$ *and an* $h \in Is(\mathfrak{A}, \mathfrak{F}\mathfrak{r}_\alpha^{(S)}\mathsf{K})$ *such that* $h^*X = \alpha / Cr_\alpha^{(S)}\mathsf{K}$.

REMARK 1.1.9. It is clear that every free algebra over K is also independent over K, and, if X K-freely generates $\mathfrak{A}$, then X K-independently generates $\mathfrak{A}$.

In the final portion of this section we discuss the notion of an algebra with a distinguished system of generators.

DEFINITION 1.1.10. *Let* I *be any set. By an* ***algebra with a distinguished system of generators,*** *or simply a* **g-algebra**, *we mean an ordered pair* $\mathfrak{A} = \langle \mathfrak{A}, \langle a_i : i \in I \rangle \rangle$ *where* $\mathfrak{A}$ *is an algebra,* I *is a set, and* $\langle a_i : i \in I \rangle$ *is a system of generators of* $\mathfrak{A}$, *i.e.,* $\mathfrak{A} = \mathfrak{Sg}\{a_i : i \in I\}$.

The set I *is called the* ***index*** *of the* **g**-*algebra* $\mathfrak{A}$ *and is denoted by* $In\, \mathfrak{A}$. *A* **g**-*algebra with index* I *is often described as an* I-***indexed*** **g**-*algebra.*

The algebra $\mathfrak{A}$ *is called the* ***underlying algebra*** *of* $\mathfrak{A}$, *and the system* $\langle a_i : i \in I \rangle$ *is called the* ***distinguished system of generators*** *of* $\mathfrak{A}$.

REMARKS 1.1.11. Recall that, given an arbitrary algebra $\mathfrak{A}$ (in the ordinary sense), the subalgebra of $\mathfrak{A}$ generated by the empty set, $\mathfrak{Sg}^{(\mathfrak{A})}0$, exists iff $\mathfrak{A}$ has at least one fundamental operation of rank 0. Thus a **g**-algebra with index 0 exists only in case algebras of the similarity type under consideration contain distinguished elements.

Both **g**-algebras and systems of **g**-algebras are represented by bold-faced German capitals $\mathfrak{A}$, $\mathfrak{B}$.... We shall extend the notational convention regarding an algebra and its universe which was described in the remarks following Definition 1.1.1. To be explicit, if a **g**-algebra is denoted by a given bold-faced letter, say $\mathfrak{A}$, it will be automatically assumed that the corresponding light-faced letter, $\mathfrak{A}$, denotes the underlying algebra of $\mathfrak{A}$ and that the corresponding lower case Roman letter, a, denotes the distinguished system of generators of $\mathfrak{A}$. Thus, $\mathfrak{A} = \langle \mathfrak{A}, a \rangle$.

We shall use the letters $\mathbf{K}$ and $\mathbf{L}$ to represent classes of **g**-algebras exclusively.

The discussion of g-algebras given here will be rather detailed since, as was mentioned at the beginning of the section, the notion is not treated in $\mathscr{HMT}$; also, apart from this fact, it seems to involve some novelty. In the following definitions some familiar notions of the general theory of algebra are applied to g-algebras, and the most important properties of g-algebras which involve these notions are then stated in a number of theorems. No proofs, however, are given since all these properties are easy consequences of well-known results of the general theory.

THEOREM 1.1.12. *Let $\mathfrak{A}$ be an I-indexed* g-*algebra.*

(i) *If f is any function such that $I \subseteq Rg\, f$, then $\langle \mathfrak{A}, a \circ f \rangle$ is a* g-*algebra with index $(f^{-1})^* I$.*

(ii) *For any $J \subseteq I$ such that $\mathfrak{Sg}^{(\mathfrak{A})} a^* J$ exists (i.e., $J \neq 0$* or *$\mathfrak{A}$ contains distinguished elements), $\langle \mathfrak{Sg}^{(\mathfrak{A})} a^* J, J \upharpoonright a \rangle$ is a* g-*algebra with index J.*

If $(f^{-1})^* I \upharpoonright f$ is one-one, the g-algebra $\langle \mathfrak{A}, a \circ f \rangle$ in (i) is said to be obtained from $\mathfrak{A}$ by *change of index*. Most of the properties of g-algebras that shall concern us are invariant under change of index, and we will often make use of this fact without explicit reference to it.

The following definition is justified by 1.1.12(ii).

DEFINITION 1.1.13. *Let $\mathfrak{A}$ be a* g-*algebra. If I is any subset of $In\, \mathfrak{A}$, then the* g-*algebra $\langle \mathfrak{Sg}^{(\mathfrak{A})} a^* I, I \upharpoonright a \rangle$ with index I is called the* ***I-indexed subalgebra,*** *or the* ***I-subalgebra,*** *of $\mathfrak{A}$ and is denoted by $\mathfrak{A}^{(I)}$.*

An *algebra $\mathfrak{B}$ is called a* ***subalgebra*** *of $\mathfrak{A}$, in symbols $\mathfrak{B} \subseteq \mathfrak{A}$* or *$\mathfrak{A} \supseteq \mathfrak{B}$,* if *$\mathfrak{B} = \mathfrak{A}^{(I)}$ for some $I \subseteq In\, \mathfrak{A}$. For any class* K *of* g-*algebras we take*

$$\mathbf{SK} = \{\mathfrak{B} : \mathfrak{B} \subseteq \mathfrak{A} \quad \textit{for some} \quad \mathfrak{A} \in \mathbf{K}\}.$$

REMARK 1.1.14. The use of the notation $\mathfrak{A}^{(I)}$ for the I-subalgebra of $\mathfrak{A}$ is not completely consistent with the notational conventions of $\mathscr{HMT}$ which we have adopted here. However, the notation can be made consistent by thinking of the symbol '$\mathfrak{A}$' as actually denoting two distinct mathematical entities – the g-algebra $\mathfrak{A}$, and also a certain function whose domain is the set of subsets of $In\, \mathfrak{A}$ (except possibly the empty set) and whose range is the set of all subalgebras of $\mathfrak{A}$. The ambiguity will never cause any confusion, and the simplification in notation that results turns out to be substantial. The simplification is so substantial, in fact, that we shall extend the notational scheme (and hence compound the ambiguity) by denoting the underlying algebra of $\mathfrak{A}^{(I)}$, $\mathfrak{Sg}^{(\mathfrak{A})} a^* I$, its universe, $Sg^{(\mathfrak{A})} a^* I$, and its distinguished system of generators, $I \upharpoonright a$, by $\mathfrak{A}^{(I)}$, $A^{(I)}$ and $a^{(I)}$ respectively. In the case of $\mathfrak{A}^{(I)}$ and $A^{(I)}$ the ambiguity takes on added dimension since the meanings of these symbols are only completely determined by the g-algebra $\mathfrak{A}$, and not by just the algebra $\mathfrak{A}$ and the set A;

however, the g-algebra under consideration will always be clear from context so that no confusion is likely to occur.

Notice that $\mathfrak{A}^{(0)}$ exists (i.e., $\mathfrak{A}$ as a function has the empty set in its domain) iff in the similarity type of $\mathfrak{A}$ there is at least one distinguished element. Notice also that, for any g-algebras $\mathfrak{A}$ and $\mathfrak{B}$, $\mathfrak{A} \subseteq \mathfrak{B}$ iff $\mathfrak{A} \subseteq \mathfrak{B}$ and $a \subseteq b$.

THEOREM 1.1.15. *Let $\mathfrak{A}$, $\mathfrak{B}$ and $\mathfrak{C}$ be g-algebras. Then:*
(i) $\mathfrak{A} \subseteq \mathfrak{A}$;
(ii) *if* $\mathfrak{A} \subseteq \mathfrak{B}$ *and* $\mathfrak{B} \subseteq \mathfrak{C}$, *then* $\mathfrak{A} \subseteq \mathfrak{C}$;
(iii) *if* $\mathfrak{A} \subseteq \mathfrak{B}$ *and* $\mathfrak{B} \subseteq \mathfrak{A}$, *then* $\mathfrak{A} = \mathfrak{B}$.

Theorem 1.1.15 says that the relation $\subseteq$ is a partial ordering of the class of g-algebras.

DEFINITION 1.1.16. *Let $\mathfrak{A}$ and $\mathfrak{B}$ be g-algebras. Then $\mathfrak{A}$ is said to be* ***homomorphic*** *to $\mathfrak{B}$, in symbols $\mathfrak{A} \geqslant \mathfrak{B}$ or $\mathfrak{B} \leqslant \mathfrak{A}$, or $\mathfrak{A}$ is said to be* ***isomorphic*** *to $\mathfrak{B}$, in symbols $\mathfrak{A} \cong \mathfrak{B}$, if $\mathfrak{A}$ and $\mathfrak{B}$ have the same index, say I, and there is an h in $Ho(\mathfrak{A}, \mathfrak{B})$, or $Is(\mathfrak{A}, \mathfrak{B})$, such that $ha_i = b_i$ for every $i \in I$.*

If $\mathbf{K}$ *is any class of g-algebras, we let*

$$\boldsymbol{H}\mathbf{K} = \{\mathfrak{B} : \mathfrak{B} \leqslant \mathfrak{A} \quad \textit{for some} \quad \mathfrak{A} \in \mathbf{K}\}$$

and

$$\boldsymbol{I}\mathbf{K} = \{\mathfrak{B} : \mathfrak{B} \cong \mathfrak{A} \quad \textit{for some} \quad \mathfrak{A} \in \mathbf{K}\}.$$

As in the case of ordinary algebras a definite function τ can be correlated with the relation $\cong$ between g-algebras such that $\tau\mathfrak{A} = \tau\mathfrak{B}$ iff $\mathfrak{A} \cong \mathfrak{B}$. For any g-algebra $\mathfrak{A}$, $\tau\mathfrak{A}$ is called the *isomorphism type* of $\mathfrak{A}$.

THEOREM 1.1.17. *Let $\mathfrak{A}$, $\mathfrak{B}$, and $\mathfrak{C}$ be g-algebras. Then:*
(i) $\mathfrak{A} \leqslant \mathfrak{A}$;
(ii) *if* $\mathfrak{A} \leqslant \mathfrak{B}$ *and* $\mathfrak{B} \leqslant \mathfrak{C}$, *then* $\mathfrak{A} \leqslant \mathfrak{C}$;
(iii) *if* $\mathfrak{A} \leqslant \mathfrak{B}$ *and* $\mathfrak{B} \leqslant \mathfrak{A}$, *then* $\mathfrak{A} \cong \mathfrak{B}$.

By 1.1.17, for each set I the relation $\leqslant$ establishes a partial ordering of the set of isomorphism types of I-indexed g-algebras.

DEFINITION 1.1.18. *Let I be a set and $\mathfrak{A} = \langle \mathfrak{A}_m : m \in M \rangle$ a system of I-indexed g-algebras. Then by the* ***canonical subdirect product*** *of $\mathfrak{A}$, denoted by $\mathsf{P}\mathfrak{A}$ or by $\mathsf{P}_{m \in M}\mathfrak{A}_m$, we mean the g-algebra $\mathfrak{B}$ with index I where*

$$\mathfrak{B} = \mathfrak{Sg}^{(\mathsf{P}_{m \in M}\mathfrak{A}_m)} \{\langle a_m(i) : m \in M \rangle : i \in I\}$$

and $b = \langle \langle a_m(i) : m \in M \rangle : i \in I \rangle$. If $M = 2$, we write $\mathsf{P}_{m \in M}\mathfrak{A} = \mathfrak{A}_0 \times \mathfrak{A}_1$.

If $\mathbf{K}$ *is a class of* g*-algebras, we take* $\mathsf{P}_S\mathbf{K}$ *to be the class of all* g*-algebras* $\mathfrak{B}$ *such that* $\mathfrak{B}$ *is isomorphic to a canonical subdirect product* $\mathsf{P}_{m\in M}\mathfrak{A}_m$ *of a system* $\langle \mathfrak{A}_m : m \in M\rangle$ *of* g*-algebras all with the same index.*

THEOREM 1.1.19. *Let I be a set, and let* $\mathfrak{A}$ *be an I-indexed* g*-algebra and* $\langle \mathfrak{B}_m : m \in M\rangle$ *a system of I-indexed* g*-algebras.*

(i) $\mathfrak{A} \succcurlyeq \mathfrak{B}_m$ *for every* $m \in M$ *iff* $\mathfrak{A} \succcurlyeq \mathsf{P}_{m\in M}\mathfrak{B}_m$.

(ii) *Suppose* $\mathfrak{A} \succcurlyeq \mathfrak{B}_m$ *for every* $m \in M$ *and, in addition, suppose that for any distinct* $x, y \in A$ *there is an* $m \in M$ *such that* $hx \neq hy$ *where h is the unique homomorphism from* $\mathfrak{A}$ *onto* $\mathfrak{B}_m$ *such that* $h \circ a = b$. *Then* $\mathfrak{A} \cong \mathsf{P}_{m\in M}\mathfrak{B}_m$.

REMARK 1.1.20. From 1.1.19 (i) we see that the isomorphism type of the product $\mathsf{P}_{m\in M}\mathfrak{B}_m$ is the least upper bound, under the partial ordering induced by $\preccurlyeq$, of the set of isomorphism types of the $\mathfrak{B}_m$ with $m \in M$. Thus we see that for any fixed set I the partial ordering of isomorphism types of g-algebras with index I is a complete lattice ordering; the corresponding complete lattice is called the *lattice of isomorphism types of I-indexed* g*-algebras*, or simply the *lattice of I-indexed* g*-algebras*. In discussing this lattice we shall often not bother to distinguish between a g-algebra and its isomorphism type.

In view of these remarks the next theorem is obvious.

THEOREM 1.1.21. *Let* $\mathfrak{A}$ *and* $\mathfrak{B}$ *be* g*-algebras with the same index. Then* $\mathfrak{A} \preccurlyeq \mathfrak{B}$ *iff* $\mathfrak{A} \times \mathfrak{B} \cong \mathfrak{B}$.

This theorem shows that the canonical subdirect product of g-algebras differs strikingly in its properties from the ordinary product of algebras.

THEOREM 1.1.22. *Let I be any set and* $\langle \mathfrak{A}_m : m \in M\rangle$ *a system of I-indexed* g*-algebras. Then for every non-empty* $J \subseteq I$ *we have*

$$(\mathsf{P}_{m\in M}\mathfrak{A}_m)^{(J)} = \mathsf{P}_{m\in M}\mathfrak{A}_m^{(J)}.$$

The conclusion also holds for $J = 0$ *provided the subalgebras in question exist.*

From this theorem we see that the function $\langle \mathfrak{A}^{(J)} : \mathfrak{A}$ an I-indexed g-algebra$\rangle$ is a complete lattice homomorphism from the lattice of I-indexed g-algebras onto the lattice of J-indexed g-algebras. This observation leads in turn to the following theorem which, however, also has a very easy direct proof.

THEOREM 1.1.23. *Let I be any set and* $\mathfrak{A}, \mathfrak{B}$ *I-indexed* g*-algebras. If* $\mathfrak{A} \preccurlyeq \mathfrak{B}$, *or* $\mathfrak{A} \cong \mathfrak{B}$, *then* $\mathfrak{A}^{(J)} \preccurlyeq \mathfrak{B}^{(J)}$, *or* $\mathfrak{A}^{(J)} \cong \mathfrak{B}^{(J)}$, *for every* $J \subseteq I$ *provided* $\mathfrak{A}^{(J)}$ *(or, equivalently,* $\mathfrak{B}^{(J)}$*) exists.*

REMARK 1.1.24. Most of the familiar terminology of the general theory of algebras can be applied directly to an algebra $\mathfrak{A}$ with a distinguished system of generators; no confusion can arise since in these cases the only possible interpretation is that we are referring to the underlying algebra $\mathfrak{A}$. For example, when we say that the g-algebras $\mathfrak{A}$ and $\mathfrak{B}$ are similar we mean that the algebras $\mathfrak{A}$ and $\mathfrak{B}$ are similar. Also, when we say that $\mathfrak{A}$ is minimal, or simple, we mean of course that $\mathfrak{A}$ is minimal, or simple, and by a congruence relation on $\mathfrak{A}$ we mean one on $\mathfrak{A}$. Notice, however, certain important exceptions to this rule: Subalgebras of $\mathfrak{A}$ and $\mathfrak{A}$ are not the same. Also, the homomorphic images of $\mathfrak{A}$ are not the same as those of $\mathfrak{A}$ (but in this case they all appear as underlying algebras of homomorphic images of $\mathfrak{A}$).

Following this line, a g-algebra $\mathfrak{A}$ is semisimple if $\mathfrak{A}$ is semisimple, i.e., if $\mathfrak{A}$ is a subdirect product of simple algebras. However, making use of the canonical subdirect product, semisimplicity can be easily defined in terms of the g-algebras themselves.

DEFINITION 1.1.25. *A* g-*algebra* $\mathfrak{A}$ *is* ***semisimple*** *if it is isomorphic to a canonical subdirect product of simple* g-*algebras.*

THEOREM 1.1.26. *Let* $\mathfrak{A}$ *be a* g-*algebra with index* I *and let* $R \in Co\ \mathfrak{A}$. *Then* $\langle \mathfrak{A}/R, \langle a_i/R : i \in I\rangle\rangle$ *is a* g-*algebra with the same index* I *as* $\mathfrak{A}$.

In view of this theorem the following definition makes sense.

DEFINITION 1.1.27. *Given any* g-*algebra, and any* $R \in Co\ \mathfrak{A}$ *we call* $\langle \mathfrak{A}/R, \langle a_i/R : i \in In\ \mathfrak{A}\rangle\rangle$ *the* ***quotient algebra*** *of* $\mathfrak{A}$ *over* R *and denote it by* $\mathfrak{A}/R$. *The distinguished system of generators of* $\mathfrak{A}/R$, $\langle a_i/R : i \in In\ \mathfrak{A}\rangle$, *is denoted by* a/R.

THEOREM 1.1.28. *Let* $\mathfrak{A}$ *be a* g-*algebra.*

(i) *If* $R \in Co\ \mathfrak{A}$, *then* $\mathfrak{A}/R \preccurlyeq \mathfrak{A}$; *conversely, if* $\mathfrak{B}$ *is a* g-*algebra and* $\mathfrak{B} \preccurlyeq \mathfrak{A}$, *then there exists a unique* $R \in Co\ \mathfrak{A}$ *such that* $\mathfrak{A}/R \cong \mathfrak{B}$.

(ii) *If* $R, S \in Co\ \mathfrak{A}$, *then* $\mathfrak{A}/R \preccurlyeq \mathfrak{A}/S$ *iff* $R \supseteq S$.

REMARK 1.1.29. The above theorem implies that the sublattice of the lattice of I-indexed g-algebras whose universe is the principal ideal (in the usual lattice-theoretic sense) determined by the isomorphism type of a given g-algebra $\mathfrak{A}$ is dually isomorphic to the lattice of congruence relations on $\mathfrak{A}$. In particular, the whole lattice of I-indexed g-algebras is dually isomorphic to the lattice of congruence relations on the absolutely free algebra $\mathfrak{Fr}_{|I|}$.

The next theorem is a generalization of the first isomorphism theorem of group theory.

THEOREM 1.1.30. *Assume* $\mathfrak{A}$ *is a* g-*algebra,* $I \subseteq In\,\mathfrak{A}$, *and* $R \in Co\,\mathfrak{A}$. *Then*

$$(\mathfrak{A}/R)^{(I)} \cong \mathfrak{A}^{(I)}/(R \cap {}^2A^{(I)}).$$

REMARK 1.1.31. It should be remarked that by specifying new distinguished elements of an algebra (and thus extending its similarity type) the theory of algebras with a distinguished system of generators can be subsumed under that of ordinary algebras. In order to do this we can replace each g-algebra

$$\mathfrak{A} = \langle\langle \mathfrak{A}, O_i\rangle_{i \in I}, \langle a_j\colon j \in In\,\mathfrak{A}\rangle\rangle$$

by the ordinary algebra

$$\mathfrak{A}^* = \langle \mathfrak{A}, \langle Q_m\colon m \in I \cup In\,\mathfrak{A}\rangle\rangle$$

where $Q_i = O_i$ for $i \in I$ and $Q_j = a_j$ for $j \in In\,\mathfrak{A}$ (we are assuming here for convenience that $I \cap In\,\mathfrak{A} = 0$). Then the fundamental notions defined in 1.1.13, 1.1.16, and 1.1.18 can be expressed in terms of the algebras $\mathfrak{A}^*$, the ordinary definitions of subalgebra, homomorphism, and direct product, and the notion of *reduct* (cf. $\mathscr{HMT}$, Section 0.5). It seems to us however, that especially for the purposes of this paper the notion of g-algebra defined in 1.1.10 is a convenient and useful one.

We close this section with a brief discussion of notions of freedom and independence for g-algebras.

DEFINITION 1.1.32. (i) *For every cardinal* $\alpha \neq 0$ *we let*

$$\mathfrak{Fr}_\alpha = \langle \mathfrak{Fr}_\alpha, \alpha \upharpoonright Id\rangle.$$

$\mathfrak{Fr}_\alpha$ *is called the* ***absolutely free*** g-***algebra with index*** α.

(ii) *For every class* $\mathbf{K}$ *of* g-*algebras and for every cardinal* $\alpha \neq 0$ *we let*

$$\mathfrak{Fr}_\alpha\mathbf{K} = \mathfrak{Fr}_\alpha/S$$

where

$$S = \bigcap\{R\colon R \in Co\,\mathfrak{Fr}_\alpha \quad \text{and} \quad \mathfrak{Fr}_\alpha/R \in \boldsymbol{ISK}\}.$$

$\mathfrak{Fr}_\alpha\mathbf{K}$ *is called the* ***free*** g-***algebra over*** $\mathbf{K}$ ***with index*** α.

REMARKS 1.1.33. Notice that, for any class $\mathbf{K}$ of g-algebras, $\mathfrak{Fr}_\alpha\mathbf{K} = \mathfrak{Fr}_\alpha\mathbf{K}'$ where $\mathbf{K}'$ is the class of g-algebras in $\mathbf{K}$ with index α; i.e., all the g-algebras of $\mathbf{K}$ with index different from α are irrelevant in determining $\mathfrak{Fr}_\alpha\mathbf{K}$. Thus it turns out that the notion of a free g-algebra with index α is much narrower in scope than the corresponding notion of free algebra with α generators. But this of course is to be expected: a proper treatment of free g-algebras would require us to define such a g-algebra (over any given class $\mathbf{K}$ of g-algebras) with index I for every non-empty set I. However, although it seems natural to attempt such a definition by substituting the set I for the set α in Definition 1.1.2, in certain special cases (e.g., if $\langle 0, i\rangle \in I$ for some $i \in I$) 1.1.2 does

not yield an algebra with the desired properties. Difficulties like this can of course be easily surmounted. On the other hand, we shall be dealing exclusively in Chapter 2 with classes of g-algebras closed under change of index, and for each such class $\mathbf{K}$ every free g-algebra over $\mathbf{K}$ (with arbitrary index) can be obtained from one of the $\mathfrak{Fr}_\alpha\mathbf{K}$ by change of index.

As opposed to the case for free g-algebras, freely generated g-algebras are easy to define and the definition actually assumes an almost elegant form:

DEFINITION 1.1.34. *Let $\mathfrak{A}$ be a* g*-algebra and $\mathbf{K}$ any class of* g*-algebras. $\mathfrak{A}$ is said to be* ***$\mathbf{K}$-freely generated*** *if $\mathfrak{B} \preccurlyeq \mathfrak{A}$ whenever $\mathfrak{B} \in \mathbf{K}$ and $In\ \mathfrak{B} = In\ \mathfrak{A}$.*

REMARK 1.1.35. For a g-algebra $\mathfrak{A}$ with index I to be $\mathbf{K}$-freely generated means simply that (the isomorphism type of) $\mathfrak{A}$ is an upper bound in the lattice of I-indexed g-algebras for the set of all (isomorphism types of) members of $\mathbf{K}$ with index I. Thus, if $\mathfrak{A}$ is both $\mathbf{K}$-freely generated and a member of $\mathbf{K}$, then $\mathfrak{A}$ is unique up to isomorphism; if, in addition, $\mathbf{K}$ is closed under the formation of canonical subdirect products, then (disregarding any set-theoretical difficulties involved in forming the direct product of a proper class of algebras) we must have $\mathfrak{A} \cong \mathsf{P}_{\mathfrak{B} \in \mathbf{L}}\mathfrak{B}$ where $\mathbf{L}$ is the class of all I-indexed g-algebras in $\mathbf{K}$.

By comparing Definitions 1.1.2, 1.1.5, 1.1.6, and 1.1.32 it is easily seen how one should define the notions of *free* g*-algebra with defining relations* and *independent* g-algebra over arbitrary classes of g-algebras. We shall not give this definition here, but certain important kinds of independent cylindric g-algebras will be defined in Section 2.1; cf. 2.1.4.

DEFINITION 1.1.36. *Let $\mathfrak{A}$ and $\mathbf{K}$ be as in* 1.1.34. *$\mathfrak{A}$ is said to be* ***$\mathbf{K}$-independently generated*** *if $\mathfrak{B} \preccurlyeq \mathfrak{A}$ whenever $\mathfrak{B} \in \mathbf{K}$, $In\ \mathfrak{B} = In\ \mathfrak{A}$, and $\mathfrak{B}^{(\{i\})} \preccurlyeq \mathfrak{A}^{(\{i\})}$ for each $i \in In\ \mathfrak{A}$.*

REMARK 1.1.37. In view of 1.1.23, if $\mathfrak{A}$ is $\mathbf{K}$-independently generated and M is any set of subsets of $In\ \mathfrak{A}$ such that $\bigcup M = In\ \mathfrak{A}$, then $\mathfrak{B} \preccurlyeq \mathfrak{A}$ whenever $\mathfrak{B} \in \mathbf{K}$, $In\ \mathfrak{B} = In\ \mathfrak{A}$, and $\mathfrak{B}^{(J)} \preccurlyeq \mathfrak{A}^{(J)}$ for every $J \in M$.

The last theorem of this section is an obvious corollary of 1.1.34 and 1.1.36.

THEOREM 1.1.38. *Let $\mathbf{K}$ be a class of* g*-algebras. Then every $\mathbf{K}$-freely generated* g*-algebra is $\mathbf{K}$-independently generated.*

1.2 Basic notions and results

From the discussion of this section it will be seen that, for any class K of algebras satisfying certain closure conditions, the question of whether K has the amalgamation

property can be reduced to a question about congruence relations on the K-independently generated algebras in K.

It turns out for our purposes to be more convenient to carry out the study of the amalgamation property in terms of arbitrary classes of g-algebras, and then to apply the whole discussion directly to a class K of ordinary algebras by simply considering in place of K the class $\mathbf{K}$ of all g-algebras $\mathfrak{A}$ such that $\mathfrak{A} \in \mathsf{K}$.

Indeed, g-algebras were introduced here expressly for the purpose of treating the amalgamation property in as convenient a way as possible. A definition of this property stated in terms of g-algebras will be given below in 1.2.1. It will be clear from this definition why in the subsequent discussion we shall restrict ourselves to those g-algebras $\mathfrak{A}$ for which $\mathfrak{A}^{(0)}$ exists, i.e., which have some distinguished elements.

DEFINITION 1.2.1. *Let* $\mathbf{K}$ *be a class of* g-*algebras.*

(i) $\mathbf{K}$ *is said to have the* ***amalgamation property****, or the* $\mathscr{AP}$ *for short, if the following condition is satisfied by any pair of sets I, J:*

(i′) *for any pair of* g-*algebras* $\mathfrak{A}, \mathfrak{B} \in \mathbf{K}$ *with indexes I and J, respectively, if* $\mathfrak{A}^{(I \cap J)}, \mathfrak{B}^{(I \cap J)} \in I\mathbf{K}$ *and* $\mathfrak{A}^{(I \cap J)} \cong \mathfrak{B}^{(I \cap J)}$*, then there is a* $\mathfrak{C} \in \mathbf{K}$ *with index* $I \cup J$ *such that*

$$\mathfrak{C}^{(I)} \cong \mathfrak{A} \quad \text{and} \quad \mathfrak{C}^{(J)} \cong \mathfrak{B}.$$

(ii) $\mathbf{K}$ *is said to have the* ***embedding property****, or the* $\mathscr{EP}$ *for short, if condition (i′) is satisfied by any pair of sets I, J such that* $I \cap J = 0$.

The terminology introduced in the following definition will prove to be very convenient in discussions involving the amalgamation property.

DEFINITION 1.2.2. *Let* $\mathbf{L}$ *be a set of* g-*algebras.*

(i) $\mathbf{L}$ *is said to be* ***compatible*** *if for all* $\mathfrak{A}, \mathfrak{B} \in \mathbf{L}$ *we have* $\mathfrak{A}^{(I)} \cong \mathfrak{B}^{(I)}$ *where* $I = In\,\mathfrak{A} \cap In\,\mathfrak{B}$.

(ii) *A* g-*algebra* $\mathfrak{A}$ *is said to* ***cover*** $\mathbf{L}$ *if* $In\,\mathfrak{A} = \bigcup_{\mathfrak{B} \in \mathbf{L}} In\,\mathfrak{B}$ *and* $\mathfrak{B} \preccurlyeq \mathfrak{A}^{(In\,\mathfrak{B})}$ *for every* $\mathfrak{B} \in \mathbf{L}$.

REMARK 1.2.3. Of particular interest among the g-algebras $\mathfrak{A}$ which cover $\mathbf{L}$ are those for which $\mathfrak{A}^{(In\,\mathfrak{B})} \cong \mathfrak{B}$ whenever $\mathfrak{B} \in \mathbf{L}$. For the purposes of this remark let us denote by $\mathbf{C}_{\mathbf{L}}$ the class of covering algebras with this special property. In order for $\mathbf{C}_{\mathbf{L}}$ to be non-empty $\mathbf{L}$ must necessarily be compatible, and it is not difficult to see that this condition is also sufficient. We arrive at an interesting problem when we take an arbitrary class $\mathbf{K}$ of g-algebras and ask for which sets $\mathbf{L}$ do we have $\mathbf{K} \cap \mathbf{C}_{\mathbf{L}} \neq 0$? In asking this question we shall restrict our attention to those sets $\mathbf{L}$ which satisfy the following condition:

(I) $\mathbf{L} \subseteq \mathbf{K}$, $\mathbf{L}$ is compatible, and $\mathfrak{A}^{(I)}, \mathfrak{B}^{(I)} \in \mathbf{K}$ whenever $\mathfrak{A}, \mathfrak{B} \in \mathbf{L}$ and $I = In\,\mathfrak{A} \cap \cap\, In\,\mathfrak{B}$.

To say that $\mathbf{K} \cap \mathbf{C}_{\mathbf{L}} \neq 0$ for every 2-element set $\mathbf{L}$ satisfying (I) is just another way of saying that $\mathbf{K}$ has the $\mathscr{AP}$, and in this case a simple inductive argument shows that $\mathbf{K} \cap \mathbf{C}_{\mathbf{L}} \neq 0$ for every finite $\mathbf{L}$ satisfying (I) as well. The question then naturally arises whether $\mathbf{K} \cap \mathbf{C}_{\mathbf{L}} \neq 0$ for arbitrary $\mathbf{L}$ satisfying (I) if $\mathbf{K}$ has the $\mathscr{AP}$. It is easy to see that this is in general not true, but it is true whenever $\mathbf{K}$ satisfies some natural closure conditions – for instance, if $\boldsymbol{H}\mathbf{K} = \boldsymbol{S}\mathbf{K} = \boldsymbol{P}_{\mathrm{S}}\mathbf{K} = \mathbf{K}$; certain weaker conditions are also known to be sufficient however. Cf. in this regard Jónsson [16].

Given an arbitrary $\mathbf{K}$ and an $\mathbf{L}$ satisfying (I) we can go further and ask if the class $\mathbf{K} \cap \mathbf{C}_{\mathbf{L}}$ contains a greatest element (under the ordering $\preccurlyeq$). Such a greatest element if its exists, is called a $\mathbf{K}$-*free amalgamated product of* $\mathbf{L}$; if $\mathbf{L}$ has the property that indexes of distinct members of $\mathbf{L}$ are disjoint, then a $\mathbf{K}$-free amalgamated product of $\mathbf{L}$ is called a $\mathbf{K}$-*free product*. It is clearly seen that any two $\mathbf{K}$-free amalgamated products of $\mathbf{L}$ are isomorphic.

Under what conditions do free amalgamated products exist? The following partial answer to this question is easily checked. If $\mathbf{K}$ has the $\mathscr{AP}$ and $\boldsymbol{P}_{\mathrm{S}}\mathbf{K} = \mathbf{K}$, then $\mathbf{K}$ must contain a $\mathbf{K}$-free amalgamated product of $\mathbf{L}$ whenever $\mathbf{L}$ is finite and (I) is satisfied; if in addition $\boldsymbol{H}\mathbf{K} = \boldsymbol{S}\mathbf{K} = \mathbf{K}$, then we can drop the condition that $\mathbf{L}$ be finite. The same result holds when we replace $\mathscr{AP}$ by $\mathscr{EP}$ and $\mathbf{K}$-free amalgamated product by $\mathbf{K}$-free product.

We shall have only one other occasion (in Remark 1.2.9) to deal with free amalgamated products in this paper; it is in connection with the fact that, if $\boldsymbol{S}\mathbf{K} = \boldsymbol{I}\mathbf{K} = \mathbf{K}$, then the $\mathbf{K}$-independently generated members of $\mathbf{K}$ can be characterized as those g-algebras $\mathfrak{A}$ such that $\mathfrak{A}$ is a $\mathbf{K}$-free product of the set $\mathbf{L} = \{\mathfrak{A}^{(\{i\})} : i \in In\,\mathfrak{A}\}$. This fact is an immediate consequence of Definition 1.1.36.

The following lemma (closely related to some results known from the literature, e.g., Sikorski [24], p. 217) will be used in the proof of 1.2.5.

LEMMA 1.2.4. *Let* $\mathbf{K}$ *be a class of* g-*algebras such that* $\boldsymbol{P}_{\mathrm{S}}\mathbf{K} = \mathbf{K}$.

(i) *In order for* $\mathbf{K}$ *to have the* $\mathscr{AP}$ *it is necessary and sufficient that the following condition be satisfied by any pair of sets* I, J:

(i′) *for any compatible pair of* g-*algebras* $\mathfrak{A}, \mathfrak{B} \in \mathbf{K}$ *with indices* I *and* J, *respectively, if* $\mathfrak{A}^{(I \cap J)}$ *(and hence* $\mathfrak{B}^{(I \cap J)}$*)* $\in \boldsymbol{I}\mathbf{K}$, *then there is a* $\mathfrak{C} \in \mathbf{K}$ *covering* $\mathfrak{A}, \mathfrak{B}$ *such that*

$$\mathfrak{C}^{(J)} \cong \mathfrak{A} \quad \text{and} \quad \mathfrak{C}^{(J)} \preccurlyeq \mathfrak{B}.$$

(ii) *In order for* $\mathbf{K}$ *to have the* $\mathscr{EP}$ *it is necessary and sufficient that condition (i′) be satisfied by any pair of sets* I, J *such that* $I \cap J = 0$.

PROOF. We first prove part (i). If $\mathbf{K}$ has the $\mathscr{AP}$, then it is clear that

(1) (i′) is satisfied by any pair of sets I, J.

Assume, conversely, that (1) holds and that I and J are any sets, $\mathfrak{A}$, $\mathfrak{B}$ is any compatible pair of g-algebras in $\mathbf{K}$, and

(2) $\mathfrak{A}$ and $\mathfrak{B}$ have indexes I and J, respectively.

Assume further that $\mathfrak{A}^{(I \cap J)} \in I\mathbf{K}$. Since (1) is assumed to hold we have g-algebras $\mathfrak{C}$ and $\mathfrak{D}$ in $\mathbf{K}$ both with index $I \cup J$ such that

(3) $$\mathfrak{C}^{(I)} \cong \mathfrak{A} \quad \text{and} \quad \mathfrak{C}^{(J)} \preccurlyeq \mathfrak{B}$$

and

(4) $$\mathfrak{D}^{(I)} \preccurlyeq \mathfrak{A} \quad \text{and} \quad \mathfrak{D}^{(J)} \cong \mathfrak{B}.$$

Let

$$\mathfrak{E} = \mathfrak{C} \times \mathfrak{D}.$$

Then by 1.1.21, 1.1.22, (3), and (4) we obtain

$$\mathfrak{E}^{(I)} \cong \mathfrak{C}^{(I)} \times \mathfrak{D}^{(I)} \cong \mathfrak{A}$$

and

$$\mathfrak{E}^{(J)} \cong \mathfrak{C}^{(J)} \times \mathfrak{D}^{(J)} \cong \mathfrak{B}.$$

Thus $\mathbf{K}$ has the $\mathscr{AP}$ and the proof of part (i) is complete.

By replacing '$\mathscr{AP}$' everywhere by '$\mathscr{EP}$' and assuming I and J are disjoint in (1) and (2), we obtain the proof of (ii) from that of (i).

THEOREM 1.2.5. *Let* $\mathbf{K}$ *be a class of* g-*algebras satisfying the following conditions:*

(i) $H\mathbf{K} = P_S\mathbf{K} = \mathbf{K}$;

(ii) *if* $\mathfrak{A}, \mathfrak{B} \in \mathbf{K}$, $\mathfrak{A} \subseteq \mathfrak{B}$, *and* $\mathfrak{B}$ *is simple, then* $\mathfrak{A}$ *is simple;*

(iii) *if* $\mathfrak{A}, \mathfrak{B}, \mathfrak{C} \in \mathbf{K}$, $\mathfrak{A}$ *is simple, and* $\mathfrak{A} \preccurlyeq \mathfrak{B} \subseteq \mathfrak{C}$, *then there exists a simple* $\mathfrak{D} \in \mathbf{K}$ *such that*

$$\mathfrak{A} \cong \mathfrak{D}^{(In\,\mathfrak{A})} \quad \textit{and} \quad \mathfrak{D} \preccurlyeq \mathfrak{C}.$$

If the class of all simple g-*algebras in* $\mathbf{K}$ *has the* $\mathscr{AP}$, *or the* $\mathscr{EP}$, *then so does the class of all semisimple* g-*algebras in* $\mathbf{K}$.

PROOF. Let

$$\mathbf{S} = \{\mathfrak{A} : \mathfrak{A} \in \mathbf{K}, \mathfrak{A} \text{ simple}\}, \quad \mathbf{L} = \{\mathfrak{B} : \mathfrak{B} \in \mathbf{K}, \mathfrak{B} \text{ semisimple}\}.$$

We first prove that $\mathbf{L}$ has the $\mathscr{AP}$ if $\mathbf{S}$ has this property. So assume that $\mathbf{S}$ has the $\mathscr{AP}$ and consider any pair of sets I, J and any pair of g-algebras $\mathfrak{A}, \mathfrak{B} \in \mathbf{L}$ with indexes I and J, respectively; assume further that $\mathfrak{A}^{(I \cap J)}$, $\mathfrak{B}^{(I \cap J)} \in \mathbf{K}$ (note that $I\mathbf{K} = \mathbf{K}$ by

(i)), and assume that $\mathfrak{A}$, $\mathfrak{B}$ are compatible, i.e.,

$$\mathfrak{A}^{(I \cap J)} \cong \mathfrak{B}^{(I \cap J)}. \tag{1}$$

Since $\mathfrak{A}$ is semisimple, there is a system $\langle \mathfrak{C}_m : m \in M \rangle$ of g-algebras such that

$$\mathfrak{C}_m \text{ is simple} \tag{2}$$

for each $m \in M$, and

$$\mathfrak{A} \cong \mathsf{P}_{m \in M} \mathfrak{C}_m. \tag{3}$$

Notice that from (3), (i), and 1.1.19(i) we have $\mathfrak{C}_m \in \mathbf{K}$ and hence, by (2),

$$\mathfrak{C}_m \in \mathbf{S} \tag{4}$$

for every $m \in M$.

By (1), (3), and 1.1.22 we obtain

$$\mathfrak{B}^{(I \cap J)} \cong \mathsf{P}_{m \in M} \mathfrak{C}_m^{(I \cap J)};$$

therefore, by 1.1.19(i),

$$\mathfrak{C}_m^{(I \cap J)} \preccurlyeq \mathfrak{B}^{(I \cap J)} \tag{5}$$

for every $m \in M$. Since each $\mathfrak{C}^{(I \cap J)}$ is simple by (4) and (ii) we can apply (iii) to obtain from (5) a J-indexed g-algebra $\mathfrak{D}_m$ for each $m \in M$ such that

$$\mathfrak{D}_m \in \mathbf{S}, \tag{6}$$

$$\mathfrak{D}_m^{(I \cap J)} \cong \mathfrak{C}_m^{(I \cap J)}, \tag{7}$$

and

$$\mathfrak{D}_m \preccurlyeq \mathfrak{B}. \tag{8}$$

From the assumption that $\mathfrak{B}^{(I \cap J)} \in \mathbf{K}$ we conclude by means of (5), (7), and (i) that, for each $m \in M$, $\mathfrak{C}_m^{(I \cap J)}, \mathfrak{D}_m^{(I \cap J)} \in \mathbf{K}$ and hence by (2), (6), and (ii),

$$\mathfrak{C}_m^{(I \cap J)}, \mathfrak{D}_m^{(I \cap J)} \in \mathbf{S}. \tag{9}$$

Because (4), (6), (7), and (9) hold for every $m \in M$, and $\mathbf{S}$ is assumed to have the $\mathscr{AP}$, we infer that there is a g-algebra $\mathfrak{E}_m$ with index $I \cup J$ for each $m \in M$ such that

$$\mathfrak{E}_m \in \mathbf{S}, \tag{10}$$

$$\mathfrak{E}_m^{(I)} \cong \mathfrak{C}_m, \tag{11}$$

and

$$\mathfrak{E}_m^{(J)} \cong \mathfrak{D}_m. \tag{12}$$

By (3), (11), and 1.1.22 we have

$$(\mathsf{P}_{m \in M}\mathfrak{E}_m)^{(I)} \cong \mathfrak{A} \tag{13}$$

and by (8), (12), 1.1.19(i), and 1.1.22 we have

$$(\mathsf{P}_{m \in M}\mathfrak{E}_m)^{(J)} \preccurlyeq \mathfrak{B}. \tag{14}$$

Since $\mathsf{P}_{m \in M}\mathfrak{E}_m \in \mathbf{K}$ and is semisimple by (10) and (i), we see from (13) and (14) that condition 1.2.4(i′) with $\mathbf{K}$ replaced by $\mathbf{L}$ is satisfied by any pair of sets I, J; in view of (i), $\mathbf{L}$ is clearly closed under the formation of canonical subdirect products, so that we may apply Lemma 1.2.4(i) to infer that $\mathbf{L}$ has the $\mathscr{AP}$.

If at the beginning of the proof just given we assume that $\mathbf{S}$ has the $\mathscr{EP}$ rather than the $\mathscr{AP}$, and consider any pair of sets I, J such that $I \cap J = 0$, we obtain a proof of the fact that condition 1.2.4(i′) with $\mathbf{K}$ replaced by $\mathbf{L}$ is satisfied by any pair of sets I, J with $I \cap J = 0$. Thus applying 1.2.4(ii) we conclude that $\mathbf{L}$ has the $\mathscr{EP}$. This completes the proof of the theorem.

REMARK 1.2.6. Theorem 1.2.5 has some interesting applications. For example, let $\mathbf{K}$ be the class of all Boolean g-algebras, i.e., all g-algebras $\mathfrak{A}$ such that $\mathfrak{A}$ is a Boolean algebra (here complementation, the zero element, and the unit element are all taken to be fundamental operations). Then condition 1.2.5(i) clearly holds, and 1.2.5(ii) holds as well since every simple Boolean algebra has just two elements. Finally, 1.2.5(iii) is an easy consequence of some simple facts about ideals in Boolean algebras. The class of all simple Boolean g-algebras trivially has the $\mathscr{AP}$, so that we can apply 1.2.5 and conclude that the class of Boolean g-algebras, and hence the class of Boolean algebras (cf. 1.2.12 below), has the $\mathscr{AP}$; cf. Remark 2.2.30 below.

Theorem 1.2.5 is closely related to Theorem 3 of Grätzer-Lakser [12], and, indeed, the two theorems prove to be equivalent when applied to algebraically closed classes of semisimple algebras. Grätzer and Lakser's theorem is more useful in application however. For example, in op. cit. it is employed to completely characterize the algebraically closed classes of pseudo-complemented distributive lattices that have the $\mathscr{AP}$; the fact that Boolean algebras have the $\mathscr{AP}$, which was seen above to be a consequence of 1.2.5, is just a special case of this result.

DEFINITION 1.2.7. *Let $\mathfrak{A}$ be a* g-*algebra* and $C \subseteq \bigcup_{\mathfrak{B} \subseteq \mathfrak{A}} Co\, \mathfrak{B}$.

(i) *$\mathfrak{A}$ is said to have the* ***congruence-extension property*** *(the $\mathscr{CP}$ for short)* ***relative to*** *C if the following condition is satisfied by any pair of sets I, J such that $I \cup J = In\, \mathfrak{A}$:*

(i′) *For every pair of congruence relations $R \in Co\, \mathfrak{A}^{(I)} \cap C$ and $S \in Co\, \mathfrak{A}^{(J)} \cap C$ such that*

$$R \cap {}^2A^{(I \cap J)} = S \cap {}^2A^{(I \cap J)} \in C$$

there is a $T \in Co\,\mathfrak{A} \cap C$ *such that*

$$T \cap {}^2A^{(I)} = R \quad \textit{and} \quad T \cap {}^2A^{(J)} = S.$$

(ii) $\mathfrak{A}$ *is said to have the* ***restricted*** $\mathscr{CP}$ ***relative to*** C *if condition (i′) is satisfied by any pair of sets* I, J *such that* $I \cup J = In\,\mathfrak{A}$ *and* $I \cap J = 0$.

In case $C = \bigcup_{\mathfrak{B} \subseteq \mathfrak{A}} Co\,\mathfrak{B}$ *we say simply that* $\mathfrak{A}$ *has the* $\mathscr{CP}$ *or the* ***restricted*** $\mathscr{CP}$.

The next theorem constitutes the principal result of this chapter; it gives a precise statement of the connection between the $\mathscr{AP}$ and the $\mathscr{CP}$.

THEOREM 1.2.8. *Assume that* $\mathbf{K}$ *is a class of* g*-algebras such that* $\mathbf{SK} = \mathbf{HK} = \mathbf{K}$.

(i) *If every compatible pair of* g*-algebras in* $\mathbf{K}$ *is covered by a member of* $\mathbf{K}$ *with the* $\mathscr{CP}$, *then* $\mathbf{K}$ *has the* $\mathscr{AP}$. *On the other hand, if* $\mathbf{K}$ *has the* $\mathscr{AP}$, *then every* $\mathbf{K}$*-independently generated* g*-algebra in* $\mathbf{K}$ *has the* $\mathscr{CP}$.

(ii) *Part (i) remains true when '*$\mathscr{CP}$*' and '*$\mathscr{AP}$*' are respectively replaced by 'restricted* $\mathscr{CP}$*' and '*$\mathscr{EP}$*'.*

PROOF. We begin by proving the first part of (i). So assume that, for any pair of sets I, J, if

(1) $\mathfrak{A}$ is an I-indexed g-algebra in $\mathbf{K}$, $\mathfrak{B}$ is a J-indexed g-algebra in $\mathbf{K}$, and $\mathfrak{A}, \mathfrak{B}$ are compatible,

then there is a g-algebra $\mathfrak{C}$ covering $\mathfrak{A}, \mathfrak{B}$ such that

(2) $$\mathfrak{C} \in \mathbf{K} \quad \text{and} \quad \mathfrak{C} \text{ has the } \mathscr{CP}.$$

Since $\mathbf{K}$ is closed under the formation of subalgebras, in order to prove that $\mathbf{K}$ has the $\mathscr{AP}$ it suffices to take arbitrary sets I, J and algebras $\mathfrak{A}, \mathfrak{B}$ satisfying (1) and show that there is a g-algebra $\mathfrak{D} \in \mathbf{K}$ with index $I \cup J$ such that

(3) $$\mathfrak{D}^{(I)} \cong \mathfrak{A} \quad \text{and} \quad \mathfrak{D}^{(J)} \cong \mathfrak{B}.$$

By assumption there is a g-algebra $\mathfrak{C}$ covering $\mathfrak{A}, \mathfrak{B}$ such that (2) holds. Then by 1.1.28(i) there are $R \in Co\,\mathfrak{C}^{(I)}$ and $S \in Co\,\mathfrak{C}^{(J)}$ such that

(4) $$\mathfrak{C}^{(I)}/R \cong \mathfrak{A}$$

and

(5) $$\mathfrak{C}^{(J)}/S \cong \mathfrak{B}.$$

From (4) and 1.1.30 we get

$$\mathfrak{C}^{(I \cap J)}/(R \cap {}^2C^{(I \cap J)}) \cong (\mathfrak{C}^{(I)}/R)^{(I \cap J)} \cong \mathfrak{A}^{(I \cap J)},$$

and in a similar manner from (5) and 1.1.30 we get

$$\mathfrak{C}^{(I\cap J)}/(S \cap {}^2C^{(I\cap J)}) \cong \mathfrak{B}^{(I\cap J)}.$$

Thus, since $\mathfrak{A}$ and $\mathfrak{B}$ are compatible by hypothesis, we have

$$\mathfrak{C}^{(I\cap J)}/(R \cap {}^2C^{(I\cap J)}) \cong \mathfrak{C}^{(I\cap J)}/(S \cap {}^2C^{(I\cap J)}).$$

In view of 1.1.28(ii) this relation of isomorphism implies that

$$R \cap {}^2C^{(I\cap J)} = S \cap {}^2C^{(I\cap J)}. \tag{6}$$

Set

$$T = Cg^{(\mathfrak{C})}(R \cup S).$$

Then by (4), (6), 1.1.30, and the premise that $\mathfrak{C}$ has the $\mathscr{CP}$ we get

$$(\mathfrak{C}/T)^{(I)} \cong \mathfrak{C}^{(I)}/(T \cap {}^2C^{(I)}) = \mathfrak{C}^{(I)}/R \cong \mathfrak{A}.$$

Similarly, but using (5) in place of (4), we get

$$(\mathfrak{C}/T)^{(J)} \cong \mathfrak{B}.$$

Therefore, (3) is satisfied by taking $\mathfrak{D} = \mathfrak{C}/T$; hence, since $\mathfrak{C}/T \in \mathbf{K}$ by (2), 1.1.28(i), and the premise $\boldsymbol{H}\mathbf{K} = \mathbf{K}$, we have shown that $\mathbf{K}$ has the $\mathscr{AP}$. This proves the first part of (i).

In order to prove the second part of (i) we assume that $\mathbf{K}$ has the $\mathscr{AP}$ and consider an arbitrary $\mathbf{K}$-independently generated $\mathfrak{A} \in \mathbf{K}$. Let I, J be arbitrary sets with

$$I \cup J = In\,\mathfrak{A}, \tag{7}$$

and consider any $R \in Co\,\mathfrak{A}^{(I)}$ and $S \in Co\,\mathfrak{A}^{(J)}$ such that

$$R \cap {}^2A^{(I\cap J)} = S \cap {}^2A^{(I\cap J)}. \tag{8}$$

Both $\mathfrak{A}^{(I)}/R$ and $\mathfrak{A}^{(J)}/S$ are contained in $\mathbf{K}$ in view of the premise $\boldsymbol{H}\mathbf{K} = \boldsymbol{S}\mathbf{K} = \mathbf{K}$. Thus by (8) and 1.1.30 (applied twice) we obtain

$$(\mathfrak{A}^{(I)}/R)^{(I\cap J)} \cong \mathfrak{A}^{(I\cap J)}/(R \cap {}^2A^{(I\cap J)}) = \mathfrak{A}^{(I\cap J)}/(S \cap {}^2A^{(I\cap J)}) \cong (\mathfrak{A}^{(J)}/S)^{(I\cap J)}$$

Therefore, since $\mathbf{K}$ is assumed to have the $\mathscr{AP}$, there exists a g-algebra

$$\mathfrak{B} \in \mathbf{K} \tag{9}$$

with the same index as $\mathfrak{A}$ such that

$$\mathfrak{B}^{(I)} \cong \mathfrak{A}^{(I)}/R \quad \text{and} \quad \mathfrak{B}^{(J)} \cong \mathfrak{A}^{(J)}/S. \tag{10}$$

Then by 1.1.17(ii) and 1.1.28(i) we have

$$\mathfrak{B}^{(I)} \preccurlyeq \mathfrak{A}^{(I)} \quad \text{and} \quad \mathfrak{B}^{(J)} \preccurlyeq \mathfrak{A}^{(J)},$$

and from this together with (7) and (9) it follows, since $\mathfrak{A}$ is $\mathbf{K}$-independently generated by hypothesis, that $\mathfrak{B} \preccurlyeq \mathfrak{A}$. Hence by 1.1.28(i) there is a $T \in Co\ \mathfrak{A}$ such that $\mathfrak{B} \cong \mathfrak{A}/T$ and, therefore, applying 1.1.30,

$$\mathfrak{B}^{(I)} \cong \mathfrak{A}^{(I)}/(T \cap {}^2A^{(I)}) \quad \text{and} \quad \mathfrak{B}^{(J)} \cong \mathfrak{A}^{(J)}/(T \cap {}^2A^{(J)}) \tag{11}$$

By (10), (11), and 1.1.28(ii) we get

$$T \cap {}^2A^{(I)} = R \quad \text{and} \quad T \cap {}^2A^{(J)} = S.$$

Since I, J are arbitrary sets satisfying (7), and R, S are arbitrary congruence relations on $\mathfrak{A}^{(I)}$, $\mathfrak{A}^{(J)}$, respectively, such that (8) holds, we see that $\mathfrak{A}$ has the $\mathscr{CP}$. This completes the proof of (i).

The proof of (ii) is obtained by making the following changes in the proof of (i) given above: everywhere change '$\mathscr{CP}$' to 'restricted $\mathscr{CP}$' and '$\mathscr{AP}$' to '$\mathscr{EP}$', and in (1) and (7) add the assumption $I \cap J = 0$.

REMARK 1.2.9. Notice that, if $\mathbf{K}$ has the $\mathscr{AP}$, then the proof just given appears to establish the $\mathscr{CP}$ for a larger class of g-algebras of $\mathbf{K}$ than just those which are $\mathbf{K}$-independently generated. Consider the following property which applies to an arbitrary g-algebra $\mathfrak{A}$ not necessarily in $\mathbf{K}$.

(I) $\mathfrak{B} \preccurlyeq \mathfrak{A}$ whenever $\mathfrak{B} \in \mathbf{K}$, $In\ \mathfrak{B} = In\ \mathfrak{A}$, and $\mathfrak{B}^{(I)} \preccurlyeq \mathfrak{A}^{(I)}$ and $\mathfrak{B}^{(J)} \preccurlyeq \mathfrak{A}^{(J)}$ for some pair of sets I, J such that $I \cup J = In\ \mathfrak{A}$.

From the proof of 1.2.8(i) we see that every $\mathfrak{A} \in \mathbf{K}$ satisfying (I) has the $\mathscr{CP}$. It is obvious that every $\mathbf{K}$-independently generated $\mathfrak{A}$ satisfies condition (I), and it is easy to show that the converse is not in general true. However, the question whether the converse holds under the hypothesis $\boldsymbol{H}\mathbf{K} = \boldsymbol{S}\mathbf{K} = \mathbf{K}$ and $\mathfrak{A} \in \mathbf{K}$ is still open. This question takes on added interest when one observes that, from the discussion in 1.2.3, a g-algebra $\mathfrak{A} \in \mathbf{K}$ satisfies (I) iff $\mathfrak{A}$ is a $\mathbf{K}$-free amalgamated product of each of the pairs $\mathfrak{A}^{(I)}$, $\mathfrak{A}^{(J)}$ such that $I \cup J = In\ \mathfrak{A}$. Thus the question at hand can be put in the following form:

(II) Suppose $\mathbf{K}$ is closed under the formation of subalgebras and homomorphic images, $\mathfrak{A} \in \mathbf{K}$, and $\mathfrak{A}$ is a $\mathbf{K}$-free amalgamated product of every pair of subalgebras $\mathfrak{B}$, $\mathfrak{C}$ with $In\ \mathfrak{B} \cup In\ \mathfrak{C} = In\ \mathfrak{A}$. Is then $\mathfrak{A}$ necessarily the $\mathbf{K}$-free product of $\{\mathfrak{A}^{\{i\}} : i \in In\ \mathfrak{A}\}$?

This problem in general seems to be open, but it can at least be shown to have an affirmative solution in case $\mathbf{K}$ has the $\mathscr{AP}$; cf. Jónsson [15]. Thus the statement of 1.2.8 would not be strengthened by replacing the words '$\mathbf{K}$-independently generated

g-algebra in $\mathbf{K}$' by the words '$\mathfrak{A} \in \mathbf{K}$ satisfying 1.2.9(I)' in condition 1.2.8(i'). Although the condition that $\mathbf{K}$ have the $\mathscr{AP}$ is sufficient for an affirmative answer to (II), it is not necessary; indeed, the answer to (II) turns out to be affirmative when $\mathbf{K}$ is taken to be the class of all g-algebras whose underlying algebras are semigroups, but it is well known that this class fails to have the $\mathscr{AP}$. Compare here Grätzer-Lakser [12], Theorem 4 and the remark immediately preceding it.

REMARK 1.2.10. Assume that $\mathbf{K}$ is any class of g-algebras such that $\boldsymbol{S}\mathbf{K} = \boldsymbol{H}\mathbf{K} = \mathbf{K}$. For each $\mathfrak{A} \in \mathbf{K}$ let $U_{\mathfrak{A}}$ be the set of all maximal proper congruence relations in $\bigcup_{\mathfrak{B} \subseteq \mathfrak{A}} Co\, \mathfrak{B}$. Then by modifying the proof of 1.2.8 in an obvious way we can show:

(I) *If every compatible pair of simple* g-*algebras in* $\mathbf{K}$ *is covered by an* $\mathfrak{A} \in \mathbf{K}$ *with the* $\mathscr{CP}$ *relative to* $U_{\mathfrak{A}}$, *then the class of simple* g-*algebras in* $\mathbf{K}$ *has the* $\mathscr{AP}$. *On the other hand, if the class of all simple* g-*algebras in* $\mathbf{K}$ *has the* $\mathscr{AP}$, *then every* $\mathbf{K}$-*independently generated* g-*algebra* $\mathfrak{A} \in \mathbf{K}$ *has the* $\mathscr{CP}$ *relative to* $U_{\mathfrak{A}}$.

(I) remains true when '$\mathscr{AP}$' and '$\mathscr{CP}$' are replaced by '$\mathscr{EP}$' and 'restricted $\mathscr{CP}$', respectively.

REMARK 1.2.11. The amalgamation property is actually a property of the pair of partial ordering relations on isomorphism types induced by the relations $\preccurlyeq$ and $\subseteq$ between g-algebras. To be more precise, let $\mathfrak{T}$ be the relational structure which consists of the class T of all isomorphism types of g-algebras along with the partial orderings of T induced by $\preccurlyeq$ and $\subseteq$; let us agree to denote the induced partial orderings by the same symbols $\preccurlyeq$ and $\subseteq$. Given any class $\mathbf{K}$ of g-algebras we let $\mathfrak{T}_{\mathbf{K}}$ be the substructure of $\mathfrak{T}$ whose universe is the class of all isomorphism types of members of $\mathbf{K}$. It is an easy matter to show that, if $\mathbf{K}$ is closed under the formation of subalgebras and homomorphic images, the amalgamation property is an elementary property of the relational structure $\mathfrak{T}_{\mathbf{K}}$; let σ be a sentence in some fixed first-order discourse language for $\mathfrak{T}_{\mathbf{K}}$ that expresses this property.

We can make use of this characterization of the $\mathscr{AP}$ in terms of the relational structure $\mathfrak{T}_{\mathbf{K}}$ to gain a deeper insight into the content of Theorem 1.2.8(i). Given any g-algebra $\mathfrak{A}$ we consider the relational structure $\mathfrak{C}_{\mathfrak{A}}$ consisting of the set $C_{\mathfrak{A}}$ of all congruence relations on all subalgebras of $\mathfrak{A}$ (i.e., the set $\bigcup_{\mathfrak{B} \subseteq \mathfrak{A}} Co\, \mathfrak{B}$) and two partial ordering relations on this set: for any $R, S \in C_{\mathfrak{A}}$ the first of the partial ordering relations holds between R and S iff they are congruence relations on the same subalgebra of $\mathfrak{A}$, and $R \supseteq S$; the second relation holds between R and S iff $R = S \cap {}^2B$ for some $\mathfrak{B} \subseteq \mathfrak{A}$. Let F be the function from $C_{\mathfrak{A}}$ into $T_{\boldsymbol{HS}\{\mathfrak{A}\}}$ which sends each $R \in C_{\mathfrak{A}}$ into the isomorphism type of $\mathfrak{B}/R$ where $\mathfrak{B}$ is the unique subalgebra of $\mathfrak{A}$ such that $R \in Co\, \mathfrak{B}$ (for convenience we assume here that $\mathfrak{B} \neq \mathfrak{B}'$ if $\mathfrak{B}, \mathfrak{B}' \subseteq \mathfrak{A}$ and $\mathfrak{B} \neq \mathfrak{B}'$). In view of

1.1.28(ii) and 1.1.30 it is easily seen that F is an isomorphism from $\mathfrak{C}_{\mathfrak{A}}$ onto $\mathfrak{T}_{\boldsymbol{HS}\{\mathfrak{A}\}}$. Theorem 1.2.8(i) is an almost immediate consequence of the fact that the structures $\mathfrak{C}_{\mathfrak{A}}$ and $\mathfrak{T}_{\boldsymbol{HS}\{\mathfrak{A}\}}$ are isomorphic: We observe that a class $\mathbf{K}$ of g-algebras with $\boldsymbol{HK} = \boldsymbol{SK} = \mathbf{K}$ has the $\mathscr{AP}$ whenever every compatible pair of members of $\mathbf{K}$ is included in a subclass of $\mathbf{K}$ of the form $\boldsymbol{HS}\{\mathfrak{A}\}$ which has the $\mathscr{AP}$. It is clear that for g-algebra $\mathfrak{A}$ to have the $\mathscr{CP}$ simply means by 1.2.7(i) that the structure $\mathfrak{C}_{\mathfrak{A}}$ is a model of the sentence σ; in view of this fact the above observation immediately leads to the first part of 1.2.8(i). On the other hand, if $\boldsymbol{HK} = \boldsymbol{SK} = \mathbf{K}$ and $\mathbf{K}$ has the $\mathscr{AP}$, then it is easily seen that $\boldsymbol{HS}\{\mathfrak{A}\}$ has the $\mathscr{AP}$ for every $\mathbf{K}$-independently generated $\mathfrak{A} \in \mathbf{K}$; thus $\mathfrak{C}_{\mathfrak{A}}$ is a model of the sentence σ for all such $\mathfrak{A}$, and hence the second part of 1.2.8(i) holds.

The entire discussion of this remark can of course be extended so as to apply to the embedding property and to its connection with the restricted $\mathscr{CP}$ as described in 1.2.8(ii).

We now define the amalgamation property, and the related properties, as they apply to classes of ordinary algebras. We still want to consider here only similarity classes whose members contain distinguished elements among their fundamental operations; see the remark immediately preceding 1.2.1.

DEFINITION 1.2.12. *Assume* K *is any class of (ordinary) algebras and let* $\mathbf{K}$ *be the class of all* g*-algebras* $\mathfrak{A}$ *such that* $\mathfrak{A} \in \mathsf{K}$. *We say that* K *has the* ***amalgamation property*** *(the* $\mathscr{AP}$*) or the* ***embedding property*** *(the* $\mathscr{EP}$*) just in case* $\mathbf{K}$ *has the corresponding property in the sense of 1.2:1.*

It is an easy matter to convince oneself that the definition of the amalgamation property given in 1.2.12 is essentially equivalent to the usual definition of this property. However, the notion of embedding property which is considered here clearly does not correspond to the usual one. (For a more or less standard definition of the $\mathscr{AP}$ and $\mathscr{EP}$ see, for instance, Jónsson [16].) It is in fact more restrictive so that, in particular, every class of algebras with the $\mathscr{AP}$ also has the $\mathscr{EP}$. In the usual conception of the $\mathscr{EP}$, where any two algebras of K are required to be simultaneously isomorphically embeddable in a third member of K, a class may have the $\mathscr{AP}$ while failing to have the $\mathscr{EP}$. The two notions of $\mathscr{EP}$ do coincide in case K is a class of algebras (with distinguished elements) any two of which have isomorphic minimal subalgebras. This condition is not satisfied by, e.g., the variety of Boolean algebras – because of 1-element algebras.

DEFINITION 1.2.13. *Assume* K *and* $\mathbf{K}$ *are as* in *1.2.12. Then for each non-zero cardinal* α *we take* $\mathfrak{Fr}_\alpha\mathsf{K}$ *equal to* $\mathfrak{Fr}_\alpha\mathbf{K}$*; we call* $\mathfrak{Fr}_\alpha\mathsf{K}$ *the free* g*-algebra over* K ***with index*** α.

By comparing Definitions 1.1.2, 1.1.32, and 1.2.13 we see that for any class K of ordinary algebras the underlying algebra of $\mathfrak{Fr}_\alpha$K is $\mathfrak{F}\mathfrak{r}_\alpha$K and the distinguished system of generators of $\mathfrak{Fr}_\alpha$K is $\langle \xi/Cr_\alpha \mathsf{K} : \xi < \alpha \rangle$.

The next theorem is an easy consequence of 1.1.33 and 1.2.8 (cf. 1.1.4 and 1.1.38).

THEOREM 1.2.14. *Assume that* K *is a class of (ordinary) algebras such that* $\mathbf{H}\mathsf{K} = \mathbf{S}\mathsf{K} = \mathbf{P}\mathsf{K} = \mathsf{K}$, *i.e.,* K *is algebraically closed. Then* K *has respectively the* $\mathscr{AP}$, *or the* $\mathscr{EP}$, *iff all free* g*-algebras* $\mathfrak{Fr}_\alpha$K *over* K *with index* α (α *a cardinal* $\neq 0$) *have the* $\mathscr{CP}$, *or the restricted* $\mathscr{CP}$.

REMARKS 1.2.15. In its application to free algebras the congruence-extension property turns out to have an interesting equivalent metamathematical formulation. The interest in this formulation stems mainly from the fact that it is quickly recognized as being a strengthened version of the so-called *equational interpolation property*. This latter property has been considered in the literature (although presumably not under this name); cf. for instance Jónsson [16]. It arises naturally when one considers possible analogues for equational logic of Craig's interpolation theorem for first-order logic. In order to state both the metamathematical form of the $\mathscr{CP}$ and the equational interpolation property, and to discuss their interrelationship, it is helpful to introduce some special metamathematical terminology (which will not be used outside of the present remarks). Let K be any class of (similar) algebras; let Φ be a set of equations in a given (first-order) discourse language Λ for K, and let ε be a single equation in Λ. We shall say that Φ K-*implies* ε, or that ε is a K-*consequence* of Φ, if f satisfies ε whenever f satisfies every member of Φ where f is any assignment of elements of an arbitrary $\mathfrak{A} \in \mathsf{K}$ to the variables of Λ. Φ may be infinite, but if it is finite, say $\Phi = \{\delta_0, \delta_1, \ldots, \delta_{\kappa-1}\}$, then Φ K-implies ε iff the formula

$$(\delta_0 \wedge \delta_1 \wedge \cdots \wedge \delta_{\kappa-1}) \rightarrow \varepsilon$$

is in the (first-order) theory of K. For any set Φ of equations in Λ we let V_Φ represent the set of all variables of Λ which occur in at least one member of Φ.

For the remainder of the present remarks we assume that K is a fixed but arbitrary class of algebras, α is any non-zero cardinal, and Λ_α is a fixed discourse language for K whose sequence of variables is of length α. Let T be the set of all terms of Λ_α. Then there is a natural way of associating with each term τ in T a unique element $f\tau$ of the free algebra over K with α generators: the function f is defined recursively in the following way. For each variable v_ξ of Λ_α with $\xi < \alpha$ we take $fv_\xi = \xi/Cr_\alpha \mathsf{K}$, and if, for example, $+$ is a fundamental binary operation of the free algebra, and if $+$ is the binary operation symbol of Λ_α corresponding to $+$, then we take $f(\tau + \sigma) = (f\tau) + (f\sigma)$ for all $\sigma, \tau \in T$. As a straightforward consequence of Theorem 1.1.4 we get the following characterization of the relation of implication between a set of equations and a

single equation in terms of congruences on the free algebra: Let Φ be an arbitrary set of equations and ε a single equation in Λ_α. Then Φ K-implies ε iff the ordered pair $\langle f\sigma, f\tau\rangle$ where $(\sigma = \tau) = \varepsilon$ is contained in the congruence relation of $\mathfrak{Fr}_\alpha\mathsf{K}$ generated by the set of pairs $\langle f\pi, f\varrho\rangle$ where $(\pi = \varrho) \in \Phi$, i.e., Φ K-implies $\varepsilon = (\sigma = \tau)$ iff

$$\langle f\sigma, f\tau\rangle \in Cg^{(\mathfrak{Fr}_\alpha\mathsf{K})}\{\langle f\pi, f\rho\rangle : (\pi = \rho) \in \Phi\}.$$

In view of the characterization just given it is easy to see what form the equivalent metamathematical formulation of the $\mathscr{CP}$ for free algebras should take: for the free g-algebra $\mathfrak{Fr}_\alpha\mathsf{K}$ to have the $\mathscr{CP}$ means that the class K satisfies the following metamathematical condition.

(I) Let Φ, Ψ be any sets of equations in Λ_α. If Φ, Ψ together satisfy the condition that every K-consequence of Φ in which only variables in $V_\Phi \cap V_\Psi$ occur is at the same time a consequence of Ψ, and vice-versa, then every K-consequence of $\Phi \cup \Psi$ in which no variables in $V_\Phi \sim V_\Psi$ occur is already a K-consequence of Ψ.

The class K is said to have the *equational interpolation property* if K satisfies the following condition.

(II) Let θ be any set of equations and ε any single equation in Λ_α. If θ K-implies ε, then there exists a set Ω of equations in Λ_α with $V_\Omega \subseteq V_\Phi \cap V_{\{\varepsilon\}}$ such that Φ K-implies δ for each $\delta \in \Omega$ and Ω in turn K-implies ε.

Assume now that $\mathfrak{Fr}_\alpha\mathsf{K}$ has the $\mathscr{CP}$; i.e., K satisfies condition (I). Then K also satisfies condition (II). In order to see this consider any set θ of equations in Λ_α and any K-consequence ε of θ. In (I) take $\Phi = \theta$ and

$$\Psi = \{\delta : \theta \text{ K-implies } \delta,\ V_{\{\delta\}} \subseteq V_\theta \cap V_{\{\varepsilon\}}\}.$$

Then from the assumption that (I) holds we see immediately that for the interpolating set Ω in (II) we can take Ψ.

From 1.2.14 we can now conclude that every algebraically closed class K with the amalgamation property has the equational interpolation property. This result appears in Jónsson [16], and is credited there (on p. 152) to unpublished work of H. J. Keisler. The essential ideas underlying 1.2.14 are apparently due to Magnus [20]; see the footnote at the bottom of p. 165. However, Magnus was concerned exclusively with the class of groups. Cf. in this connection also Lyndon [18] and [19].

Chapter 2

AMALGAMATION AND INTERPOLATION PROPERTIES IN CYLINDRIC ALGEBRAS

In this chapter we apply to cylindric algebras the general results on the amalgamation and congruence-extension properties obtained in the first chapter. In its application to cylindric algebras the $\mathscr{CP}$ has important metalogical significance, and this is explained in Section 2.1. In Sections 2.2 and 2.3 we obtain new results of both a positive and negative nature concerning the amalgamation properties of various classes of cylindric algebras. All these results together with some easy consequences are summarized in tabular form in Section 2.4.

2.1 Main problem

We begin by reproducing the formal definition of cylindric algebras which is given in $\mathscr{HMT}$. Throughout this chapter unless otherwise indicated, α and β will always represent arbitrary ordinal numbers.

DEFINITION 2.1.1. *By a* ***cylindric algebra of dimension*** α *we mean an algebra*

$$\mathfrak{A} = \langle A, +, \cdot, -, 0, 1, \mathsf{c}_\kappa, \mathsf{d}_{\kappa\lambda}\rangle_{\kappa,\lambda<\alpha}$$

such that $0, 1$, *and* $\mathsf{d}_{\kappa\lambda}$ *are distinguished elements of* A *(for all* $\kappa, \lambda < \alpha$*),* $-$ *and* c_κ *are unary operations on* A *(for all* $\kappa < \alpha$*),* $+$ *and* $\cdot$ *are binary operations on* A*, and such that the following postulates are satisfied for all* $x, y \in A$ *and any* $\kappa, \lambda, \mu < \alpha$*:*

(C_0) *the structure* $\langle A, +, \cdot, -, 0, 1\rangle$ *is a Boolean algebra;*
(C_1) $\mathsf{c}_\kappa 0 = 0$;
(C_2) $x \leq \mathsf{c}_\kappa x$ *(i.e.,* $x + \mathsf{c}_\kappa x = \mathsf{c}_\kappa x$*);*
(C_3) $\mathsf{c}_\kappa(x \cdot \mathsf{c}_\kappa y) = \mathsf{c}_\kappa x \cdot \mathsf{c}_\kappa y$;
(C_4) $\mathsf{c}_\kappa \mathsf{c}_\lambda x = \mathsf{c}_\lambda \mathsf{c}_\kappa x$;
(C_5) $\mathsf{d}_{\kappa\kappa} = 1$;
(C_6) *if* $\kappa \neq \lambda, \mu$*, then* $\mathsf{d}_{\lambda\mu} = \mathsf{c}_\kappa(\mathsf{d}_{\lambda\kappa} \cdot \mathsf{d}_{\kappa\mu})$;
(C_7) *if* $\kappa \neq \lambda$*, then* $\mathsf{c}_\kappa(\mathsf{d}_{\kappa\lambda} \cdot x) \cdot \mathsf{c}_\kappa(\mathsf{d}_{\kappa\lambda} \cdot -x) = 0$.

The class of all cylindric algebras is denoted by CA*, and the class of all cylindric algebras of dimension* α *by* CA_α*. The elements* $\mathsf{d}_{\kappa\lambda}$ *are called* ***diagonal elements****, and the operations* c_κ *are called* ***cylindrifications****. The structure* $\mathfrak{B}\mathfrak{l}\,\mathfrak{A} = \langle A, +, \cdot, -, 0, 1\rangle$ *is the* ***Boolean part*** *of* $\mathfrak{A}$.

In the present chapter we often use the symbols 'CA' and 'CA_α' simply as ab-

breviations for the words 'cylindric algebra' and 'cylindric algebra of dimension α'; analogous abbreviations will be used without further comment.

DEFINITION 2.1.2. *By a* ***cylindric*** **g**-***algebra***, *or a* ***cylindric*** **g**-***algebra of dimension*** α, *we mean any* **g**-*algebra* $\mathfrak{A}$ *such that* $\mathfrak{A} \in \mathsf{CA}$, *or* $\mathfrak{A} \in \mathsf{CA}_\alpha$. *The classes of all cylindric* **g**-*algebras and cylindric* **g**-*algebras of dimension* α *are denoted respectively by* $\mathbf{CA}$ *and* $\mathbf{CA}_\alpha$.

The preceding definition is an example of a convention that is applied often in this chapter – usually without warning. Thus if a given class of ordinary algebras is represented by some letter or string of letters, it is automatically assumed that the corresponding string of bold-faced letters represents the class of all **g**-algebras $\mathfrak{A}$ with $\mathfrak{A}$ in the original class.

REMARK 2.1.3. To understand this chapter the reader will have to have a good understanding of the basic notions of the theory of cylindric algebras and of the fundamental properties of these notions. In addition, he will have to be familiar with a good part of the notation and terminology that is employed in $\mathscr{HMT}$, Chapters 1 and 2 – the same notation will be used in this paper with only minor changes. With the exception of cylindric algebras themselves we do not reproduce the definitions of any of the various special kinds of algebras, or any of the other special notions, from $\mathscr{HMT}$ that are used in this chapter, but when such an algebra or notion appears here for the first time the reader is referred by number to the appropriate definition in $\mathscr{HMT}$. The same situation applies in the case of theorems and technical remarks found in $\mathscr{HMT}$, and we shall freely refer to these by number whenever we have occasion to use them. Numbers of definitions, theorems, and enumerated remarks in $\mathscr{HMT}$ will almost always be preceded by the letters '$\mathscr{HMT}$' to distinguish them from enumerated items of the present work.

It has been impossible to make this paper self-contained, but of course only a small part of the entire theory of CA's as it is developed in $\mathscr{HMT}$ is actually required in order to read it. In the following list the reader will find portions of Chapters 1 and 2 of $\mathscr{HMT}$ that we feel represent the minimum part of the theory of CA's necessary for understanding the subsequent discussion: all of Section 1.1; Section 1.2 through 1.2.12; Section 1.3 through 1.3.9; Section 1.5 through 1.5.11; Section 1.11 through 1.11.3; Section 2.1 through 2.1.7; Section 2.3 through 2.3.8; Section 2.4 through 2.4.5; Section 2.5 through 2.5.2, also the remark preceding 2.5.31 through 2.5.38(*i*); Section 2.6 through 2.6.2, also 2.6.26 through 2.6.33.

The cylindric **g**-algebras we next define will play an important role in this chapter. In this definition, and throughout the entire chapter except where indicated otherwise, γ represents an arbitrary non-zero cardinal.

DEFINITION 2.1.4. *Consider any function* $\Delta \in {}^{\gamma}Sb\,\alpha$. *We let*

$$Cr_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha} = \bigcap \{R : R \in Co\,\mathfrak{Fr}_{\gamma},\ \mathfrak{Fr}_{\gamma}/R \in \mathbf{IS}\mathsf{CA}_{\alpha},\ (\mathsf{c}_{\kappa}^{(\mathfrak{Fr}_{\gamma})}\xi)/R = \xi/R \text{ for each } \xi < \gamma \text{ and each } \kappa \in \alpha \sim \Delta\xi\}$$

and

$$\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha} = \mathfrak{Fr}_{\gamma}/Cr_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha}.$$

The g*-algebra* $\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha}$ *is called a* ***dimension-restricted free cylindric*** g***-algebra with index*** γ.

The underlying algebra $\mathfrak{Fr}_{\gamma}/Cr_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha}$ of $\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha}$ and its universe $Fr_{\gamma}/Cr_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha}$ *are denoted by* $\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha}$ *and* $Fr_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha}$, *respectively.*

REMARK 2.1.5. The underlying algebra $\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha}$ is defined in $\mathscr{HMT}$ 2.5.31. It also corresponds in a natural way to a *polyadic algebra (with equality) freely generated by predicates*; see Daigneault [8]. The underlying algebra is easily seen to be a special case of the independent algebras that we defined in 1.1.6. It is also easily seen from 1.1.8 that, for any class $\mathbf{K} \subseteq \mathbf{CA}_{\alpha}$, $\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha}$ is $\mathbf{K}$-independently generated in the sense of 1.1.36; cf. also $\mathscr{HMT}$ 2.5.36.

The class Lf_{α} of *locally finite* CA_{α}'s is defined in $\mathscr{HMT}$ 1.11.1(i). As indicated in the introduction, locally finite dimension-restricted free CA's of infinite dimension have strong metalogical connections: To each first-order language Λ satisfying certain very general conditions there corresponds an $\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha} \in \mathsf{Lf}_{\alpha}$ such that $\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha} \cong \cong \mathfrak{Fm}^{(\Lambda)}/\equiv_0$ where $\mathfrak{Fm}^{(\Lambda)}/\equiv_0$ is the CA of formulas of Λ associated with the theory of all logically valid sentences (cf. Definition $\mathscr{HMT}$ 1.1.11); conversely, given $\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha} \in \mathsf{Lf}_{\alpha}$ with $\alpha \geq \omega$ there is a language Λ to which $\mathfrak{Fr}^{(\Delta)}\mathsf{CA}_{\alpha}$ corresponds in this way. This correspondence is discussed in general terms in $\mathscr{HMT}$ 2.5.46; however, we now state a stronger version which involves isomorphism between g-algebras.

THEOREM 2.1.6. *Let* $\alpha \geq \omega$ *and let* Λ_{α} *be any first-order language with an* α*-termed variable sequence and with no operation symbols. Let* φ *be any sequence of atomic formulas of* Λ_{α} *indexed by a non-zero cardinal* γ *and satisfying the following two conditions:*

(i) *for each relation symbol* π *of* Λ_{α} *there is exactly one* $\xi < \gamma$ *such that*

$$\varphi_{\xi} = \langle \pi \rangle \frown \langle v_{\kappa_0} \rangle \frown \langle v_{\kappa_1} \rangle \frown \cdots \frown \langle v_{\kappa_{\gamma-1}} \rangle$$

for some sequence of variables $v_{\kappa_0}, v_{\kappa_1}, \ldots, v_{\kappa_{\lambda-1}}$;

(ii) *for each* $\xi < \gamma$, *if* φ_{ξ} *is given by the formula in (i), then the variables* $v_{\kappa_0}, v_{\kappa_1}, \ldots, v_{\kappa_{\lambda-1}}$ *are all distinct.*

Then the algebra of formulas $\mathfrak{Fm}^{(\Lambda_{\alpha})}/\equiv_0$ *is generated by the set* $\{\varphi_{\xi}/\equiv_0 : \xi < \gamma\}$ *and we have*

$$\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha} \cong \langle \mathfrak{Fm}^{(\Lambda_{\alpha})}/\equiv_0, \langle \varphi_{\xi}/\equiv_0 : \xi < \gamma \rangle \rangle$$

where Δ *is the function from* γ *into* $Sb\,\alpha$ *such that* $\Delta\xi = \{\kappa_0, \kappa_1, \ldots, \kappa_{\lambda-1}\}$ *if* $\xi < \gamma$, *and* φ_{ξ} *is given by the formula in (i).*

We do not give the proof of 2.1.6 even though it cannot be found in $\mathscr{HMT}$; the proof will be given in Part II of Henkin, Monk, and Tarski's work which is now in preparation. The theorem is also an easy consequence of Daigneault [8], Theorem 3.6.

By $\mathscr{HMT}$ 2.5.44, $\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha \in \mathsf{Lf}_\alpha$ iff $|\Delta\xi| < \omega$ for each $\xi < \omega$. Thus 2.1.6 can be used to describe the correspondence between languages Λ and $\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha \in \mathsf{Lf}_\alpha$ that was mentioned in Remark 2.1.5.

REMARK 2.1.7. An *ideal* of a CA_α $\mathfrak{A}$ is defined in $\mathscr{HMT}$ 2.3.5. The set of all ideals of $\mathfrak{A}$ is denoted by $Il\,\mathfrak{A}$, and for each $X \subseteq A$ the ideal $M = \bigcap \{N : X \subseteq N \in Il\,\mathfrak{A}\}$ is called the *ideal generated by* X and is denoted by $Ig\,X$. By Zorn's lemma any proper ideal is included in a maximal proper ideal. It is discussed in $\mathscr{HMT}$ 2.3.6 how a natural one-one correspondence between the congruence relations and ideals of a CA allows us to substitute the latter in the role that would normally be played by the former in the theory of CA's. This applies in particular to the notion of congruence-extension property defined in Section 1.2, and it is obvious how Definition 1.2.7, in application to cylindric g-algebras, is to be re-formulated in terms of ideals. With this re-formulation in mind one can use the simple intrinsic characterization of ideals given in $\mathscr{HMT}$ 2.3.7 to obtain a correspondingly simple condition on $\mathbf{CA}$'s which proves to be equivalent to the $\mathscr{CP}$; this condition is given in 2.1.8.

Theorem $\mathscr{HMT}$ 2.3.7 says that, if $\mathfrak{A}$ is any CA_α, then $M \in Il\,\mathfrak{A}$ iff M is a Boolean ideal of $\mathfrak{A}$ (i.e., $0 \in M$ and $x \in M$ whenever $x \leq y + z$ where $y, z \in M$) and $\mathsf{c}_\kappa^* M \subseteq M$ for every $\kappa < \alpha$. This intrinsic characterization of ideals, together with its consequence that the restriction of a proper ideal to a subalgebra is always proper, will be used constantly throughout this chapter without explicit reference.

In regard to the following definition see $\mathscr{HMT}$ 1.7.1 and $\mathscr{HMT}$ 1.7.8 for an explanation of the symbols $\mathsf{c}_{(\Gamma)}$ and $\mathsf{c}_{(\Gamma)}{}^\partial$; the latter operation corresponds to a generalized universal quantifier.

DEFINITION 2.1.8. *Let $\mathfrak{A}$ be a $\mathbf{CA}_\alpha$, i.e., a cylindric* g*-algebra of dimension α.*

(i) *$\mathfrak{A}$ is said to have the* ***interpolation property*** *(the $\mathscr{IP}$ for short) if the following condition is satisfied by any pair of sets $I, J \subseteq In\,\mathfrak{A}$:*

(i′) *under the hypothesis $x \in A^{(I)}, z \in A^{(J)}$, and $x \leq z$, we can always find a $y \in A^{(I \cap J)}$ and a finite $\Gamma \subseteq \alpha$ such that*

$$x \leq y \leq \mathsf{c}_{(\Gamma)} z\,.$$

$\mathfrak{A}$ has the ***restricted*** *$\mathscr{IP}$ if (i′) is satisfied by any pair of sets $I, J \subseteq In\,\mathfrak{A}$ such that $I \cap J = 0$.*

(ii) *$\mathfrak{A}$ is said to have the* ***weak*** *$\mathscr{IP}$ and the* ***weak restricted*** *$\mathscr{IP}$ if condition (i′) with*

the formula $x \leq y \leq \mathsf{c}_{(\Gamma)}z$ *replaced by the formula*

$$\mathsf{c}_{(\Gamma)}{}^{\partial}x \leq y \leq \mathsf{c}_{(\Gamma)}z$$

is satisfied by any $I, J \subseteq In\, \mathfrak{A}$ *and any mutually disjoint* $I, J \subseteq In\, \mathfrak{A}$, *respectively.*

The next lemma follows easily from $\mathscr{HMT}$ 2.3.8.

LEMMA 2.1.9. *Let* $\mathfrak{A}, \mathfrak{B} \in \mathsf{CA}_\alpha$ *with* $\mathfrak{B} \subseteq \mathfrak{A}$. *Suppose* $M \in Il\, \mathfrak{B}$. *We then have:*

(i) $Ig^{(\mathfrak{A})}M = \{x : x \in A,\ x \leq b$ *for some* $b \in M\}$;

(ii) $M = (Ig^{(\mathfrak{A})}M) \cap B$;

(iii) *if* $\mathfrak{C} \subseteq \mathfrak{A}$ *and* $N \in Il\, \mathfrak{C}$, *then* $Ig^{(\mathfrak{A})}(M \cup N) = \{x : x \leq b + c$ *for some* $b \in M$ *and* $c \in N\}$.

Taking $\alpha = 0$ we see that 2.1.9 holds when $\mathfrak{A}$ and $\mathfrak{B}$ are taken to be Boolean algebras; the same remark applies to the next lemma.

LEMMA 2.1.10. *Let* $\mathfrak{A}, \mathfrak{B} \in \mathsf{CA}_\alpha$ *with* $\mathfrak{B} \subseteq \mathfrak{A}$. *Suppose* $M \in Il\, \mathfrak{B}$. *Then for every* $N \in Il\, \mathfrak{A}$ *such that*

$$N \cap B \subseteq M$$

we can find an $N' \in Il\, \mathfrak{A}$ *such that*

$$N' \supseteq N \quad \textit{and} \quad N' \cap B = M.$$

Moreover, if M *is a maximal proper ideal of* $\mathfrak{B}$, *then* N' *can be taken to be a maximal proper ideal of* $\mathfrak{A}$.

PROOF. In the special case $N = \{0\}$ the desired result follows trivially from 2.1.9(ii). The general result is easily obtained from this special case by considering $\mathfrak{A}/N$, $\mathfrak{B}/(N \cap B)$, and $M/(N \cap B)$ in place of $\mathfrak{A}$, $\mathfrak{B}$, and M, respectively.

THEOREM 2.1.11. *An arbitrary* CA_α $\mathfrak{A}$ *has the* $\mathscr{CP}$ *iff it has the* $\mathscr{IP}$, *and it has the restricted* $\mathscr{CP}$ *iff it has the restricted* $\mathscr{IP}$.

PROOF. We shall only prove the equivalence of the $\mathscr{CP}$ and the $\mathscr{IP}$; the equivalence of the restricted versions of these properties is proved similarly.

We first prove that the $\mathscr{CP}$ implies the $\mathscr{IP}$, so assume that $\mathfrak{A}$ has the $\mathscr{CP}$. We must show that 2.1.8(i′) is satisfied for all $I, J \subseteq In\, \mathfrak{A}$. It is clear that we may assume without loss of generality that $I \cup J = In\, \mathfrak{A}$, so take any sets I, J with this property.

Let x and z be as in the hypothesis of 2.1.8(i′); i.e., assume that

(1) $$x \in A^{(I)}, \quad z \in A^{(J)}, \quad \text{and } x \leq z.$$

Then

(2) $$x \in (Ig^{(\mathfrak{A})}\{z\}) \cap A^{(I)}.$$

Setting

$$M = Ig^{(\mathfrak{A}(J))}\{z\} \quad \text{and} \quad N = Ig^{(\mathfrak{A}(I))}(M \cap A^{(I \cap J)}) \tag{3}$$

we have by 2.1.9(ii) that $M \cap A^{(I \cap J)} = N \cap A^{(I \cap J)}$. Thus, since $\mathfrak{A}$ is assumed to have the $\mathscr{CP}$, there is a $P \in Il\,\mathfrak{A}$ such that $P \cap A^{(I)} = N$ and $P \cap A^{(J)} = M$; hence

$$Ig^{(\mathfrak{A})}(N \cup M) \cap A^{(I)} \subseteq P \cap A^{(I)} = N.$$

The left-most member of this formula clearly equals $(Ig^{(\mathfrak{A})}\{z\}) \cap A^{(I)}$, so that $(Ig^{(\mathfrak{A})}\{z\}) \cap A^{(I)} \subseteq N$. Thus from (2) and (3) we get

$$x \in Ig^{(\mathfrak{A}(I))}[(Ig^{(\mathfrak{A}(J))}\{z\}) \cap A^{(I \cap J)}].$$

This formula together with 2.1.9(i) implies the existence of an element y such that

$$x \leq y \in A^{(I \cap J)} \tag{4}$$

and $y \in Ig^{(\mathfrak{A}(J))}\{z\}$. By this last formula and $\mathscr{HMT}$ 2.3.10(i) there is a finite subset $\Gamma \subseteq \alpha$ such that

$$y \leq \mathsf{c}_{(\Gamma)} z. \tag{5}$$

Since x, y are arbitrary elements of A satisfying (1), we see from (4) and (5) that $\mathfrak{A}$ has the $\mathscr{IP}$.

We now assume that $\mathfrak{A}$ has the $\mathscr{IP}$ and proceed to show that it has the $\mathscr{CP}$. Take any $I, J \subseteq In\,\mathfrak{A}$ such that $I \cup J = In\,\mathfrak{A}$ and any $M \in Il\,\mathfrak{A}^{(I)}$ and $N \in Il\,\mathfrak{A}^{(J)}$ such that

$$M \cap A^{(I \cap J)} = N \cap A^{(I \cap J)}. \tag{6}$$

Suppose

$$x \in [Ig^{(\mathfrak{A})}(M \cup N)] \cap A^{(I)}. \tag{7}$$

Then by 2.1.9(iii) there are elements

$$b \in M \quad \text{and} \quad c \in N \tag{8}$$

such that $x \leq b + c$. Hence we have

$$x \cdot - b \leq c. \tag{9}$$

By (7) and the first formula of (8) we get $x \cdot - b \in A^{(I)}$, while from the second formula of (8) we get $c \in A^{(J)}$; hence (9) together with the assumption that $\mathfrak{A}$ has the $\mathscr{IP}$ implies the existence of a $d \in A^{(I \cap J)}$ and a finite $\Gamma \subseteq \alpha$ such that

$$x \cdot - b \leq d \leq \mathsf{c}_{(\Gamma)} c. \tag{10}$$

By the second formula of (8) and the second inclusion of (10) we have $d \in N$, and thus,

by (6),

$$d \in M. \tag{11}$$

By the first inclusion of (10) we get $x \leq d + b$, and thus (11) and the first formula of (8) together imply $x \in M$. Therefore, since x is an arbitrary element satisfying (7), we have

$$[Ig^{(\mathfrak{A})}(M \cup N)] \cap A^{(I)} \subseteq M,$$

and in a similar manner we obtain

$$[Ig^{(\mathfrak{A})}(M \cup N)] \cap A^{(J)} \subseteq N.$$

The inclusions in the opposite direction are obvious; thus, since M and N are arbitrary ideals of $\mathscr{A}^{(I)}$ and $\mathscr{A}^{(J)}$, respectively, such that (6) holds, we see that $\mathfrak{A}$ has the $\mathscr{CP}$. This completes the proof of the theorem.

REMARK 2.1.12. The $\mathscr{IP}$ in its application to free **CA**'s has a natural and simple equivalent metamathematical formulation which we shall describe. In view of 2.1.11 this gives a new metamathematical formulation of the $\mathscr{CP}$ in application to free **CA**'s which is much simpler than the one given in 1.2.15.

In this remark and elsewhere in this chapter where we deal with the metatheory of CA's we shall denote each operation symbol of a discourse language for CA_α's by the bold-faced version of the corresponding symbol in the ordinary theory. Thus $\mathbf{c}_\kappa$ denotes the operation symbol corresponding to c_κ. A similar remark applies to all the various defined notions such as $\leq$, s^κ_γ, $\mathsf{c}_{(\Gamma)}$, etc.

Let γ be any non-zero cardinal and let Λ_γ be a discourse language for CA_α's whose sequence of variables is of length γ. Let K be any class of CA_α's. Then the free g-algebra $\mathfrak{Fr}_\gamma\mathsf{K}$ has the $\mathscr{IP}$ iff every inclusion $\sigma \leq \tau$ between terms of Λ_γ which holds in every member of K can be interpolated by a term ϱ containing only variables common to both σ and τ. More precisely, there is a term ϱ of Λ_γ in which only those variables may occur which occur simultaneously in σ and τ, and there is a finite $\Gamma \subseteq \alpha$ such that both the inclusions $\sigma \leq \varrho$ and $\varrho \leq \mathbf{c}_{(\Gamma)}\tau$ hold in every member of K.

$\mathfrak{Fr}_\gamma\mathsf{K}$ has the restricted $\mathscr{IP}$ iff every inclusion $\sigma \leq \tau$ holding in all members of K such that no variable occurs simultaneously in σ and τ can be interpolated by a term ϱ without variables, i.e., a constant term of Λ_γ.

THEOREM 2.1.13. *Let $\mathfrak{A} \in \mathbf{CA}_\alpha$ and let U be the class of all sets M such that M is a maximal proper ideal of some subalgebra of $\mathfrak{A}$. Then $\mathfrak{A}$ has the $\mathscr{CP}$ relative to U iff it has the weak $\mathscr{IP}$, and it has the restricted $\mathscr{CP}$ relative to U iff it has the weak restricted $\mathscr{IP}$.*

PROOF. As in the proof of 2.1.11 we will not bother proving the part of the theorem involving the restricted properties.

We first prove that the $\mathscr{CP}$ relative to U implies the weak $\mathscr{IP}$. So assume that $\mathfrak{A}$ has the $\mathscr{CP}$ relative to U, and let I, J be any given subsets of $In\,\mathfrak{A}$ such that $I \cup J = In\,\mathfrak{A}$. Consider any elements

$$x \in A^{(I)} \quad \text{and} \quad z \in A^{(J)} \tag{1}$$

such that

$$\mathsf{c}_{(\Gamma)}{}^{\partial} x \leq y \leq \mathsf{c}_{(\Gamma)} z \text{ fails to hold for all } y \in A^{(I \cap J)} \text{ and finite } \Gamma \subseteq \alpha. \tag{2}$$

Then, by $\mathscr{HMT}$ 1.7.8.,

$$\mathsf{c}_{(\Gamma)} - x + y < 1 \quad \text{or} \quad \mathsf{c}_{(\Gamma)} z + - y < 1$$

whenever $y \in A^{(I \cap J)}$ and Γ is a finite subset of α. It then follows easily that whenever Δ, θ are finite subsets of α we have $u + w < 1$ for all $u, w \in A^{(I \cap J)}$ such that $u \leq \mathsf{c}_{(\Delta)} - x$ and $w \leq \mathsf{c}_{(\theta)} z$. Thus setting

$$P = Ig^{(\mathfrak{A}(I \cap J))} [[(Ig^{(\mathfrak{A}(I))} \{-x\}) \cap A^{(I \cap J)}] \cup [(Ig^{(\mathfrak{A}(J))} \{z\}) \cap A^{(I \cap J)}]] \tag{3}$$

we have, in view of (1), $\mathscr{HMT}$ 2.3.10(i), and 2.1.9(iii), that P is a proper ideal of $\mathfrak{A}^{(I \cap J)}$. Now choose any maximal proper $P' \in Il\,\mathfrak{A}^{(I \cap J)}$ such that $P' \supseteq P$. Then by (3) and 2.1.10 there are maximal proper ideals $M \in Il\,\mathfrak{A}^{(I)}$ and $N \in Il\,\mathfrak{A}^{(J)}$ such that

$$M \supseteq Ig^{(\mathfrak{A}(I))} \{-x\}, \quad N \supseteq Ig^{(\mathfrak{A}(J))} \{z\}, \tag{4}$$

and $M \cap A^{(I \cap J)} = P' = N \cap A^{(I \cap J)}$. It follows then from the assumption that $\mathfrak{A}$ has the $\mathscr{CP}$ relative to U that $Ig^{(\mathfrak{A})}(M \cup N) \cap A^{(I)} = M$ and thus, *a fortiori*, that $Ig^{(\mathfrak{A})}(M \cup N)$ is a proper ideal of $\mathfrak{A}$. Hence it cannot be the case that $x \leq z$ since, if this inclusion did hold, $Ig^{(\mathfrak{A})}(M \cup N)$ would not be proper in light of (4). x and y are arbitrary elements satisfying (1) and (2). Thus we have shown that condition 2.1.8(i′) as modified in 2.1.8(ii) holds for all $I, J \subseteq In\,\mathfrak{A}$ such that $I \cup J = In\,\mathfrak{A}$, and from this it is easily seen that the same condition holds for all subsets I, J of $In\,\mathfrak{A}$ without qualification. Hence $\mathfrak{A}$ has the weak $\mathscr{IP}$.

Now assume that $\mathfrak{A}$ has the weak $\mathscr{IP}$. Take any sets I, J where $I \cup J = In\,\mathfrak{A}$ and consider maximal proper ideals $M \in Il\,\mathfrak{A}^{(I)}$, $N \in Il\,\mathfrak{A}^{(J)}$ such that

$$M \cap A^{(I \cap J)} = N \cap A^{(I \cap J)}. \tag{5}$$

Suppose that

$$Ig^{(\mathfrak{A})}(M \cup N) = A. \tag{6}$$

Then by 2.1.9(iii) there are elements $x \in M$ and $z \in N$ such that $x + z = 1$, i.e., such that $-x \leq z$. Therefore, from the assumption that $\mathfrak{A}$ has the weak $\mathscr{IP}$ we conclude

that there is an element

$$y \in A^{(I \cap J)} \tag{7}$$

and a finite $\Gamma \subseteq \alpha$ such that $\mathsf{c}_{(\Gamma)}{}^{\partial} - x \leq y \leq \mathsf{c}_{(\Gamma)} z$. Hence, by $\mathscr{HMT}$ 1.7.8,

$$-y \leq \mathsf{c}_{(\Gamma)} x \in Ig^{(\mathfrak{A}(I))}\{x\} \subseteq M \quad \text{and} \quad y \leq \mathsf{c}_{(\Gamma)} z \in Ig^{(\mathfrak{A}(J))}\{z\} \subseteq N,$$

and these two formulas together with (7) imply that $-y \in M \cap A^{(I \cap J)}$ and $y \in N \cap A^{(I \cap J)}$. Thus in light of (5) we have that $1 = -y + y \in M$, which contradicts the fact that M is proper by assumption. This shows that (6) cannot hold, and hence $Ig^{(\mathfrak{A})}(M \cup N)$ is a proper ideal of $\mathfrak{A}$. The formulas

$$Ig^{(\mathfrak{A})}(M \cup N) \cap A^{(I)} = M \quad \text{and} \quad Ig^{(\mathfrak{A})}(M \cup N) \cap A^{(J)} = N$$

are now easy consequences of the maximality of M and N, and for the same reason we may conclude that $P \cap A^{(I)} = M$ and $P \cap A^{(J)} = N$ for any maximal proper ideal of $\mathfrak{A}$ which includes $Ig^{(\mathfrak{A})}(M \cup N)$. Since $I, J \subseteq In\, \mathfrak{A}$ with $I \cup J = In\, \mathfrak{A}$ are arbitrary, we see that $\mathfrak{A}$ has the $\mathscr{CP}$ relative to U. This completes the proof of the theorem.

THEOREM 2.1.14. *For any given $\mathfrak{A} \in \mathbf{Lf}_\alpha$ the $\mathscr{IP}$ and the weak $\mathscr{IP}$ are equivalent conditions.*

PROOF. It is obvious from definition that the $\mathscr{IP}$ implies the weak $\mathscr{IP}$ for any $\mathbf{CA}_\alpha$. Suppose that $\mathfrak{A} \in \mathbf{Lf}_\alpha$ has the weak $\mathscr{IP}$ and consider any $I, J \subseteq In\, \mathfrak{A}$ and $x \in A^{(I)}$, $z \in A^{(J)}$ such that $x \leq z$. Then by $\mathscr{HMT}$ 1.2.7 we have

$$x \leq \mathsf{c}_{(\Delta x)} x \leq \mathsf{c}_{(\Delta x)} z \tag{1}$$

where Δx is the *dimension set* of x defined in $\mathscr{HMT}$ 1.6.1. From (1) and the hypothesis that $\mathfrak{A}$ has the weak $\mathscr{IP}$ we conclude that there is a $y \in A^{(I \cap J)}$ and a finite $\theta \subseteq \Gamma$ such that

$$\mathsf{c}_{(\theta)}{}^{\partial} \mathsf{c}_{(\Delta x)} x \leq y \leq \mathsf{c}_{(\theta)} \mathsf{c}_{(\Delta x)} z, \tag{2}$$

but $\mathsf{c}_{(\theta)}{}^{\partial} \mathsf{c}_{(\Delta x)} x = \mathsf{c}_{(\Delta x)} x$ by $\mathscr{HMT}$ 1.6.7(i), $\mathscr{HMT}$ 1.7.8, and $\mathscr{HMT}$ 1.7.11(i). Thus taking $\Gamma = \theta \cup \Delta x$ and using $\mathscr{HMT}$ 1.7.3 we see from (1) and (2) that $x \leq y \leq \mathsf{c}_{(\Gamma)} z$. Therefore, $\mathfrak{A}$ has the $\mathscr{IP}$. This completes the proof.

REMARK 2.1.15. 2.1.14 does not apply to every $\mathfrak{A} \in \mathbf{CA}_\alpha$ – not even every freely generated $\mathbf{CA}_\alpha$. In fact, it will be seen in 2.2.25 and 2.3.4 that, while $\mathfrak{Fr}_\gamma \mathsf{CA}_\alpha$ has the weak $\mathscr{IP}$ whenever $\alpha \geq \omega$, this algebra fails to have the $\mathscr{IP}$ if $\gamma \geq 4$.

If one compares the definitions of the $\mathscr{IP}$ and the weak $\mathscr{IP}$, a third property cannot help but come to mind.

DEFINITION 2.1.16. *A* $\mathbf{CA}_\alpha\,\mathfrak{A}$ *is said to have the* ***strong interpolation property*** *(the* ***strong*** $\mathscr{IP}$*) if the following condition holds for all subsets I, J of In* $\mathfrak{A}$*:*

(i) *under the hypothesis* $x \in A^{(I)}$, $z \in A^{(J)}$, *and* $x \leq z$, *we can always find a* $y \in A^{(I \cap J)}$ *such that*

$$x \leq y \leq z.$$

REMARKS 2.1.17. It is obvious that, if $\mathfrak{A}$ has the strong $\mathscr{IP}$, then every g-algebra obtained from $\mathfrak{A}$ by change of index also has the property, and this fact is implicitly used in much of the subsequent discussion; this remark applies of course to the weak and ordinary $\mathscr{IP}$ as well. Cf. 1.1.12 and the following remark.

Clearly every $\mathbf{CA}_\alpha\,\mathfrak{A}$ with the strong $\mathscr{IP}$ has the ordinary $\mathscr{IP}$. The converse is not true even for $\mathbf{Lf}_\alpha$'s: by $\mathscr{HMT}$ 2.3.14 every simple $\mathbf{CA}_\alpha$ has the $\mathscr{IP}$, but it is a trivial matter to find simple $\mathbf{Lf}_\alpha$'s without the strong $\mathscr{IP}$. On the other hand, we know of no $\mathbf{CA}_\alpha$ which is $\mathbf{K}$-freely generated, or even $\mathbf{K}$-independently generated, for some reasonably inclusive class $\mathbf{K}$ of $\mathbf{CA}_\alpha$'s which has the $\mathscr{IP}$ but not the strong $\mathscr{IP}$; compare here 2.1.15.

REMARK 2.1.18. In its metalogical interpretation the strong interpolation property turns out to be a very interesting and important notion. In fact, from Theorems 2.1.6 and $\mathscr{HMT}$ 2.5.44 it follows immediately that the well-known interpolation theorem of Craig [5] is simply the translation into metalogical terms of the following statement:

(I) every $\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha \in \mathbf{Lf}_\alpha$ with $\alpha \geq \omega$ has the strong $\mathscr{IP}$.

It is also easily seen that the theorem of Craig in its restriction to sentences is the translation of the statement:

(II) every $\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha \in \mathbf{Lf}_\alpha$ with $\alpha \geq \omega$ has the $\mathscr{IP}$ or, equivalently (in view of 2.1.14), the weak $\mathscr{IP}$.

Another result from the theory of models which is often compared to Craig's theorem is the general consistency result of Robinson [23], p. 54, Theorem 2.9. Robinson's and Craig's theorems are known to be equivalent in the sense that each one is derivable from the other in a natural way. Their equivalence can also be demonstrated algebraically, and in order to do this we first determine the algebraic version of Robinson's theorem.

We begin by stating the theorem in essentially the same form as it appears in Robinson's paper. Let Λ be any first-order language. By a sublanguage of Λ we mean of course any language Λ' which has the same logical constants of Λ and whose set Γ

of non-logical constants is some subset of the set of non-logical constants of Λ; for convenience Λ' will be denoted by $\Lambda^{(\Gamma)}$. Then Robinson's theorem says the following:

(III) Assume Λ is a first-order language with infinitely many variables. Let Γ and Δ be sets of non-logical constants of Λ, and let Π and Σ be consistent sets of sentences of $\Lambda^{(\Gamma)}$ and $\Lambda^{(\Delta)}$, respectively. Finally, let θ be a complete and consistent theory of $\Lambda^{(\Gamma \cap \Delta)}$ such that $\theta \subseteq \Pi \cap \Sigma$. Then $\Pi \cup \Sigma$ is a consistent set of sentences of Λ.

Let H be the function which takes any formula φ of Λ into the element $H\varphi = (\neg\varphi/\equiv_0)$ of $\mathfrak{Fm}^{(\Lambda)}/\equiv_0$, the CA of formulas associated with the theory of logically valid sentences (cf. $\mathscr{HMT}$ 1.1.11). H clearly includes every element of $Fm^{(\Lambda)}/\equiv_0$ in its range, and in light of the discussion of $\mathscr{HMT}$ 2.3.13 the function $\langle Ig\, H^*\theta : \theta$ a theory of $\Lambda\rangle$ is an isomorphism between the lattice of theories of Λ and the lattice of ideals of $\mathfrak{Fm}^{(\Lambda)}/\equiv_0$. We immediately get from this fact that Σ is a consistent set of sentences of Λ iff $Ig\, H^*\Sigma$ is a proper ideal of $\mathfrak{Fm}^{(\Lambda)}/\equiv_0$, and that a theory θ is complete and consistent iff $Ig\, H^*\theta$ is a maximal proper ideal. Applying first this correspondence between consistent sets of sentences and proper ideals and then the isomorphism of 2.1.6 we see that Robinson's theorem when translated directly into algebraic terms takes the following form: Assume $\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha \in \mathbf{L}_\alpha$ with $\alpha \geq \omega$ and set $\mathfrak{A} = \mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha$. Then the following condition is satisfied by any pair of sets $\Gamma, \Delta \subseteq \gamma$.

(IV) If $R \in Il\,\mathfrak{A}^{(I)}$ and $S \in Il\,\mathfrak{A}^{(J)}$ are proper ideals and there is a maximal proper ideal $M \in Il\,\mathfrak{A}^{(I \cap J)}$ such that $M \subseteq R$ and $M \subseteq S$, then $R \cup S$ is included in a proper ideal $N \in Il\,\mathfrak{A}$.

Observe that, since the restriction of a proper ideal to a subalgebra is always proper, the inclusions $M \subseteq R$, $M \subseteq S$ in (IV) can be replaced by the equalities $R \cap A^{(I \cap J)} = M = S \cap A^{(I \cap J)}$; furthermore, we may assume that R, S, and N are all maximal proper ideals since every ideal can be extended to one with this property. Therefore, we arrive at the conclusion that Robinson's theorem is the translation into metalogical terms of the statement:

(V) every $\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha \in \mathbf{Lf}_\alpha$ with $\alpha \geq \omega$ has the $\mathscr{CP}$ relative to the set of all maximal proper ideals of subalgebras of $\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha$.

By comparing (II) and (V) we see that, in their algebraic formulations, the equivalence of Craig's and Robinson's theorems is established in Theorem 2.1.13.

The question naturally arises whether there is a natural property like the $\mathscr{AP}$ to which the strong $\mathscr{IP}$ is related in the same sense that the $\mathscr{AP}$ is related to the $\mathscr{IP}$ by 1.2.8 and 2.1.11, and the $\mathscr{AP}$ for simple algebras is related to the weak $\mathscr{IP}$ by

1.2.10 and 2.1.13. This question is still open, but we shall now define a stronger version of the amalgamation property, and we will show in Theorem 2.1.20 that it is closely related to the strong interpolation property.

DEFINITION 2.1.19. (i) *Let* $\mathbf{K}$ *be a class of* $\mathbf{CA}_\alpha$'s. $\mathbf{K}$ *is said to have the* ***strong amalgamation property, or the strong*** $\mathscr{AP}$ *for short, if the following condition is satisfied by any pair of sets* I, J:

(i′) *for any pair of* $\mathbf{CA}_\alpha$'s $\mathfrak{A}, \mathfrak{B} \in \mathbf{K}$ *with indexes* I *and* J, *respectively, if* $\mathfrak{A}^{(I \cap J)}, \mathfrak{B}^{(I \cap J)} \in I\mathbf{K}$ *and* $\mathfrak{A}^{(I \cap J)} \cong \mathfrak{B}^{(I \cap J)}$ *(i.e.,* $\mathfrak{A}, \mathfrak{B}$ *are compatible), then there is a* $\mathfrak{C} \in \mathbf{K}$ *with index* $I \cup J$ *such that*

$$\mathfrak{C}^{(I)} \cong \mathfrak{A}, \quad \mathfrak{C}^{(J)} \cong \mathfrak{B},$$

and

$$C^{(I)} \cap C^{(J)} = C^{(I \cap J)}.$$

(ii) *Let* K *be a class of (ordinary)* CA_α*'s and take* $\mathbf{K}$ *to be the class of all* $\mathbf{CA}_\alpha$*'s* $\mathfrak{A}$ *such that* $\mathfrak{A} \in \mathsf{K}$. *We say that* K *has the* ***strong amalgamation property (the strong*** $\mathscr{AP}$***)*** *just in case* $\mathbf{K}$ *has the strong* $\mathscr{AP}$ *in the sense of part (i).*

THEOREM 2.1.20. *Assume that* $\mathbf{K}$ *is a class of* $\mathbf{CA}_\alpha$*'s such that* $\mathbf{SK} = \mathbf{HK} = \mathbf{K}$. *If every compatible pair of* $\mathbf{CA}_\alpha$*'s in* $\mathbf{K}$ *is covered by a member of* $\mathbf{K}$ *with the strong* $\mathscr{IP}$, *then* $\mathbf{K}$ *has the strong* $\mathscr{AP}$.

PROOF. Assume $\mathfrak{A}, \mathfrak{B} \in \mathbf{K}$ and $\mathfrak{A}^{(I \cap J)} \cong \mathfrak{B}^{(I \cap J)}$ where I, J are the indexes of $\mathfrak{A}, \mathfrak{B}$, respectively. Then $\mathfrak{A}, \mathfrak{B}$ are compatible, and so by hypothesis there is a $\mathbf{CA}_\alpha$ $\mathfrak{C}$ covering $\mathfrak{A}, \mathfrak{B}$ such that $\mathfrak{C} \in \mathbf{K}$ and $\mathfrak{C}$ has the strong $\mathscr{IP}$. Then by 1.1.28(i) there are ideals

$$M \in Il\,\mathfrak{C}^{(I)} \quad \text{and} \quad N \in Il\,\mathfrak{C}^{(J)} \tag{1}$$

such that $\mathfrak{C}^{(I)}/M \cong \mathfrak{A}$ and $\mathfrak{C}^{(J)}/N \cong \mathfrak{B}$. Set

$$P = Ig^{(\mathfrak{C})}(M \cup N). \tag{2}$$

Then by essentially repeating that part of the argument in the proof of 1.2.8 between formulas (5) and (7) (with the ideals M, N in place of the congruence relations R, S) we can show that $(\mathfrak{C}/P)^{(I)} \cong \mathfrak{A}$ and $(\mathfrak{C}/P)^{(J)} \cong \mathfrak{B}$. Thus, since $\mathfrak{C}/P \in \mathbf{K}$ because of the premise $\mathbf{HK} = \mathbf{K}$, we see that in order to complete the proof it remains only to show

$$D^{(I)} \cap D^{(J)} = D^{(I \cap J)} \tag{3}$$

where

$$\mathfrak{D} = \mathfrak{C}/P. \tag{4}$$

For this purpose suppose

$$y \in D^{(I)} \cap D^{(J)}. \tag{5}$$

It is easily seen that $D^{(I)} = C^{(I)}/P$ $(=\{x/P : x \in C^{(I)}\})$ and $D^{(J)} = C^{(J)}/P$, so that (5) implies the existence of elements

$$x \in C^{(I)} \quad \text{and} \quad z \in C^{(J)} \tag{6}$$

such that

$$x/P = y = z/P. \tag{7}$$

From this equality we conclude that

$$(x \cdot - z)/P = (x/P) \cdot^{(\mathfrak{D})} - {}^{(\mathfrak{D})} (z/P) = 0,$$

and hence that $x \cdot - z \in P$. Thus by (2) and 2.1.9(iii) we get elements

$$r \in M \quad \text{and} \quad s \in N \tag{8}$$

such that $x \cdot - z \leq r + s$, and hence such that

$$x \cdot - r \leq z + s. \tag{9}$$

From (1), (6), and (8) we have $x \cdot - r \in C^{(I)}$ and $z + s \in C^{(J)}$, and therefore, using the fact that $\mathfrak{C}$ has the strong $\mathscr{I}\mathscr{P}$, we conclude from (9) that there is an element

$$u \in C^{(I \cap J)} \tag{10}$$

such that $x \cdot - r \leq u \leq z + s$. Thus by (2), (4), and (8) we have

$$x/P = (x/P) \cdot^{(\mathfrak{D})} - {}^{(\mathfrak{D})} (r/P) = (x \cdot - r)/P \leq u/P$$

and

$$u/P \leq (z + s)/P = (z/P) + {}^{(\mathfrak{D})} (s/P) = z/P.$$

Because of (7) these two inclusions give $y = u/P$, and hence $y \in D^{(I \cap J)}$ by (4) and (10). y is an arbitrary element satisfying (5), so that we have shown that $D^{(I)} \cap D^{(J)} \subseteq D^{(I \cap J)}$; because the inclusion in the opposite direction is obvious, (3) holds. This completes the proof of the theorem.

REMARK 2.1.21. We do not know if the strong $\mathscr{A}\mathscr{P}$ implies the strong $\mathscr{I}\mathscr{P}$ in the obvious sense. I.e., we do not know if Theorem 1.2.8(i) continues to hold when '$\mathscr{A}\mathscr{P}$' and '$\mathscr{C}\mathscr{P}$' are replaced everywhere by 'strong $\mathscr{A}\mathscr{P}$' and 'strong $\mathscr{I}\mathscr{P}$', respectively (and $\mathbf{K}$ is taken to be a class of $\mathbf{CA}_\alpha$'s). However, it seems highly unlikely to us that this is true.

To one who is acquainted with the well-known proof of the definability theorem of Beth [1] via the theorem of Craig (see, for example, Craig [6]) the proof of 2.1.20 may have looked familiar. It is in fact little more than an algebraic translation of that

metalogical proof. We remark finally that one of the algebraic forms ascribed to Beth's result is simply the statement that the class Lf_ω has the strong amalgamation property; cf. in this regard Daigneault [8].

REMARK 2.1.22. Notions of strong restricted $\mathscr{IP}$ and strong $\mathscr{EP}$ can be defined by modifying 2.1.16 and 2.1.19(i) in the obvious way: A **CA** $\mathfrak{A}$ has the *strong restricted* $\mathscr{IP}$ if condition 2.1.16(i) holds for all mutually disjoint subsets I, J of $In\,\mathfrak{A}$. A class **K** of $\mathbf{CA}_\alpha$'s has the *strong* $\mathscr{EP}$ if condition 2.1.19(i′) holds for any pair of mutually disjoint sets I, J, and a class K of (ordinary) CA_α's has the strong $\mathscr{EP}$ if the class **K** of all g-algebras $\mathfrak{A}$ with $\mathfrak{A} \in \mathsf{K}$ has the property in the sense just defined.

The same relation holds between the strong restricted $\mathscr{IP}$ and the strong $\mathscr{EP}$ as is shown in 2.1.20 to hold between the strong $\mathscr{IP}$ and the strong $\mathscr{AP}$.

Finally, we remark that the strong $\mathscr{IP}$, the strong restricted $\mathscr{IP}$, the weak $\mathscr{IP}$, and the weak restricted $\mathscr{IP}$, in their application to free **CA**'s, all have equivalent metamathematical formulations similar to the ones for the $\mathscr{IP}$ and restricted $\mathscr{IP}$ which are found in 2.1.12. These formulations are obvious and will not be given here.

2.2 Positive results

In the main theorem of this section, 2.2.6, we shall prove that for $\alpha \geq \omega$ every dimension-restricted free $\mathbf{CA}_\alpha$ which is dimension-complemented has the strong interpolation property. (See $\mathscr{HMT}$ 1.11.1(ii) for the definition of the class Dc_α of *dimension-complemented* CA_α's.) From this single theorem (together with one or two others of a non-trivial but still essentially auxiliary character) there follows via the general results of Sections 1.2 and 2.1 a variety of positive results concerning both amalgamation and interpolation properties. These results involve almost exclusively cylindric algebras and classes of cylindric algebras of infinite dimension. For this reason, in this section and throughout the remaining part of the paper, α and β will always represent arbitrary infinite ordinals – unless it is explicitly indicated to the contrary. As before, γ is intended in all cases to represent an arbitrary non-zero cardinal.

The notion of a *neat reduct* which is defined in $\mathscr{HMT}$ 2.6.28 plays an essential role in Theorem 2.2.6 and throughout the entire section. The importance of this notion for the theory of CA's lies in its close connection with representability; cf. Remark $\mathscr{HMT}$ 2.6.26. The neat α-reduct $\mathfrak{Nr}_\alpha\mathfrak{A}$ of $\mathfrak{A}$ is defined for any CA_β $\mathfrak{A}$ such that $\beta \geq \alpha$; $Nr_\alpha\mathfrak{A}$ is the universe of $\mathfrak{Nr}_\alpha\mathfrak{A}$ and $\mathsf{Nr}_\alpha\mathsf{CA}_\beta$ is the class of all neat α-reducts of CA_β's. Thus, for fixed α, β with $\beta \geq \alpha$, $\mathfrak{Nr}_\alpha$ and Nr_α can be thought of as unary operations with the class CA_β as common domain and with ranges included in CA_α and the class of universes of CA_α's, respectively. We now want to define three related operations whose common domain is now a certain special subclass of $\mathbf{CA}_\beta$.

DEFINITION 2.2.1. *Assume that α, β are finite or infinite ordinals with $\beta \geq \alpha$. Assume that $\mathfrak{A} \in \mathbf{CA}_\beta$ and $a_i \in Nr_\alpha\mathfrak{A}$ for each $i \in In\,\mathfrak{A}$. Then the* ***neat α-reduct of*** *$\mathfrak{A}$, in symbols $\mathfrak{Nr}_\alpha\mathfrak{A}$, is the* g*-algebra*

$$\langle \mathfrak{Sg}^{(\mathfrak{Nr}_\alpha\mathfrak{A})}a^*(In\,\mathfrak{A}), a\rangle .$$

The underlying algebra $\mathfrak{Sg}^{(\mathfrak{Nr}_\alpha\mathfrak{A})}a^*(In\,\mathfrak{A})$ is denoted by $\mathfrak{Nr}_\beta\mathfrak{A}$ and its universe $Sg^{(\mathfrak{Nr}_\alpha\mathfrak{A})}a^*(In\,\mathfrak{A})$ by $Nr_\alpha\mathfrak{A}$.

Notice that $\mathfrak{Nr}_\alpha\mathfrak{A}$ is not defined for every $\mathfrak{A} \in \mathbf{CA}_\beta$, but only those $\mathfrak{A}$ whose distinguished generators are all elements of the neat α-reduct of the underlying algebra of $\mathfrak{A}$. Notice also that the symbols $\mathfrak{Nr}_\alpha$ and Nr_α are each used for two separate operations – one on ordinary CA_α's and one on $\mathbf{CA}_\alpha$'s (the importance of keeping this distinction in mind can be seen by observing Theorems 2.2.2(i) and 2.2.3(i) below); it will always be clear from context however which operation is intended.

The following theorem collects some useful properties of, and relations between, the various neat-reduct operations.

THEOREM 2.2.2. *Assume that α, β, and $\mathfrak{A}$ are as in 2.2.1.*

(i) *$\mathfrak{Nr}_\alpha\mathfrak{A} \subseteq \mathfrak{Nr}_\alpha\mathfrak{A}$ and $Nr_\alpha\mathfrak{A} \subseteq Nr_\alpha\mathfrak{A}$; in fact, $\mathfrak{Nr}_\alpha\mathfrak{A}$ is a generating neat subreduct of $\mathfrak{A}$ in the sense of $\mathscr{HMT}$ 2.6.62.*

In addition, for any $I \subseteq In\,\mathfrak{A}$ we have:

(ii) *$\mathfrak{Nr}_\alpha\mathfrak{A}^{(I)} = (\mathfrak{Nr}_\alpha\mathfrak{A})^{(I)}$, $\mathfrak{Nr}_\alpha\mathfrak{A}^{(I)} = (\mathfrak{Nr}_\alpha\mathfrak{A})^{(I)}$, and $Nr_\alpha\mathfrak{A}^{(I)} = (Nr_\alpha\mathfrak{A})^{(I)}$;*

(iii) *$Nr_\alpha\mathfrak{A}^{(I)} = Nr_\alpha\mathfrak{A} \cap A^{(I)}$.*

PROOF. (i) is obvious from Definition 2.2.1. To obtain (ii) simply apply 2.2.1 to get

$$\mathfrak{Nr}_\alpha\mathfrak{A}^{(I)} = \langle \mathfrak{Sg}^{(\mathfrak{Nr}_\alpha\mathfrak{A}^{(I)})}a^*I, I \upharpoonleft a\rangle$$

and $(\mathfrak{Nr}_\alpha\mathfrak{A})^{(I)} = \langle \mathfrak{Sg}^{(\mathfrak{B})}a^*I, I \upharpoonleft a\rangle$ where $\mathfrak{B} = \mathfrak{Sg}^{(\mathfrak{Nr}_\alpha\mathfrak{A})}Rg\,a$. The first equality of (ii) then follows from the fact that $\mathfrak{Nr}_\alpha\mathfrak{A}^{(I)}$ and $\mathfrak{B}$ are both subalgebras of the same algebra $\mathfrak{Nr}_\alpha\mathfrak{A}$; the remaining equalities of (ii) are trivial consequences of the first one. (iii) is obvious from $\mathscr{HMT}$ 2.6.28.

We see from the next two theorems that $\mathbf{CA}_\beta$'s whose neat α-reducts are dimension-complemented have very special properties; these properties are crucial for some of our later results.

THEOREM 2.2.3. *Assume that α, β, and $\mathfrak{A}$ are as in 2.2.1, except that α (and hence also β) are now taken to be infinite. Assume in addition that $\mathfrak{Nr}_\alpha\mathfrak{A} \in \mathbf{Dc}_\alpha$.*

(i) *$\mathfrak{Nr}_\alpha\mathfrak{A} = \mathfrak{Nr}_\alpha\mathfrak{A}$ and $Nr_\alpha\mathfrak{A} = Nr_\alpha\mathfrak{A}$.*

(ii) *If $\mathfrak{B} \in \mathbf{CA}_\beta$ and $\mathfrak{Nr}_\alpha\mathfrak{B} \cong \mathfrak{Nr}_\alpha\mathfrak{A}$, then $\mathfrak{B} \cong \mathfrak{A}$.*

PROOF. (i) and (ii) are immediate consequences of $\mathscr{HMT}$ 2.6.67 and $\mathscr{HMT}$ 2.6.72, respectively.

THEOREM 2.2.4. *If* $\Delta \in {}^{\gamma}Sb\,\alpha$, $\beta \geq \alpha (\geq \omega)$, *and* $\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha} \in \mathbf{Dc}_{\alpha}$, *then* $\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha} \cong \cong \mathfrak{Nr}_{\alpha}\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\beta}$.

PROOF. Consider any $\mathfrak{A} \in \mathbf{CA}_{\alpha}$ such that $In\,\mathfrak{A} = \gamma$ and $\Delta\xi$ includes the dimension set of a_{ξ} for every $\xi < \gamma$, i.e.,

$$\Delta a_{\xi} \subseteq \Delta\xi \text{ for each } \xi < \gamma. \tag{1}$$

By $\mathscr{HMT}$ 2.1.7, $\mathscr{HMT}$ 2.5.33(ii), and the premise $\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\alpha} \in \mathbf{Dc}_{\alpha}$ we have $|\alpha \sim \bigcup_{\xi \in \Gamma} \Delta\xi| \geq \omega$ for each finite $\Gamma \subseteq \gamma$. Thus, applying $\mathscr{HMT}$ 2.1.7 again, we see from (1) that $\mathfrak{A} \in \mathsf{Dc}_{\alpha}$. Hence

$$\mathfrak{A} \in \mathbf{S}\mathsf{Nr}_{\alpha}\mathsf{CA}_{\beta} \tag{2}$$

by $\mathscr{HMT}$ 2.6.34, $\mathscr{HMT}$ 2.6.35, and $\mathscr{HMT}$ 2.6.49.

Theorems $\mathscr{HMT}$ 2.5.36 and $\mathscr{HMT}$ 2.6.45 together tell us that $\mathfrak{Nr}_{\alpha}\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\beta}$ is $\mathsf{Nr}_{\alpha}\mathsf{CA}_{\beta}$-freely generated under Δ by its distinguished system of generators (cf. Definition $\mathscr{HMT}$ 2.5.34). Therefore, if $\mathfrak{A}$ is any $\mathbf{CA}_{\alpha}$ such that $In\,\mathfrak{A} = \gamma$ and (1) holds, we see from (1), (2), and $\mathscr{HMT}$ 2.5.34 that

$$\mathfrak{Nr}_{\alpha}\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\beta} \succcurlyeq \mathfrak{A}.$$

This shows that $\langle \xi/Cr_{\gamma}^{(\Delta)}\mathsf{CA}_{\beta} : \xi < \gamma \rangle$ actually CA_{α}-freely generates $\mathfrak{Nr}_{\alpha}\mathfrak{Fr}_{\gamma}^{(\Delta)}\mathsf{CA}_{\beta}$ under Δ, and hence the conclusion of the theorem follows in view of $\mathscr{HMT}$ 2.5.37.

We now turn to the principal result of this section. Theorem 2.2.6 is an algebraic version of a metalogical theorem found in Henkin [10]; more precisely, it generalizes such an algebraic version since, strictly speaking, Henkin's theorem in its algebraic form would apply to locally finite rather than dimension-complemented cylindric algebras. Henkin's theorem has as rather easy corollaries both the completeness theorem of first-order predicate logic, and the interpolation theorem of Craig. Correspondingly, the proof of 2.2.6 given here leads rather easily to both the representability of Dc_{α}'s, and to the strong $\mathscr{IP}$ for dimension-restricted free $\mathbf{CA}_{\alpha}$'s which are $\mathbf{Dc}_{\alpha}$'s. Because we are only interested here in the latter result, we shall not bother to formulate the former one explicitly. However, the reader will certainly have no trouble in discerning the representability result just below the surface of the following discussion.

The essential ideas of the following lemma and its proof are due to Henkin. They generalize ideas of Tarski used in the proof of the representation theorem for locally finite CA's of infinite dimension which was announced in Tarski [25] but never published. In connection with the statement of this lemma see $\mathscr{HMT}$ 1.1.5 for the definition of a *full cylindric set algebra*, $\mathscr{HMT}$ 1.5.1 for the definition of the *λ-for-κ substitution* $\mathsf{s}_{\lambda}^{\kappa}$, and $\mathscr{HMT}$ 1.11.9 for the definition of the *substitution operator* s.

LEMMA 2.2.5. *Let F be the set of all finite transformations of α, i.e.,*

$$F = \{\sigma : \sigma \in {}^{\alpha}\alpha, |\{\xi : \sigma\xi \neq \xi\}| < \omega\}.$$

(i) *Assume R is any equivalence relation on α. Set*

$$\bar{R} = \{\langle \sigma, \tau \rangle : \sigma, \tau \in F, (\sigma\xi)\, R(\tau\xi) \text{ for each } \xi < \alpha\}$$

and let $\langle Sb^{\alpha}\alpha, \cup, \cap, {}_{\alpha\alpha}\sim, 0, {}^{\alpha}\alpha, \mathsf{C}_{\kappa}, \mathsf{D}_{\kappa\lambda}\rangle_{\kappa,<\alpha}$ be the full cylindric set algebra in the space ${}^{\alpha}\alpha$. Then the following conditions hold: $\bar{R}$ is an equivalence relation on F; the set

$$B = \{X : X \subseteq F, \bar{R}^{*}X = X\}$$

is closed under set-theoretical union, intersection, and complementation with respect to F, and it is also closed under C_{κ} for each $\kappa < \alpha$; finally, the algebra

$$\mathfrak{B} = \langle B, \cup, \cap, {}_{F}\sim, 0, F, \mathsf{C}_{\kappa}, \bar{R}^{*}(\mathsf{D}_{\kappa\lambda} \cap F)\rangle_{\kappa,\lambda<\alpha}$$

is a CA_{α}.

(ii) *Assume that $\mathfrak{A} \in \mathsf{Dc}_{\alpha}$ and M is a maximal proper ideal of $\mathfrak{B}\mathfrak{l}\,\mathfrak{A}$ satisfying the following condition:*

(ii′) *for every $\kappa < \alpha$ and every $x \in A$ such that $\mathsf{c}_{\kappa}x \notin M$ there is a $\lambda < \alpha$ such that $\mathsf{s}^{\kappa}_{\lambda}x \notin M$.*

Set

$$R = \{\langle \kappa, \lambda \rangle : \kappa, \lambda < \alpha, \mathsf{d}_{\kappa\lambda} \notin M\}.$$

Then R is an equivalence relation on α, and, if

$$H = \langle \{\sigma : \sigma \in F, \mathsf{s}_{\sigma}x \notin M\} : x \in A\rangle,$$

and $\bar{R}$ and $\mathfrak{B}$ are as in part (i), then $H \in Hom(\mathfrak{A}, \mathfrak{B})$.

PROOF. It is a straightforward matter to check that (i) holds, so we shall go directly to the proof of (ii). Notice that R is obviously reflexive and symmetric; since $\mathsf{d}_{\kappa\lambda} \cdot \mathsf{d}_{\lambda\mu} \leq \mathsf{c}_{\lambda}(\mathsf{d}_{\kappa\lambda} \cdot \mathsf{d}_{\lambda\mu}) = \mathsf{d}_{\kappa\mu}$ for all $\kappa, \lambda, \mu < \alpha$ with $\lambda \neq \kappa, \mu$, and M is a maximal proper Boolean ideal by hypothesis, it can easily be seen that R is transitive. Thus R is an equivalence relation, and hence $\mathfrak{B} \in \mathsf{CA}_{\alpha}$ by part (i).

We now want to prove that the following condition holds for all $\sigma, \tau \in F$:

(1) $$\text{if } \sigma\bar{R}\tau \text{ and } x \in A, \text{ then } \mathsf{s}_{\sigma}x \notin M \text{ iff } \mathsf{s}_{\tau}x \notin M.$$

This is proved by induction on the cardinality of the finite set $\Gamma = \{\xi : \xi < \alpha, \sigma\xi \neq \tau\xi\}$. If $\Gamma = 0$, the result is obvious, so assume $\kappa \in \Gamma$. Take any

(2) $$\lambda \in \{\xi : (\sigma^{-1})^{*}\{\xi\} = (\tau^{-1})^{*}\{\xi\} = \{\xi\}\} \sim \Delta x.$$

By (2), $\mathscr{HMT}$ 1.11.11(i), (iv), and $\mathscr{HMT}$ 1.11.12(v) we have

(3) $$\mathsf{s}_{\sigma}x = \mathsf{s}^{\lambda}_{\sigma\kappa}\mathsf{s}_{(\alpha\sim\{\kappa\})1\sigma\cup\{\langle\kappa,\lambda\rangle\}}x.$$

On the other hand, using (2), $\mathscr{HMT}$ 1.11.11(i),(iii),(iv), and $\mathscr{HMT}$ 1.1.12(v),(vii) we have

$$\mathsf{s}^{\lambda}_{\tau\kappa}(\mathsf{d}_{\lambda,\sigma\kappa}\cdot\mathsf{s}_{\sigma}x) = \mathsf{d}_{\tau\kappa,\sigma\kappa}\cdot\mathsf{s}_{\sigma}x\,.$$

and

$$\mathsf{s}^{\lambda}_{\tau\kappa}(\mathsf{d}_{\lambda,\sigma\kappa}\cdot\mathsf{s}_{(\alpha\sim\{\kappa\})\restriction\sigma\cup\{\langle\kappa,\lambda\rangle\}}x) = \mathsf{d}_{\tau\kappa,\sigma\kappa}\cdot\mathsf{s}_{(\alpha\sim\{\kappa\})\restriction\sigma\cup\{\langle\kappa,\tau\kappa\rangle\}}x\,.$$

Thus from (3) and $\mathscr{HMT}$ 1.5.5 we get

$$\mathsf{d}_{\tau\kappa,\sigma\kappa}\cdot\mathsf{s}_{\sigma}x = \mathsf{d}_{\tau\kappa,\sigma\kappa}\cdot\mathsf{s}_{(\alpha\sim\{\kappa\})\restriction\sigma\cup\{\langle\kappa,\tau\kappa\rangle\}}x\,. \tag{4}$$

Using the premise that M is a maximal proper ideal, together with the fact that $(\tau\kappa)R(\sigma\kappa)$ and hence $\mathsf{d}_{\tau\kappa,\sigma\kappa}\notin M$, we conclude from (4) that $\mathsf{s}_{\sigma}x\notin M$ iff $\mathsf{s}_{(\sigma\sim\{\kappa\})\restriction\sigma\cup\{\langle\kappa,\tau\kappa\rangle\}}x\notin M$; the desired result now follows by the inductive hypothesis.

(1) directly implies that $Rg\,H\subseteq B$, so it only remains for us to show that for all $x, y\in A$ and all $\kappa, \lambda<\alpha$ we have

$$H(x+y) = Hx\cup Hy\,, \tag{5}$$

$$H-x = {}_F\sim Hx\,, \tag{6}$$

$$H\mathsf{d}_{\kappa\lambda} = R^*(\mathsf{D}_{\kappa\lambda}\cap F)\,, \tag{7}$$

$$H\mathsf{c}_{\kappa}x = \mathsf{C}_{\kappa}Hx\,. \tag{8}$$

For the rest of the proof let σ be a fixed but arbitrary element of F. Then using $\mathscr{HMT}$ 1.11.11(iii) and the premise that M is a maximal proper ideal of $\mathfrak{Bl}\,\mathfrak{A}$ we get $\sigma\in H(x+y)$ iff $\mathsf{s}_{\sigma}(x+y)\notin M$ iff $(\mathsf{s}_{\sigma}x+\mathsf{s}_{\sigma}y)\notin M$ iff $(\mathsf{s}_{\sigma}x\notin M$ or $\mathsf{s}_{\sigma}y\notin M)$ iff $(\sigma\in Hx$ or $\sigma\in Hy)$ iff $\sigma\in Hx\cup Hy$. Thus (5) holds. Again applying $\mathscr{HMT}$ 1.11.11(iii) and the assumption about M we get $\sigma\in H-x$ iff $\mathsf{s}_{\sigma}(-x)\notin M$ iff $-\mathsf{s}_{\sigma}x\notin M$ iff $\mathsf{s}_{\sigma}x\in M$ iff $\sigma\notin Hx$; thus (6) holds. $\sigma\in H\mathsf{d}_{\kappa\lambda}$ iff $\mathsf{s}_{\sigma}\mathsf{d}_{\kappa\lambda}\notin M$; hence, by $\mathscr{HMT}$ 1.11.12(vii), $\sigma\in H\mathsf{d}_{\kappa\lambda}$ iff $\mathsf{d}_{\sigma\kappa,\sigma\lambda}\notin M$ iff $(\sigma\kappa)R(\sigma\lambda)$ iff $\sigma\in\bar{R}^*(\mathsf{D}_{\kappa\lambda}\cap F)$. Thus (7) holds.

Finally, assume $\sigma\in\mathsf{C}_{\kappa}Hx$. Then $((\alpha\sim\{\kappa\})\restriction\sigma\cup\{\langle\kappa,\lambda\rangle\})\in Hx$ and hence $\mathsf{s}_{(\sigma\sim\{\kappa\})\restriction\sigma\cup\{\langle\kappa,\lambda\rangle\}}x\notin M$ for some $\lambda<\alpha$. Therefore, using $\mathscr{HMT}$ 1.11.11(iii) and the inclusion $x\leq\mathsf{c}_{\kappa}x$,

$$\mathsf{s}_{(\alpha\sim\{\kappa\})\restriction\sigma\cup\{\langle\kappa,\lambda\rangle\}}\mathsf{c}_{\kappa}x\notin M\,;$$

from this we conclude by $\mathscr{HMT}$ 1.11.12(iv) that $\mathsf{s}_{\sigma}\mathsf{c}_{\kappa}x\notin M$ and, consequently, that $\sigma\in H\mathsf{c}_{\kappa}x$. We have thus shown that $H\mathsf{c}_{\kappa}x\supseteq\mathsf{C}_{\kappa}Hx$. To show the inclusion in the opposite direction assume that $\sigma\in H\mathsf{c}_{\kappa}x$. Then $\mathsf{s}_{\sigma}\mathsf{c}_{\kappa}x\notin M$. Let

$$\lambda\in\{\xi\colon\xi<\alpha, (\sigma^{-1})^*\{\xi\}=\{\xi\}\}\sim\Delta x \tag{9}$$

and set

$$\tau=(\alpha\sim\{\kappa,\lambda\})\restriction\sigma\cup\{\langle\kappa,\lambda\rangle,\langle\lambda,\kappa\rangle\}\,. \tag{10}$$

Then, by $\mathscr{HMT}$ 1.11.12(v),(vi),

$$\mathsf{c}_\lambda \mathsf{s}_\tau x = \mathsf{s}_\tau \mathsf{c}_\kappa x = \mathsf{s}_\sigma \mathsf{c}_\kappa x \notin M,$$

and hence by condition (ii′) there is a $\mu < \alpha$ such that $\mathsf{s}^\lambda_\mu \mathsf{s}_\tau x \notin M$. We can now apply (9), (10), $\mathscr{HMT}$ 1.11.11(i),(iv), and $\mathscr{HMT}$ 1.11.12(v) to show that $\mathsf{s}_{(\alpha \sim \{\kappa\}) \upharpoonright \sigma \cup \{\langle \kappa, \mu \rangle\}} \times x \notin M$. Hence $(\alpha \sim \{\kappa\}) \upharpoonright \sigma \cup \{\langle \kappa, \mu \rangle\} \in Hx$ and, consequently, $\sigma \in \mathsf{C}_\kappa Hx$. Therefore, (8) holds and the proof of the lemma is complete.

THEOREM 2.2.6. *If $\Delta \in {}^\gamma Sb\ \alpha$ and $\mathfrak{Fr}^{(\Delta)}_\gamma \mathsf{CA}_\alpha \in \mathbf{D}\mathsf{c}_\alpha$, then $\mathfrak{Fr}^{(\Delta)}_\gamma \mathsf{CA}_\alpha$ has the strong interpolation property.*

PROOF. Set $\mathfrak{A} = \mathfrak{Fr}^{(\Delta)}_\gamma \mathsf{CA}_\alpha$. Let $\Gamma, \theta \subseteq \gamma$, $a \in A^{(\Gamma)}$, and $c \in A^{(\theta)}$ such that $a \leq c$. (Notice that here a does not denote the distinguished system of generators of $\mathfrak{A}$.) We must show that there exists a $b \in A^{(\Gamma \cap \theta)}$ such that

$$a \leq b \leq c. \tag{1}$$

By taking $\Gamma \cup (\gamma \sim \theta)$ in place of Γ we see that we may suppose without loss of generality that $\Gamma \cup \theta = \gamma$.

Assume by way of contradiction that for every $b \in A^{(\Gamma \cap \theta)}$ at least one of the inclusions of (1) fails to hold. Also assume, for the time being, that α is a regular cardinal such that

$$\alpha > |\Delta x| \quad \text{for every} \quad x \in A, \tag{2}$$

and assume that $\alpha \geq \gamma$ so that $|A| = \alpha$ by $\mathscr{HMT}$ 2.5.38(ii). Then $\alpha \times A^{(\Gamma)}$ and $\alpha \times A^{(\theta)}$ can be arranged into α-termed sequences $\langle \langle \kappa_\xi, x_\xi \rangle : \xi < \alpha \rangle$ and $\langle \langle \lambda_\xi, y_\xi \rangle : \xi < \alpha \rangle$, respectively. Because α is regular and (2) holds we can define simultaneously by transfinite recursion α-termed sequences $\langle \mu_\xi : \xi < \alpha \rangle$ and $\langle \nu_\xi : \xi < \alpha \rangle$ such that, for every $\eta < \alpha$,

$$\mu_\eta \in \alpha \sim [\Delta a \cup \Delta c \cup \textstyle\bigcup_{\xi \leq \eta} (\Delta x_\xi \cup \Delta y_\xi) \cup \{\mu_\xi : \xi < \eta\} \cup \{\nu_\xi : \xi < \eta\}] \tag{3}$$

and

$$\nu_\eta \in \alpha \sim [\Delta a \cup \Delta c \cup \textstyle\bigcup_{\xi \leq \eta} (\Delta x_\xi \cup \Delta y_\xi) \cup \{\mu_\xi : \xi \leq \eta\} \cup \{\nu_\xi : \xi < \eta\}]. \tag{4}$$

We set

$$M = Ig^{(\mathfrak{BI}\ \mathfrak{A}^{(\Gamma)})} (\{-a\} \cup \{\mathsf{c}_{\kappa_\xi} x_\xi \cdot - \mathsf{s}^{\kappa_\xi}_{\mu_\xi} x_\xi : \xi < \alpha\}), \tag{5}$$

$$N = Ig^{(\mathfrak{BI}\ \mathfrak{A}^{(\theta)})} (\{c\} \cup \{\mathsf{c}_{\lambda_\xi} y_\xi \cdot - \mathsf{s}^{\lambda_\xi}_{\nu_\xi} y_\xi : \xi < \alpha\}), \tag{6}$$

and

$$P = Ig^{(\mathfrak{BI}\ \mathfrak{A}^{(\Gamma \cap \theta)})} [(M \cap A^{(\Gamma \cap \theta)}) \cup (N \cap A^{(\Gamma \cap \theta)})]. \tag{7}$$

We want to show that P is a proper Boolean ideal of $\mathfrak{A}^{(\Gamma \cap \theta)}$. To do this it is sufficient

in view of the Boolean version of $\mathscr{HMT}$ 2.3.8 to consider any pair of finite, strictly increasing sequences of ordinals

(8) $$\xi 0 < \xi 1 < \cdots < \xi(\varphi - 1) < \alpha \quad \text{and} \quad \eta 0 < \eta 1 < \cdots < \eta(\psi - 1) < \alpha$$

and to prove that the following condition holds:

(9) for any $b_0, b_1 \in A^{(\Gamma \cap \theta)}$ such that

$$b_0 \leq -a + \textstyle\sum_{\pi<\varphi}(\mathsf{c}_{\kappa_{\xi\pi}} x_{\xi\pi} \cdot {}_{\nu} \mathsf{s}^{\kappa_{\xi\pi}}_{\mu_{\xi\pi}} x_{\xi\pi})$$

and

$$b_1 \leq c + \textstyle\sum_{\rho<\psi}(\mathsf{c}_{\lambda_{\eta\rho}} y_{\eta\rho} \cdot - \mathsf{s}^{\lambda_{\eta\rho}}_{\nu_{\eta\rho}} y_{\eta\rho}),$$

we have

$$b_0 + b_1 < 1.$$

This is proved by simple induction on $\varphi + \psi$. If $\varphi + \psi = 0$, then (9) just expresses our assumption that (1) does not hold for any $b \in A^{(\Gamma \cap \theta)}$. Assume now that $\varphi + \psi > 0$, and distinguish two cases: $\xi(\varphi - 1) > \eta(\psi - 1)$ and $\xi(\varphi - 1) \leq \eta(\psi - 1)$. In the first case we apply $\mathsf{c}^{\partial}_{\mu_{\xi(\varphi-1)}}$ to both sides of the first inclusion of (9), and then using the first formula of (8), condition (3) with $\eta = \xi(\varphi - 1)$, the dual of axiom (C_3), $\mathscr{HMT}$ 1.5.8(i), (ii), and $\mathscr{HMT}$ 1.5.9(i) we obtain

$$\begin{aligned}
\mathsf{c}^{\partial}_{\mu_{\xi(\varphi-1)}} b_0 &\leq -a + \textstyle\sum_{\pi<\varphi-1}(\mathsf{c}_{\kappa_{\xi\pi}} x_{\xi\pi} \cdot - \mathsf{s}^{\kappa_{\xi\pi}}_{\mu_{\xi\pi}} x_{\xi\pi}) + \\
&\qquad + [\mathsf{c}_{\kappa_{\xi(\varphi-1)}} x_{\xi(\varphi-1)} \cdot \mathsf{c}^{\partial}_{\mu_{\xi(\varphi-1)}} - \mathsf{s}^{\kappa_{\xi(\varphi-1)}}_{\mu_{\xi(\varphi-1)}} x_{\xi(\varphi-1)}] \\
&\leq -a + \textstyle\sum_{\pi<\varphi-1}(\mathsf{c}_{\kappa_{\xi\pi}} x_{\xi\pi} \cdot - \mathsf{s}^{\kappa_{\xi\pi}}_{\mu_{\xi\pi}} x_{\xi\pi}) + [\mathsf{c}_{\kappa_{\xi(\varphi-1)}} x_{\xi(\varphi-1)} \cdot - \mathsf{c}_{\kappa_{\xi(\varphi-1)}} x_{\xi(\varphi-1)}] \\
&= -a + \textstyle\sum_{\pi<\varphi-1}(\mathsf{c}_{\kappa_{\xi\pi}} x_{\xi\pi} \cdot - \mathsf{s}^{\kappa_{\xi\pi}}_{\mu_{\xi\pi}} x_{\xi\pi});
\end{aligned}$$

we then apply $\mathsf{c}_{\mu_{\xi(\varphi-1)}}$ to both sides of the second inclusion of (9), and we use the second formula of (8), condition (3) with $\eta = \xi(\varphi - 1)$, the assumption $\xi(\varphi - 1) > \eta(\psi - 1)$, $\mathscr{HMT}$ 1.2.6(i), and $\mathscr{HMT}$ 1.5.8(ii) to get

$$\mathsf{c}_{\mu_{\xi(\varphi-1)}} b_1 \leq c + \textstyle\sum_{\rho<\psi}(\mathsf{c}_{\lambda_{\eta\rho}} y_{\eta\rho} \cdot - \mathsf{s}^{\lambda_{\eta\rho}}_{\nu_{\eta\rho}} y_{\eta\rho}).$$

Now making use of the inductive hypothesis we conclude that $\mathsf{c}^{\partial}_{\mu_{\xi(\varphi-1)}} b_0 + \mathsf{c}_{\mu_{\xi(\varphi-1)}} b_1 \neq 1$ and hence, by the dual of $\mathscr{HMT}$ 1.2.5, that $b_0 + \mathsf{c}_{\mu_{\xi(\varphi-1)}} b_1 \neq 1$. Thus we get

$$b_0 + b_1 \leq b_0 + \mathsf{c}_{\mu_{\xi(\varphi-1)}} b_1 < 1.$$

The second case, where $\xi(\varphi - 1) \leq \eta(\psi - 1)$, is treated similarly. Therefore, we have shown that (9) holds for every pair of sequences ξ, η satisfying (8), and as we have noted this tells us that P is a proper Boolean ideal of $\mathfrak{A}^{(\Gamma \cap \theta)}$.

Let P' be a maximal proper Boolean ideal of $\mathfrak{A}^{(\Gamma \cap \theta)}$ such that $P' \supseteq P$; then applying 2.1.10 we see that there are maximal proper Boolean ideals M' and N' of $\mathfrak{A}^{(\Gamma)}$ and

$\mathfrak{A}^{(\theta)}$, respectively, such that

(10) $$M' \supseteq M, \quad N' \supseteq N,$$

and

(11) $$M' \cap A^{(\Gamma \cap \theta)} = P' = N' \cap A^{(\Gamma \cap \theta)}.$$

Let

$$R = \{\langle \kappa, \lambda \rangle : \kappa, \lambda < \alpha, \mathsf{d}_{\kappa\lambda} \notin P'\};$$

then R is an equivalence relation on α by 2.2.5(ii). Let $\mathfrak{B}$ be the algebra defined in 2.2.5(i). Recalling that $Rg\, x = A^{(\Gamma)}$ and $Rg\, y = A^{(\theta)}$, we see that (5), (6), and (10) imply that M' and N' both satisfy condition (ii') of 2.2.5 (with $\mathfrak{A}$, M respectively replaced by $\mathfrak{A}^{(\Gamma)}$, M' and then by $\mathfrak{A}^{(\theta)}$, N'); and from (11) we see that

$$\{\langle \kappa, \lambda \rangle : \kappa, \lambda < \alpha, \mathsf{d}_{\kappa\lambda} \notin M'\} = \{\langle \kappa, \lambda \rangle : \kappa, \lambda < \alpha, \mathsf{d}_{\kappa\lambda} \notin N'\} = R.$$

Thus 2.2.5 tells us that

$$\langle \{\sigma : \sigma \in F, \mathsf{s}_\sigma x \notin M'\} : x \in A^{(\Gamma)} \rangle \in Hom(\mathfrak{A}^{(\Gamma)}, \mathfrak{B})$$

and

$$\langle \{\sigma : \sigma \in F, \mathsf{s}_\sigma x \notin N'\} : x \in A^{(\theta)} \rangle \in Hom(\mathfrak{A}^{(\theta)}, \mathfrak{B})$$

where $F = \{\sigma : \sigma \in {}^\alpha\alpha, |\{\xi : \sigma\xi \neq \xi\}| < \omega\}$; furthermore, again using (11) we have that these two functions agree on $A^{(\Gamma \cap \theta)}$. Therefore, since $\{\xi / Cr_\gamma^{(\Delta)}\mathsf{CA}_\alpha : \xi < \alpha\}$ CA_α-independently generates $\mathfrak{A} = \mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha$ by 1.1.8, there is an $H \in Hom(\mathfrak{A}, \mathfrak{B})$ such that

(12) $$H \supseteq \langle \{\sigma : \sigma \in F, \mathsf{s}_\sigma x \notin M'\} : x \in A^{(\Gamma)} \rangle \cup \langle \{\sigma : \sigma \in F, \mathsf{s}_\sigma x \notin N'\} : x \in A^{(\theta)} \rangle.$$

From (5), (6), (10), and the fact that M', N' are proper ideals we see that $a \notin M'$ and $-c \notin N'$. Hence by (12) and $\mathscr{HMT}$ 1.11.12(i) we get

$$\alpha \upharpoonright Id \in Ha \cap H - c = H(a \cdot - c).$$

However, this contradicts the premise that $a \leq c$ and hence that $H(a \cdot - c) = H0 = 0$. Thus we have succeeded in proving, at least in case α is a regular cardinal $\geq \gamma$ satisfying (2), that (1) holds for some $b \in A^{(\Gamma \cap \theta)}$; hence the theorem holds for all α's satisfying these special conditions.

We now turn to the case of an arbitrary (infinite) α. Let β be a regular cardinal such that $\beta \geq \gamma \cup \alpha$ and

(13) $$\beta > |\Delta\xi| \quad \text{for every} \quad \xi < \gamma.$$

Set

$$\mathfrak{A} = \mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\beta.$$

In view of the premise $\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha \in \mathbf{Dc}_\alpha$ we have by 2.2.4 that

$$\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha \cong \mathfrak{Nr}_\alpha\mathfrak{A}. \tag{14}$$

Thus the proof of the theorem will be complete when we show that $\mathfrak{Nr}_\alpha\mathfrak{A}$ has the strong $\mathscr{IP}$.

From (14) we see that $\mathfrak{Nr}_\alpha\mathfrak{A} \in \mathbf{Dc}_\alpha$ and hence, by 2.2.2(ii) and $\mathscr{HMT}$ 2.1.3(ii), $\mathfrak{Nr}_\alpha\mathfrak{A}^{(\Phi)} = (\mathfrak{Nr}_\alpha\mathfrak{A})^{(\Phi)} \in \mathbf{Dc}_\alpha$ for every $\Phi \subseteq \gamma$. Applying 2.2.2(ii) again and 2.2.3(i) with $\mathfrak{A}^{(\Phi)}$ in place of $\mathfrak{A}$ we then have

$$(Nr_\alpha\mathfrak{A})^{(\Phi)} = Nr_\alpha\mathfrak{A}^{(\Phi)} = Nr_\alpha\mathfrak{A}^{(\Phi)} \tag{15}$$

for every $\Phi \subseteq \gamma$. In order to prove that $\mathfrak{Nr}_\alpha\mathfrak{A}$ has the strong $\mathscr{IP}$ consider any Γ, $\theta \subseteq \gamma$ and elements

$$a \in (Nr_\alpha\mathfrak{A})^{(\Gamma)} \quad \text{and} \quad c \in (Nr_\alpha\mathfrak{A})^{(\theta)} \tag{16}$$

such that

$$a \leq c \tag{17}$$

Condition (13), $\mathscr{HMT}$ 2.1.4, and $\mathscr{HMT}$ 2.5.33 together imply that $\beta > |\Delta x|$ for every $x \in A$; therefore, $\mathfrak{A}$ has the strong $\mathscr{IP}$ by the special case of the theorem just proved. Thus, from (17) and the fact that $a \in A^{(\Gamma)}$, $b \in A^{(\theta)}$ by (16), we see that there is an element

$$b \in A^{(\Gamma \cap \theta)} \tag{18}$$

such that

$$a \leq b \leq c. \tag{19}$$

Set $\Omega = (\beta \sim \alpha) \cap \Delta b$. Then, since $Rg\ \Delta \subseteq Sb\ \alpha$, $\mathscr{HMT}$ 2.1.4 gives us $|\Omega| < \omega$ and from (19) we conclude by $\mathscr{HMT}$ 1.2.7 and (16) that

$$a \leq c_{(\Omega)}b \leq c. \tag{20}$$

Obviously, in light of (18), $c_{(\Omega)}b \in Nr_\alpha\mathfrak{A}^{(\Gamma \cap \theta)}$, and applying (15) with $\Phi = \Gamma \cap \theta$ we get

$$\mathsf{c}_{(\Omega)}b \in (Nr_\alpha\mathfrak{A})^{(\Gamma \cap \theta)}. \tag{21}$$

Because the sets Γ, $\theta \subseteq \gamma$ and the elements a, c satisfying (16) and (17) were chosen arbitrarily, we see from (20) and (21) that $\mathfrak{Nr}_\alpha\mathfrak{A}$ has the strong $\mathscr{IP}$. In view of (14) this completes the proof of the theorem.

COROLLARY 2.2.7. *If $\Delta \in {}^\gamma Sb\ \alpha$ and $\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha \in \mathbf{Lf}_\alpha$, then $\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha$ has the strong $\mathscr{IP}$*

PROOF: by 2.2.6 and $\mathscr{HMT}$ 1.11.3(i).

From observations made in Remark 2.1.18 we see that 2.2.7 provides a new algebraic proof of Craig's interpolation theorem; an algebraic proof was first given in Daigneault [8].

THEOREM 2.2.8. Lf_α *has the strong* $\mathscr{AP}$.

PROOF. Let $\mathfrak{A}, \mathfrak{B}$ be any compatible pair of $\mathbf{Lf}_\alpha$'s and let $I = In\,\mathfrak{A} \cup In\,\mathfrak{B}$ (recall the definition of compatibility given in 1.2.2(i)). Set $\gamma = |I|$. Let f be a one-one function from γ onto I and define $\Delta \in {}^\gamma Sb\,\alpha$ by setting $\Delta\xi = \Delta^{(\mathfrak{A})}a_{f\xi}$ if $f\xi \in In\,\mathfrak{A}$ and $\Delta\xi = \Delta^{(\mathfrak{B})}b_{f\xi}$ if $f\xi \in In\,\mathfrak{B}$. (The compatibility of $\mathfrak{A}, \mathfrak{B}$ guarantees that Δ is well-defined.) $\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha \in \mathbf{Lf}_\alpha$ by $\mathscr{HMT}$ 2.5.44 and it is obvious that the $\mathbf{Lf}_\alpha$ $\langle \mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha, \langle f^{-1}i/Cr_\gamma^{(\Delta)}\mathsf{CA}_\alpha : i \in I\rangle\rangle$ covers $\mathfrak{A}, \mathfrak{B}$. The desired result now follows from 2.1.19(ii), 2.1.20, 2.2.7, $\mathscr{HMT}$ 2.1.3(ii), and $\mathscr{HMT}$ 2.3.3.

REMARK 2.2.9. Theorem 2.2.8 was first proved in Daigneault [7]. But there the theorem was formulated in terms of polyadic algebras with equality rather than cylindric algebras.

2.2.6 cannot be used to show that Dc_α has the strong $\mathscr{AP}$ since it is easy to find a compatible pair $\mathfrak{A}, \mathfrak{B}$ of $\mathbf{Dc}_\alpha$'s which is not covered by any g-algebra obtained from a $\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_\alpha \in \mathbf{Dc}_\alpha$ by change of index. To see this, observe $\mathscr{HMT}$ 2.1.7 and then simply choose any $\mathfrak{A}, \mathfrak{B}$ such that $\mathfrak{A}, \mathfrak{B}$ have isomorphic minimal subalgebras, $In\,\mathfrak{A} \cap In\,\mathfrak{B} = 0$, and there is an $x \in A$ and a $y \in B$ such that $\Delta^{(\mathfrak{A})}x \cup \Delta^{(\mathfrak{B})}y = \alpha$. Of course, such a compatible pair $\mathfrak{A}, \mathfrak{B}$ cannot be covered by any $\mathbf{Dc}_\alpha$, so we actually see that Dc_α does not have the strong $\mathscr{AP}$, and, in fact, not even the ordinary $\mathscr{AP}$ or the $\mathscr{EP}$:

However, Theorems 2.1.20 and 2.2.6 can be used to show that, for any α, whether infinite or not, there is a comprehensive subclass of $\mathsf{Dc}_{\alpha+\omega}$'s which does have the strong $\mathscr{AP}$.

THEOREM 2.2.10. *For any finite or infinite* α, *the class*

$$\mathsf{K} = \{\mathfrak{A} : \mathfrak{A} \in \mathsf{CA}_{\alpha+\omega}, |\Delta x \sim \alpha| < \omega \quad \textit{for every} \quad x \in A\}$$

has the strong $\mathscr{AP}$.

PROOF. Let $\mathfrak{A}, \mathfrak{B}$ be any compatible pair of g-algebras in $\mathbf{K}$ and let $I = In\,\mathfrak{A} \cup In\,\mathfrak{B}$. From the compatibility of these g-algebras we conclude that

$$\Delta^{(\mathfrak{A})}a_i = \Delta^{(\mathfrak{B})}b_i \quad \text{for every} \quad i \in In\,\mathfrak{A} \cap In\,\mathfrak{B}. \tag{1}$$

Let ξ be a one-one function from I onto a cardinal γ, and let $\Delta \in {}^\gamma Sb(\alpha+\omega)$ such that $\Delta\xi i = \Delta^{(\mathfrak{A})}a_i$ for $i \in In\,\mathfrak{A}$ and $\Delta\xi j = \Delta^{(\mathfrak{B})}b_j$ for $j \in In\,\mathfrak{B} \sim In\,\mathfrak{A}$. Then Theorem $\mathscr{HMT}$ 2.1.4 implies that $\mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_{\alpha+\omega} \in \mathbf{K}$; also, in view of (1) it is clear that

$$\langle \mathfrak{Fr}_\gamma^{(\Delta)}\mathsf{CA}_{\alpha+\omega}, \langle \xi i/Cr_\gamma^{(\Delta)}\mathsf{CA}_{\alpha+\omega} : i \in I\rangle\rangle$$

covers $\mathfrak{A}, \mathfrak{B}$. Thus, since it is obvious that $\mathbf{SK} = \mathbf{HK} = \mathbf{K} \subseteq \mathbf{Dc}_{\alpha+\omega}$ (cf. $\mathscr{HMT}$ 2.3.2), we conclude from 2.1.19(ii), 2.1.20, and 2.2.6 that K has the strong $\mathscr{AP}$.

Theorem 2.2.10 has as consequences some interesting results concerning the amalgamation and embedding properties for familiar classes of CA's of infinite dimension. However, in formulating some of these results it is convenient to have at our disposal relativized versions of both the strong $\mathscr{AP}$ and the ordinary $\mathscr{AP}$ for classes of cylindric algebras.

DEFINITION 2.2.11. *Let* K, K', *and* L *be classes of* CA_α's.

(i) K *is said to have the* ***strong*** L***-amalgamation property with respect to*** K' *whenever the following condition holds:*

(i′) *Assume* $\mathfrak{A}, \mathfrak{B} \in \mathbf{K}$ *and let* I *and* J *be their respective indexes. Assume that* $\mathfrak{C}$ *is a* g*-algebra in* $\mathbf{L}$ *with index* $I \cap J$, *and that*

$$\mathfrak{A}^{(I \cap J)} \cong \mathfrak{C} \cong \mathfrak{B}^{(I \cap J)}.$$

Then there is a $\mathfrak{D} \in \mathbf{K}'$ *such that* $\mathfrak{D}$ *has index* $I \cup J$,

$$\mathfrak{D}^{(I)} \cong \mathfrak{A}, \quad \mathfrak{D}^{(J)} \cong \mathfrak{B},$$

and

$$D^{(I)} \cap D^{(J)} = D^{(I \cap J)}.$$

(ii) K *is said to have the* L***-amalgamation property with respect to*** K' *if condition (i′) holds when the last formula is deleted.*

If K *has the (strong)* L*-amalgamation property with respect to itself, we say simply that* K *has the (**strong**)* L***-amalgamation property***; *if* K *has the (strong)* K*-amalgamation property with respect to* K' *we say simply that* K *has the (**strong**)* ***amalgamation property with respect to*** K'.

We next prove a lemma which has a rather complicated statement but plays an important role in the proofs of the main theorems of the present part of this section. In the statement of the lemma RCA_α denotes the class of all representable CA_α's; cf. $\mathscr{HMT}$ 1.1.13.

LEMMA 2.2.12. *Let* α *be any finite or infinite ordinal, and let* K *and* L *be classes of* CA_α*'s. Assume that there exists a function* $\mathfrak{H}$ *from* $\mathbf{K} \cup \mathbf{L}$ *into* $\mathbf{CA}_{\alpha+\omega}$ *satisfying the following two conditions:*

(i) $\mathfrak{A} \cong \mathfrak{Nr}_\alpha \mathfrak{H}\mathfrak{A}$ *for every* $\mathfrak{A} \in \mathbf{K} \cup \mathbf{L}$;

(ii) *for all* $\mathfrak{A}, \mathfrak{B} \in \mathbf{K} \cup \mathbf{L}$ *and* $I \subseteq In\,\mathfrak{A} \cap In\,\mathfrak{B}$, *if* $\mathfrak{A}^{(I)} \cong \mathfrak{B}^{(I)} \in I\mathbf{L}$, *then* $(\mathfrak{H}\mathfrak{A})^{(I)} \cong \cong (\mathfrak{H}\mathfrak{B})^{(I)}$.

Then K *has the* L*-amalgamation property with respect to* RCA_α.

Assume in addition to (i) and (ii) that the following condition is satisfied.

(iii) $Nr_\alpha \mathfrak{H}\mathfrak{A} = Nr_\alpha \mathfrak{H}\mathfrak{A}$ *for every* $\mathfrak{A} \in \mathbf{L}$.

($\mathfrak{H}\mathfrak{A}$ *is taken to be the underlying algebra of* $\mathfrak{H}\mathfrak{A}$.) *Then* K *has the strong* L*-amalgamation property with respect to* RCA_α.

PROOF. Assume that (i) and (ii) hold. Consider $\mathfrak{A}$, $\mathfrak{B}$, and $\mathfrak{C}$ such that

$$\mathfrak{A}, \mathfrak{B} \in \mathbf{K} \quad \text{and} \quad \mathfrak{C} \in \mathbf{L}. \tag{1}$$

Set $I = In\, \mathfrak{A}$ and $J = In\, \mathfrak{B}$, and assume that $In\, \mathfrak{C} = I \cap J$ and

$$\mathfrak{A}^{(I \cap J)} \cong \mathfrak{C} \cong \mathfrak{B}^{(I \cap J)}. \tag{2}$$

Then by (1) and (ii) we have

$$(\mathfrak{H}\mathfrak{A})^{(I \cap J)} \cong \mathfrak{H}\mathfrak{C} \cong (\mathfrak{H}\mathfrak{A})^{(I \cap J)}. \tag{3}$$

By (i) and $\mathscr{HMT}$ 2.1.4 we get $|\Delta x \sim \alpha| < \omega$ for every x in the universe of $\mathfrak{H}\mathfrak{A}$, and similarly for $\mathfrak{H}\mathfrak{B}$. So by (3) and 2.2.10 there is a $\mathfrak{D} \in \mathbf{CA}_{\alpha+\omega}$ with $In\, \mathfrak{D} = I \cup J$ such that

$$\mathfrak{D}^{(I)} \cong \mathfrak{H}\mathfrak{A}, \quad \mathfrak{D}^{(J)} \cong \mathfrak{H}\mathfrak{B}, \tag{4}$$

and

$$D^{(I)} \cap D^{(J)} = D^{(I \cap J)}. \tag{5}$$

From (1), (i), 2.2.2(ii), and the first formula of (4) we get

$$(\mathfrak{Nr}_\alpha \mathfrak{D})^{(I)} = \mathfrak{Nr}_\alpha \mathfrak{D}^{(I)} \cong \mathfrak{Nr}_\alpha \mathfrak{H}\mathfrak{A} \cong \mathfrak{A}; \tag{6}$$

similarly, but using the second formula of (4), we get

$$(\mathfrak{Nr}_\alpha \mathfrak{D})^{(J)} \cong \mathfrak{B}. \tag{7}$$

Since $\mathfrak{D} \in \mathbf{CA}_{\alpha+\omega}$ we see from $\mathscr{HMT}$ 2.6.26 that

$$\mathfrak{Nr}_\alpha \mathfrak{D} \in \mathbf{RCA}_\alpha. \tag{8}$$

We may stop at this point and conclude from (6)–(8) and the fact that $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$ are arbitrary g-algebras satisfying (1) and (2) that K has the L-amalgamation property with respect to RCA_α.

Now in addition to (i) and (ii) let us assume that (iii) holds. From (3) and the first formula of (4) we have $\mathfrak{D}^{(I \cap J)} \cong \mathfrak{H}\mathfrak{C}$. Therefore, taking $\mathfrak{C}$ for $\mathfrak{A}$ in (iii) and recalling (1) we get

$$Nr_\alpha \mathfrak{D}^{(I \cap J)} = Nr_\alpha \mathfrak{D}^{(I \cap J)}. \tag{9}$$

We now compute a sequence of inclusions and equalities:

$$\begin{aligned}
&(Nr_\alpha\mathfrak{D})^{(I)} \cap (Nr_\alpha\mathfrak{D})^{(J)} \\
&\quad = Nr_\alpha\mathfrak{D}^{(I)} \cap Nr_\alpha\mathfrak{D}^{(J)} && \text{by 2.2.2 (ii)} \\
&\quad \subseteq Nr_\alpha\mathfrak{D}^{(I)} \cap Nr_\alpha\mathfrak{D}^{(J)} && \text{by 2.2.2 (i)} \\
&\quad = Nr_\alpha\mathfrak{D} \cap D^{(I)} \cap D^{(J)} && \text{by 2.2.2 (iii)} \\
&\quad = Nr_\alpha\mathfrak{D} \cap D^{(I\cap J)} && \text{by (5)} \\
&\quad = Nr_\alpha\mathfrak{D}^{(I\cap J)} && \text{by 2.2.2 (iii) ;}
\end{aligned}$$

finally, applying (9) and 2.2.2(ii) we obtain

$$(Nr_\alpha\mathfrak{D})^{(I)} \cap (Nr_\alpha\mathfrak{D})^{(J)} \subseteq (Nr_\alpha\mathfrak{D})^{(I\cap J)}.$$

The inclusion in the opposite direction is obvious, and from the resulting equality together with (6)–(8) we conclude that K has the strong L-amalgamation property with respect to RCA_α. This completes the proof.

THEOREM 2.2.13. RCA_α *has the strong* Dc_α*-amalgamation property.*

PROOF. In view of $\mathscr{HMT}$ 2.6.26, $\mathsf{RCA}_\alpha \subseteq \mathsf{SNr}_\alpha\mathsf{CA}_{\alpha+\omega}$. For each $\mathfrak{A} \in \mathbf{RCA}_\alpha$ choose an algebra $\mathfrak{B}\mathfrak{A} \in \mathsf{CA}_{\alpha+\omega}$ such that $\mathfrak{A}$ is isomorphic to a subalgebra of $\mathfrak{Nr}_\alpha\mathfrak{B}\mathfrak{A}$, and then choose $f_{\mathfrak{A}} \in Ism(\mathfrak{A}, \mathfrak{Nr}_\alpha\mathfrak{B}\mathfrak{A})$. Finally, let $\mathfrak{H}$ be the function with domain $\mathbf{RCA}_\alpha$ such that

$$\mathfrak{H}\mathfrak{A} = \langle \mathfrak{Sg}^{(\mathfrak{B}\mathfrak{A})} f_{\mathfrak{A}}^* A, f_{\mathfrak{A}} \circ a\rangle$$

for each $\mathfrak{A} \in \mathbf{RCA}_\alpha$. It is clear that $\mathfrak{H}$ satisfies condition 2.2.12(i) with with $\mathsf{K} = \mathsf{RCA}_\alpha$ and $\mathsf{L} = \mathsf{Dc}_\alpha$ (recall that $\mathsf{Dc}_\alpha \subseteq \mathsf{RCA}_\alpha$ by $\mathscr{HMT}$ 2.6.26 and $\mathscr{HMT}$ 2.6.49).

Consider any $\mathfrak{A}, \mathfrak{B} \in \mathbf{RCA}_\alpha$ and any $I \subseteq In\,\mathfrak{A} \cap In\,\mathfrak{B}$ such that

$$\mathfrak{A}^{(I)} \cong \mathfrak{B}^{(I)} \in \mathbf{Dc}_\alpha. \tag{1}$$

By 2.2.2(ii) and 2.2.12(i) we have

$$\mathfrak{Nr}_\alpha(\mathfrak{H}\mathfrak{A})^{(I)} = (\mathfrak{Nr}_\alpha\mathfrak{H}\mathfrak{A})^{(I)} \cong \mathfrak{A}^{(I)}$$

and, similarly, $\mathfrak{Nr}(\mathfrak{H}\mathfrak{B})^{(I)} \cong \mathfrak{B}^{(I)}$. Therefore, by (1),

$$\mathfrak{Nr}_\alpha(\mathfrak{H}\mathfrak{A})^{(I)} \cong \mathfrak{Nr}_\alpha(\mathfrak{H}\mathfrak{B})^{(I)} \in \mathbf{Dc}_\alpha.$$

Thus $(\mathfrak{H}\mathfrak{A})^{(I)} \cong (\mathfrak{H}\mathfrak{B})^{(I)}$ by 2.2.3(ii), and hence 2.2.12(ii) holds with $\mathsf{K} = \mathsf{RCA}_\alpha$ and $\mathsf{L} = \mathsf{Dc}_\alpha$.

Finally, if $\mathfrak{A} \in \mathbf{Dc}_\alpha$, then $\mathfrak{Nr}_\alpha\mathfrak{H}\mathfrak{A} \in \mathbf{Dc}_\alpha$ by 2.2.12(i), and thus $Nr_\alpha\mathfrak{H}\mathfrak{A} = Nr_\alpha\mathfrak{H}\mathfrak{A}$ by 2.2.3(i). Therefore, 2.2.12(iii) holds when $\mathsf{K} = \mathsf{RCA}_\alpha$ and $\mathsf{L} = \mathsf{Dc}_\alpha$. Apply 2.2.12 to obtain the theorem.

COROLLARY 2.2.14. Dc_α *has the strong* $\mathscr{AP}$ *with respect to* RCA_α.

PROOF: by 2.2.13 and the fact that $\mathsf{Dc}_\alpha \subseteq \mathsf{RCA}_\alpha$; cf. $\mathscr{HMT}$ 2.6.26 and $\mathscr{HMT}$ 2.6.49.

COROLLARY 2.2.15. RCA_α *has the embedding property.*

PROOF: by 2.2.13 and the fact that every minimal CA_α is a Dc_α; cf. $\mathscr{HMT}$ 1.11.3(i) and $\mathscr{HMT}$ 2.1.16.

REMARK 2.2.16. Actually, it is easily seen that RCA_α has the strong $\mathscr{EP}$ as defined in 2.1.22.

In view of 1.2.8(ii) and 2.1.11, Corollary 2.2.15 immediately gives:

THEOREM 2.2.17. $\mathfrak{Fr}_\gamma\mathsf{RCA}_\alpha$ *has the restricted* $\mathscr{IP}$.

REMARK 2.2.18. We do not know if $\mathfrak{Fr}_\gamma\mathsf{RCA}_\alpha$ has the strong restricted $\mathscr{IP}$, but we shall see in the next section that it fails to have the ordinary $\mathscr{IP}$ for any given $\gamma > 3$. In view of the general correspondence stated in Theorem 1.2.8(i) this will show that RCA_α fails to have the $\mathscr{AP}$, so that Corollary 2.2.14 cannot be improved so as to apply to the whole class RCA_α – even if its statement is weakened by deleting the word 'strong'. However, in this weakened form 2.2.14 can be improved substantially as we now show.

$\mathfrak{A}$ is a *diagonal cylindric algebra of dimension* α, in symbols $\mathfrak{A} \in \mathsf{Di}_\alpha$, if for every finite $\Gamma \subseteq \alpha$ and every non-zero $x \in A$ there exist distinct $\kappa, \lambda \in \alpha \sim \Gamma$ such that $x \cdot \mathsf{d}_{\kappa\lambda} \neq 0$. It is known that $\mathsf{Ss}_\alpha \cup \mathsf{Dc}_\alpha \subseteq \mathsf{Di}_\alpha \subseteq \mathsf{RCA}_\alpha$ where Ss_α is the class of all semisimple CA_α's. See $\mathscr{HMT}$ 2.6.50 and the immediately following remark.

THEOREM 2.2.19. Di_α *has the* $\mathscr{AP}$ *with respect to* RCA_α.

PROOF. We will prove the theorem by applying 2.2.12, taking both K and L to be the class Di_α.

Let R be the set of all ordered quadruples $\langle \Gamma, \mu, \kappa, \lambda \rangle$ such that Γ is a finite subset of α, μ is a finite ordinal, and κ, λ are sequences without repeating terms such that

$$\kappa, \lambda \in {}^{\mu}(\alpha \sim \Gamma) \quad \text{and} \quad Rg\,\kappa \cap Rg\,\lambda = 0. \tag{1}$$

Let M be the filter on R which is generated by the family of all subsets of the form

$$X_{\Gamma,\mu} = \{\langle \Delta, \nu, \kappa, \lambda \rangle : \langle \Delta, \nu, \kappa, \lambda \rangle \in R,\ \Delta \supseteq \Gamma,\ \nu \geq \mu\}$$

where $\Gamma \subseteq \alpha$, $|\Gamma| < \omega$, and $\mu < \omega$. Then

$$M = \{Y : Y \subseteq R \text{ and } Y \supseteq X_{\Gamma,\mu} \text{ for some } \Gamma, \mu \text{ with } \Gamma \subseteq \alpha, |\Gamma| < \omega, \text{ and } \mu < \omega\}.$$

Clearly M is a proper filter.

We next choose a function ρ from R into the set of all one-one functions from $\alpha+\omega$ onto α; we choose ρ such that for any $\langle\Gamma, \mu, \kappa, \lambda\rangle \in R$ we have

$$\Gamma \upharpoonright \rho\langle\Gamma, \mu, \kappa, \lambda\rangle \subseteq Id \tag{2}$$

and

$$(\rho\langle\Gamma, \mu, \kappa, \lambda\rangle)(\alpha+\nu) = \kappa_\nu \quad \text{for every} \quad \nu < \mu. \tag{3}$$

Let $\mathfrak{F}$ be the function whose domain is the class Di_α and such that

$$\mathfrak{F}\mathfrak{A} = \mathsf{P}_{\varphi \in R}\mathfrak{Rd}^{(\rho\varphi)}\mathfrak{A} \tag{4}$$

for each $\mathfrak{A} \in \mathsf{Di}_\alpha$. ($\mathfrak{Rd}^{(\rho\varphi)}\mathfrak{A}$, the *$\rho\varphi$-reduct of* $\mathfrak{A}$, is defined in *HMT* 2.6.1.) Notice that the universe of $\mathfrak{Rd}^{(\rho\varphi)}\mathfrak{A}$ is just A, the universe of $\mathfrak{A}$; thus the universe of $\mathfrak{F}\mathfrak{A}$ is simply ${}^RA$. By *HMT* 2.6.2(i), $\mathfrak{Rd}^{(\rho\varphi)}\mathfrak{A} \in \mathsf{CA}_{\alpha+\omega}$, and hence

$$\mathfrak{F}\mathfrak{A} \in \mathsf{CA}_{\alpha+\omega} \quad \text{for every} \quad \mathfrak{A} \in \mathsf{CA}_\alpha \tag{5}$$

(cf. *HMT* 2.4.1). Recall that the relation

$$\bar{M}^{(\mathfrak{F}\mathfrak{A})} = \bar{M} = \{\langle x, y\rangle : x, y \in {}^RA, \{\varphi : \varphi \in R, x_\varphi = y_\varphi\} \in M\} \tag{6}$$

is a congruence relation on $\mathfrak{F}\mathfrak{A}$. For each $\mathfrak{A} \in \mathsf{Di}_\alpha$ let $f_{\mathfrak{A}}$ be the function from A into ${}^RA/\bar{M}$ such that

$$f_{\mathfrak{A}}x = \langle \mathsf{s}^{\kappa_0}_{\lambda_0}\mathsf{s}^{\kappa_1}_{\lambda_1}\ldots\mathsf{s}^{\kappa_{\mu-1}}_{\lambda_{\mu-1}}x : \langle\Gamma, \mu, \kappa, \lambda\rangle \in R\rangle/\bar{M} \tag{7}$$

for each $x \in A$. By *HMT* 1.5.3 $\mathsf{s}^{\kappa_0}_{\lambda_0}\circ\cdots\circ\mathsf{s}^{\kappa_{\mu-1}}_{\lambda_{\mu-1}}$ is a Boolean homomorphism from $\mathfrak{A}$ into $\mathfrak{Rd}^{(\rho\langle\Gamma, \mu, \kappa, \lambda\rangle)}\mathfrak{A}$ whenever $\langle\Gamma, \mu, \kappa, \lambda\rangle \in R$. Thus it is easily seen that

$$f_{\mathfrak{A}} \in Hom(\mathfrak{Bl}\,\mathfrak{A}, \mathfrak{Bl}(\mathfrak{F}\mathfrak{A}/\bar{M})). \tag{8}$$

Consider any $\xi < \alpha$. Then for each $\langle\Gamma, \mu, \kappa, \lambda\rangle \in R$ such that $\xi \in \Gamma$ we have

$$\mathsf{s}^{\kappa_0}_{\lambda_0}\ldots\mathsf{s}^{\kappa_{\mu-1}}_{\lambda_{\mu-1}}\mathsf{c}_\xi x = \mathsf{c}_\xi\mathsf{s}^{\kappa_0}_{\lambda_0}\ldots\mathsf{s}^{\kappa_{\mu-1}}_{\lambda_{\mu-1}}x$$

for every $x \in A$. This equality follows from *HMT* 1.5.8(ii) since $\xi \notin Rg\,\kappa \cup Rg\,\lambda$ by (1). Thus from (2) and the obvious fact that $\{\langle\Gamma, \mu, \kappa, \lambda\rangle : \langle\Gamma, \mu, \kappa, \lambda\rangle \in R, \xi \in \Gamma\} \in M$ we conclude that

$$f_{\mathfrak{A}}\mathsf{c}_\xi x = \mathsf{c}^{(\mathfrak{F}\mathfrak{A})}_\xi f_{\mathfrak{A}}x \quad \text{for all} \quad x \in A \quad \text{and} \quad \xi < \alpha. \tag{9}$$

In a similar manner we can show with the aid of *HMT* 1.5.4(ii) that

$$f_{\mathfrak{A}}\mathsf{d}_{\xi\eta} = \mathsf{d}^{(\mathfrak{F}\mathfrak{A})}_{\xi\eta} \quad \text{for all} \quad \xi, \eta < \alpha. \tag{10}$$

Now consider any $\eta \in (\alpha+\omega) \sim \alpha$. Then for each $\langle\Gamma, \mu, \kappa, \lambda\rangle \in R$ such that $\eta < \alpha+\mu$ we have

$$\mathsf{c}_{(\rho\langle\Gamma, \mu, \kappa, \lambda\rangle)\eta}\mathsf{s}^{\kappa_0}_{\lambda_0}\ldots\mathsf{s}^{\kappa_{\mu-1}}_{\lambda_{\mu-1}}x = \mathsf{s}^{\kappa_0}_{\lambda_0}\ldots\mathsf{s}^{\kappa_{\mu-1}}_{\lambda_{\mu-1}}x$$

for every $x \in A$; this equality follows from $\mathscr{HMT}$ 1.5.8(ii) and $\mathscr{HMT}$ 1.5.9(ii) since $(\rho\langle\Gamma, \mu, \kappa, \lambda\rangle)\eta \in Rg\,\kappa \sim Rg\,\lambda$ by (1) and (3). Therefore, because $\{\langle\Gamma, \mu, \kappa, \lambda\rangle : \langle\Gamma, \mu, \kappa, \lambda\rangle \in R, \eta < \alpha + \mu\} \in M$, we see that $f_{\mathfrak{A}}{}^*A \subseteq Nr_\alpha\mathfrak{F}\mathfrak{A}$. This inclusion together with (8)–(10) gives

$$f_{\mathfrak{A}} \in Hom(\mathfrak{A}, \mathfrak{Nr}_\alpha(\mathfrak{F}\mathfrak{A}/\bar{M})). \tag{11}$$

Consider now any $x \in A$ such that $x \neq 0$. Then for any given finite $\Gamma \subseteq \alpha$ and any $\mu < \omega$ there exist sequences $\kappa, \lambda \in {}^\mu(\alpha \sim \Gamma)$ without repeating terms with the property that $Rg\,\kappa \cap Rg\,\lambda = 0$ and

$$\mathsf{s}^{\kappa_0}_{\lambda_0} \ldots \mathsf{s}^{\kappa_{\mu-1}}_{\lambda_{\mu-1}} x \neq 0\,;$$

that such κ, λ must always exist follows directly from the definition of diagonal algebras by a simple inductive argument. Thus we have $f_{\mathfrak{A}}x \neq 0^{(\mathfrak{F}\mathfrak{A})}/\bar{M}$ for every $x \in A \sim \{0\}$, and this result together with (11) shows that

$$f_{\mathfrak{A}} \in Ism(\mathfrak{A}, \mathfrak{Nr}_\alpha(\mathfrak{F}\mathfrak{A}/\bar{M})). \tag{12}$$

We are now ready to define the function $\mathfrak{H}$ to be used in the application of 2.2.12. For each $\mathfrak{A} \in \mathbf{Di}_\alpha$ let

$$\mathfrak{H}\mathfrak{A} = \langle \mathfrak{Sg}^{(\mathfrak{F}\mathfrak{A}/\bar{M})} f_{\mathfrak{A}}^* A, f_{\mathfrak{A}} \circ a\rangle \tag{13}$$

it is then an immediate consequence of (12) that 2.2.12(i) holds when $\mathsf{K} = \mathsf{L} = \mathsf{Di}_\alpha$.

For condition 2.2.12(ii) to hold when $\mathsf{K} = \mathsf{L} = \mathsf{Di}_\alpha$ it is easily seen to be sufficient (and also necessary) that both of the following two conditions hold:

$$\text{for all } \mathfrak{A}, \mathfrak{B} \in \mathbf{Di}_\alpha, \text{ if } \mathfrak{A} \cong \mathfrak{B}, \text{ then } \mathfrak{H}\mathfrak{A} \cong \mathfrak{H}\mathfrak{B}\,; \tag{14}$$

$$\text{for all } \mathfrak{A} \in \mathbf{Di}_\alpha \text{ and } I \subseteq In\,\mathfrak{A}, \mathfrak{H}\mathfrak{A}^{(I)} \cong (\mathfrak{H}\mathfrak{A})^{(I)}. \tag{15}$$

Condition (14) obviously holds. To show that (15) holds consider any $\mathfrak{A} \in \mathbf{Di}_\alpha$ and $I \subseteq In\,\mathfrak{A}$. That $\mathfrak{Rd}^{(\rho\varphi)}\mathfrak{A}^{(I)} \subseteq \mathfrak{Rd}^{(\rho\varphi)}\mathfrak{A}$ for each $\varphi \in R$ is obvious from the definition of a reduct, and hence $\mathfrak{F}\mathfrak{A}^{(I)} \subseteq \mathfrak{F}\mathfrak{A}$ by (4). It is also immediately seen from (6) that

$$\bar{M}^{(\mathfrak{F}\mathfrak{A}^{(I)})} = \bar{M}^{(\mathfrak{F}\mathfrak{A})} \cap ({}^R A^{(I)} \times {}^R A^{(I)}).$$

Thus, if

$$g = \{\langle x/\bar{M}^{(\mathfrak{F}\mathfrak{A}^{(I)})}, x/\bar{M}^{(\mathfrak{F}\mathfrak{A})}\rangle : x \in {}^R A^{(I)}\}, \tag{16}$$

then

$$g \in Ism(\mathfrak{F}\mathfrak{A}^{(I)}/\bar{M}^{(\mathfrak{F}\mathfrak{A}^{(I)})}, \mathfrak{F}\mathfrak{A}/\bar{M}^{(\mathfrak{F}\mathfrak{A})}). \tag{17}$$

Therefore $\mathfrak{H}\mathfrak{A}^{(I)}$ is isomorphic with some subalgebra of $\mathfrak{H}\mathfrak{A}$. It is obvious from the definition of the substitution operator that

$$(\mathsf{s}^\kappa_\lambda)^{(\mathfrak{F}\mathfrak{A}^{(I)})} = {}^R A^{(I)} \upharpoonright (\mathsf{s}^\kappa_\lambda)^{(\mathfrak{F}\mathfrak{A})}$$

for all $\kappa, \lambda < \alpha$; hence we see from (7) and (16) that

$$g[(f_{\mathfrak{A}(I)}a_i)/\bar{M}^{(\mathfrak{F}\mathfrak{A}(I))}] = (f_{\mathfrak{A}}a_i)/\bar{M}^{(\mathfrak{F}\mathfrak{A})} \quad \text{for} \quad i \in I. \tag{18}$$

Combining (13), (17), and (18) we obtain the conclusion of (15). Thus (15) holds and hence also 2.2.12(ii) with $\mathsf{K} = \mathsf{L} = \mathsf{Di}_\alpha$. We now apply 2.2.12 to get the theorem.[3])

THEOREM 2.2.20. *The class* Di_α *has the* $\mathscr{AP}$.

PROOF. Let $\mathfrak{A}$ and $\mathfrak{B}$ be any pair of compatible $\mathbf{Di}_\alpha$'s. Let I and J be the indexes of $\mathfrak{A}$ and $\mathfrak{B}$, respectively. Then by 2.2.19 there is a $\mathfrak{C}$ with index $I \cup J$ such that

$$\mathfrak{C}^{(I)} \cong \mathfrak{A}, \quad \mathfrak{C}^{(J)} \cong \mathfrak{B}, \tag{1}$$

and $\mathfrak{C} \in \mathbf{RCA}_\sigma$. (We must have $\mathfrak{A}^{(I \cap J)}$, $\mathfrak{B}^{(I \cap J)} \in \mathbf{Di}_\alpha$ since Di_α is obviously closed under the formation of subalgebras.)

We now define a transfinite sequence of ideals M_ξ of $\mathfrak{C}$ where ξ ranges over arbitrary ordinals. The sequence is defined recursively in the following way: $M_0 = \{0\}$; for each ordinal $\xi > 0$ we take M_ξ to be the set of all elements $x \in A$ such that, for some finite $\Gamma \subseteq \alpha$ and all distinct $\kappa, \lambda \in \alpha \sim \Gamma$,

$$x \cdot \mathsf{d}_{\kappa\lambda} \in \bigcup_{\eta<\xi} M_\xi.$$

Using transfinite induction it is an easy matter to prove that $M_\xi \in Il\,\mathfrak{C}$ for each ξ; also, because $\mathfrak{C}^{(I)}, \mathfrak{C}^{(J)} \in \mathsf{Di}_\alpha$ by (1), we can show that

$$M_\xi \cap C^{(I)} = M_\xi \cap C^{(J)} = 0 \tag{2}$$

for every ξ. Since $M_\xi \subseteq M_\eta$ whenever $\xi < \eta$ we must have $M_\xi = M_{\xi+1}$ for all sufficiently large ξ; let β be the least ξ with this property. Then it follows almost immediately from the definition of diagonal algebras that

$$\mathfrak{C}/M_\beta \in \mathsf{Di}_\alpha. \tag{3}$$

From (1), (2) with $\xi = \beta$, and 1.1.30 we conclude that

$$(\mathfrak{C}/M)^{(I)} \cong \mathfrak{C}^{(I)}/(M \cap C^{(I)}) \cong \mathfrak{A},$$
$$(\mathfrak{C}/M)^{(J)} \cong \mathfrak{C}^{(J)}/(M \cap C^{(J)}) \cong \mathfrak{B}.$$

Hence, because of (3) and the fact that $\mathfrak{A}$, $\mathfrak{B}$ are arbitrary compatible $\mathbf{Di}_\alpha$'s, the theorem is proved.

REMARK 2.2.21. We do not know if the class of all diagonal CA_α's has the strong $\mathscr{AP}$. We also do not know if it has the strong $\mathscr{AP}$ with respect to RCA_α. In particular,

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[3]) In its essential ideas the construction of the algebra $\mathfrak{H}\mathfrak{A}$ given in this proof was used for the first time in Monk [21] to prove that every diagonal CA_α (with $\alpha \geq \omega$) is representable.

we do not know if condition 2.2.12(iii) holds when $\mathsf{L} = \mathsf{Di}_\alpha$ and $\mathfrak{H}$ is defined as in (13) of the proof of 2.2.19 (see Lemma 2.2.12); however, this seems highly unlikely to us.

The following is an immediate corollary of 2.2.19 and the fact that every semisimple CA_α is a diagonal algebra; cf. $\mathscr{HMT}$ 2.6.50(i),(ii).

THEOREM 2.2.22. *The class* Ss_α *of all semisimple* CA_α*'s has the* $\mathscr{AP}$ *with respect to* RCA_α.

Actually, a slightly stronger result holds, as we shall see in Theorem 2.2.24 below.

THEOREM 2.2.23. *The class of all simple* CA_α*'s has the* $\mathscr{AP}$.

PROOF. Consider any pair $\mathfrak{A}, \mathfrak{B}$ of simple $\mathbf{CA}_\alpha$'s. Let $I = In\,\mathfrak{A}$ and $J = In\,\mathfrak{B}$. By $\mathscr{HMT}$ 2.3.16(i) and 2.2.22 there is a $\mathfrak{C} \in \mathbf{CA}_\alpha$ (actually, $\mathfrak{C} \in \mathbf{RCA}_\alpha$) such that $In\,\mathfrak{C} = I \cup J$ and

$$\mathfrak{C}^{(I)} \cong \mathfrak{A} \quad , \quad \mathfrak{C}^{(J)} \cong \mathfrak{B}. \tag{1}$$

Let M be any maximal proper ideal of $\mathfrak{C}$. Then $\mathfrak{C}/M$ is a simple $\mathbf{CA}_\alpha$ and, since we must have $M \cap C^{(I)} = M \cap C^{(J)} = \{0\}$, we conclude from (1) and 1.1.30 that $(\mathfrak{C}/M)^{(I)} \cong \mathfrak{A}$, $(\mathfrak{C}/M)^{(J)} \cong \mathfrak{B}$. This completes the proof.

THEOREM 2.2.24. *The class* Ss_α *has the* $\mathscr{AP}$.

PROOF. Take $\mathbf{K} = \mathbf{CA}_\alpha$. Then conditions 1.2.5(i),(ii) hold (cf. $\mathscr{HMT}$ 2.3.16(i)). To show that 1.2.5(iii) also holds consider any $\mathfrak{A}, \mathfrak{B}, \mathfrak{C} \in \mathbf{CA}_\alpha$ such that $\mathfrak{A}$ is simple and $\mathfrak{A} \preccurlyeq \mathfrak{B} \subseteq \mathfrak{C}$. Then by 1.1.28(i) there is an ideal $M \in Il\,\mathfrak{B}$ such that $\mathfrak{A} \cong \mathfrak{B}/M$, and M is a maximal proper ideal since $\mathfrak{A}$ is simple. Thus by 2.1.10 there is a maximal proper ideal N' of $\mathfrak{C}$ such that $N' \cap B = M$. Hence $\mathfrak{C}/N'$ is simple, and by 1.1.30 we have

$$\mathfrak{A} \cong \mathfrak{B}/M \cong (\mathfrak{C}/N')^{(In\,\mathfrak{A})};$$

of course, $\mathfrak{C}/N' \preccurlyeq \mathfrak{C}$. Therefore, we have shown that 1.2.5(iii) holds. Applying 1.2.5 we conclude from 2.2.23 that Ss_α has the $\mathscr{AP}$.

THEOREM 2.2.25. *Let* K *be any class of* CA_α*'s such that* $\mathsf{K} \supseteq \mathsf{RCA}_\alpha$. *Then* $\mathfrak{Fr}_\gamma\mathsf{K}$ *has the weak* $\mathscr{IP}$.

PROOF. Since $\mathfrak{Fr}_\gamma\mathsf{K} = \mathfrak{Fr}_\gamma \boldsymbol{HSP}\mathsf{K}$ we can assume that K is algebraically closed. Every simple CA_α is representable by $\mathscr{HMT}$ 2.6.26 and $\mathscr{HMT}$ 2.6.50(i),(iv). Thus the class of simple algebras in K coincides with the class of all simple CA_α's, and hence the theorem follows immediately by 1.2.10, 2.1.13, and 2.2.23 (cf. also 1.1.38).

REMARK 2.2.26. There is an alternative proof of 2.2.25 that is of independent interest because it appeals directly to 2.2.6 and does not involve any notion of amal-

gamation property. This new proof, which we now outline in some detail, has a metamathematical character. (The proof can be translated into a mathematical one, but we shall not do this here.)

In these remarks we shall be dealing with the first-order theories of cylindric algebras whose dimensions range from α up to $\alpha+\omega$. Therefore, for convenience let us agree that, for each $\xi \leq \omega$, $\Lambda_\gamma^{(\xi)}$ shall represent a discourse language for $\mathsf{CA}_{\alpha+\xi}$'s whose sequence of variables is of length γ; furthermore, we shall write Λ_γ for $\Lambda_\gamma^{(0)}$. Clearly we may assume that $\Lambda_\gamma^{(\xi)}$ is included in $\Lambda_\gamma^{(\eta)}$ whenever $\xi \leq \eta \leq \omega$. By this we mean that the variables and non-logical constants of $\Lambda_\gamma^{(\xi)}$ and $\Lambda_\gamma^{(\eta)}$ are the same, and that the operation symbol of $\Lambda_\gamma^{(\eta)}$ which corresponds to any one of the fundamental operations $+, \cdot, \ldots, \mathsf{c}_\kappa$ for $\kappa < \alpha+\xi$, and $\mathsf{d}_{\kappa\lambda}$ for $\kappa, \lambda < \alpha+\xi$ coincides with the operation symbol of $\Lambda_\gamma^{(\xi)}$ corresponding to the same fundamental operation.

Expressed in metamathematical terms Theorem 2.2.25 says the following: Let K be any class of CA_α's such that $\mathsf{K} \supseteq \mathsf{RCA}_\alpha$. Then any inclusion $\sigma \leq \tau$ between terms of Λ_γ which holds in every member of K can be *weakly interpolated relative to* K; that is to say, there is a finite subset Γ of α and a term ρ of Λ_γ containing only occurrences of variables which occur simultaneously in σ and τ such that the two inclusions

$$\mathsf{c}_{(\Gamma)}{}^{\partial}\sigma \leq \rho \leq \mathsf{c}_{(\Gamma)}\tau$$

are identically satisfied in every member of K (compare here 2.1.12).

For reasons of convenience we shall assume in the present proof that the cardinal γ is equal to ω, and we shall then write everywhere $\Lambda^{(\xi)}$ in place of $\Lambda_\gamma^{(\xi)}$; it should be clear that this assumption really does not result in any loss of generality. Suppose, then, that $\sigma \leq \tau$ is an inclusion which holds in every $\mathfrak{A} \in \mathsf{K}$. Since $\mathsf{K} \supseteq \mathsf{RCA}_\alpha \supseteq \mathsf{Nr}_\alpha\mathsf{CA}_{\alpha+\omega}$ (cf. $\mathscr{HMT}$ 2.6.26), we see that for every $\mathfrak{B} \in \mathsf{CA}_{\alpha+\omega}$ the inclusion $\tau \leq \sigma$ is satisfied in $\mathfrak{B}$ by every assignment of variables to elements of $\mathfrak{B}$ – if we restrict ourselves to those assignments which take as values only elements whose dimensions sets are included in α. I.e., $\sigma \leq \tau$ is satisfied by every assignment whose range is included in $Nr_\alpha\mathfrak{B}$. From this fact we are able to conclude using $\mathscr{HMT}$ 2.1.7, $\mathscr{HMT}$ 2.5.33(ii), and 2.2.6 (or rather the re-formulation of 2.2.6 in metamathematical terms) that there is a term π of $\Lambda^{(\omega)}$ which contains only occurrences of variables which occur in both σ and τ and which satisfies the following condition for every $\mathfrak{B} \in \mathsf{CA}_{\alpha+\omega}$.

(1) The two inclusions $\sigma \leq \pi$ and $\pi \leq \tau$ are satisfied in $\mathfrak{B}$ by every assignment whose range is included in $Nr_\alpha\mathfrak{B}$.

Now π may contain occurrences of cylindrification symbols $\mathbf{c}_\kappa$ and diagonal symbols $\mathbf{d}_{\kappa\lambda}$ which have as indices ordinals $\geq \alpha$, but this obviously can happen for only a finite number of such ordinals. (π cannot in general be chosen so that the indices of all cylindrification symbols and diagonal symbols occurring in it are $< \alpha$. This is a con-

sequence of the fact that the free g-algebra $\mathfrak{Fr}_\omega \mathsf{RCA}_\alpha$ fails to have the strong $\mathscr{IP}$; cf. Theorem 2.3.4(i) below.)

For each $\xi \leq \omega$ set $\Sigma^{(\xi)}$ equal to the set of all instances of the axiom schemas (C_0)–(C_7) of 2.1.1 which define $\mathsf{CA}_{\alpha+\xi}$'s. Observe that

$$\Sigma^{(\xi)} \subseteq \Sigma^{(\eta)} \quad \text{whenever} \quad \xi \leq \eta \leq \omega \tag{2}$$

and

$$\Sigma^{(\omega)} = \bigcup_{\kappa<\omega} \Sigma^{(\kappa)}. \tag{3}$$

We now extend each of the languages $\Lambda^{(\xi)}$, $\xi \leq \omega$, by adjoining to its set of non-logical constants all the terms of a fixed ω-termed sequence $\boldsymbol{a} = \langle \boldsymbol{a}_0, \boldsymbol{a}_1, \boldsymbol{a}_2, \ldots \rangle$ of distinct individual constant symbols. Let σ', τ', and π' be the terms of the language extending $\Lambda^{(\omega)}$ that are obtained respectively from σ, τ, and π by replacing each variable $\boldsymbol{v}_\kappa$ in all of its occurrences by the corresponding constant symbol $\boldsymbol{a}_\kappa$. For each $\kappa < \omega$ let $\Pi^{(\kappa)}$ be the set of all identities of the form

$$\mathbf{c}_\mu \boldsymbol{a}_\nu = \boldsymbol{a}_\nu$$

where $\alpha \leq \mu < \alpha + \kappa$ and $\nu < \omega$; of course we have

$$\Pi^{(\kappa)} \subseteq \Pi^{(\lambda)} \quad \text{whenever} \quad \kappa \leq \lambda < \omega. \tag{4}$$

To say that (1) holds for every $\mathfrak{B} \in \mathsf{CA}_{\alpha+\omega}$ means exactly that the double inclusion $\sigma' \leq \pi' \leq \tau'$ is a consequence of $\Sigma^{(\omega)} \cup \bigcup_{\kappa<\omega} \Pi^{(\kappa)}$, i.e., that

$$\Sigma^{(\omega)} \cup \bigcup_{\kappa<\omega} \Pi^{(\kappa)} \models (\sigma' \leq \pi') \wedge (\pi' \leq \tau').$$

Therefore, by the compactness theorem of first-order logic there is a finite subset θ of $\Sigma^{(\omega)} \cup \bigcup_{\kappa<\omega} \Pi^{(\kappa)}$ such that

$$\theta \models (\sigma' \leq \pi') \wedge (\pi' \leq \tau'). \tag{5}$$

In view of (2)–(4) we see that there is an ordinal $\delta < \omega$ such that $\theta \subseteq \Sigma^{(\delta)} \cup \Pi^{(\delta)}$. Hence, if we assume (as we obviously may in light of (3) and (4)) that the indices of all cylindrification and diagonal symbols occurring in π are ordinals $< \alpha + \delta$, we can conclude from (5) that condition (1) holds for every $\mathfrak{B} \in \mathsf{CA}_{\alpha+\delta}$.

Choose two disjoint sets $\Gamma, \Delta \subseteq \alpha$ such that $|\Gamma| = |\Delta| = \delta$ and such that neither Γ nor Δ contains an ordinal which appears as the index of a cylindrification or diagonal symbol occurring in σ, τ, or π (recall that $\alpha \geq \omega$). Let μ, ν be two sequences of length δ which enumerate the elements of Γ and Δ, respectively (and hence are necessarily one-one). Let $\bar{\sigma}$ and $\bar{\tau}$ be the terms of Λ that are obtained from σ and τ, respectively, by replacing each variable $\boldsymbol{v}_\kappa$ in all its occurrences by the term

$$\mathbf{s}_{\nu_0}^{\mu_0} \mathbf{s}_{\nu_1}^{\mu_1} \ldots \mathbf{s}_{\nu_{\delta-1}}^{\mu_{\delta-1}} \boldsymbol{v}_\kappa.$$

Let $\bar{\pi}$ be obtained from π by making these same replacements in addition to the following ones: every index λ of a cylindrification or diagonal symbol occurring in π such that $\lambda \in (\alpha + \delta) \sim \alpha$ is changed to an ordinal in Γ; the only restriction on how these changes are made is that different ordinals are to be substituted for different ordinals. Notice that $\bar{\pi}$, as well as $\bar{\sigma}$ and $\bar{\tau}$, is a term of Λ. Finally, let ρ be any one-one function from $\alpha + \delta$ onto α such that $\delta\xi = \xi$ for every $\xi < \alpha$ which appears as an index of a cylindrification or diagonal symbol occurring in σ, τ, or π and such that

$$\rho(\alpha + \kappa) = \mu_\kappa \quad \text{for every} \quad \kappa < \delta .$$

Then for every $\mathfrak{C} \in \mathsf{CA}_\alpha$ we have, by $\mathscr{HMT}$ 2.6.2(i),

$$\mathfrak{Rd}^{(\rho)}\mathfrak{C} \in \mathsf{CA}_{\alpha+\delta} \tag{6}$$

and, by $\mathscr{HMT}$ 1.5.8(ii) and $\mathscr{HMT}$ 1.5.9(ii),

$$\mathsf{s}^{\mu_0}_{\nu_0}\mathsf{s}^{\mu_1}_{\nu_1} \dots \mathsf{s}^{\mu_{\delta-1}}_{\nu_{\delta-1}} x \in Nr_\alpha \mathfrak{Rd}^{(\rho)}\mathfrak{C} \quad \text{for every} \quad x \in C . \tag{7}$$

We have seen that (1) holds for every $\mathfrak{B} \in \mathsf{CA}_{\alpha+\delta}$. Hence, since (6) and (7) hold for every CA_α $\mathfrak{C}$, we conclude that the inclusions $\bar{\sigma} \leq \bar{\pi}$ and $\bar{\pi} \leq \bar{\tau}$ are identically satisfied in every CA_α. On the other hand, since neither Γ nor Δ contains an ordinal which is the index of a cylindrification or diagonal symbol occurring in σ, τ, or π, we readily see that the equations

$$\bar{\sigma} = \mathsf{s}^{\mu_0}_{\nu_0}\mathsf{s}^{\mu_1}_{\nu_1} \dots \mathsf{s}^{\mu_{\delta-1}}_{\nu_{\delta-1}}\sigma$$

and

$$\bar{\tau} = \mathsf{s}^{\mu_0}_{\nu_0}\mathsf{s}^{\mu_1}_{\nu_1} \dots \mathsf{s}^{\mu_{\delta-1}}_{\nu_{\delta-1}}\tau$$

are both identically satisfied in every CA_α. Combining these results we get that

$$\mathsf{s}^{\mu_0}_{\nu_0}\mathsf{s}^{\mu_1}_{\nu_1} \cdots \mathsf{s}^{\mu_{\delta-1}}_{\nu_{\delta-1}}\sigma \leq \bar{\pi} \leq \mathsf{s}^{\mu_0}_{\nu_0}\mathsf{s}^{\mu_1}_{\nu_1} \cdots \mathsf{s}^{\mu_{\delta-1}}_{\nu_{\delta-1}}\tau$$

is identically satisfied in every CA_α and hence, in particular, in every member of K; using $\mathscr{HMT}$ 1.5.1, $\mathscr{HMT}$ 1.2.7, $\mathscr{HMT}$ 1.4.4(vi), and the dual of $\mathscr{HMT}$ 1.2.7 we easily show that the same is true of

$$\mathsf{c}^{\partial}_{\mu_0}\mathsf{c}^{\partial}_{\mu_1} \dots \mathsf{c}^{\partial}_{\mu_{\delta-1}}\sigma \leq \bar{\pi} \leq \mathsf{c}_{\mu_0}\mathsf{c}_{\mu_1} \dots \mathsf{c}_{\mu_{\delta-1}}\tau .$$

Therefore, since $\bar{\pi}$ is a term of Λ and it contains like π only occurrences of variables which occur at the same time in both σ and τ, we have shown that the inclusion $\sigma \leq \tau$ can indeed be weakly interpolated relative to K.

REMARK 2.2.27. We now discuss an interesting result concerning equations holding in all RCA_α's which appears as an easy consequence of the proof outlined in 2.2.26. (It should be noted however that this result also has a simple direct proof which does not involve 2.2.6 or any notion of amalgamation or interpolation.) In the course of

the proof in 2.2.26 it was shown that, if $\sigma \leq \tau$ is any inclusion between terms in Λ which is identically satisfied in all RCA_α's, then there is a finite ordinal δ and a pair of sequences $\mu, \nu \in {}^\delta\alpha$ such that the inclusion

$$\mathbf{s}^{\mu_0}_{\nu_0}\mathbf{s}^{\mu_1}_{\nu_1}\ldots\mathbf{s}^{\mu_{\delta-1}}_{\nu_{\delta-1}}\sigma \leq \mathbf{s}^{\mu_0}_{\nu_0}\mathbf{s}^{\mu_1}_{\nu_1}\ldots\mathbf{s}^{\mu_{\delta-1}}_{\nu_{\delta-1}}\tau$$

is identically satisfied in all CA_α's; moreover, it was shown that μ, ν may be taken to be any sequences in ${}^\delta\alpha$ satisfying the conditions:

(1) μ, ν have no repeating terms, their respective ranges are disjoint, and no term of either of them appears as an index of a cylindrification or diagonal symbol occurring in σ or τ.

Using now only $\mathscr{HMT}$ 1.2.1, $\mathscr{HMT}$ 1.5.1, $\mathscr{HMT}$ 1.5.3, and $\mathscr{HMT}$ 1.5.4(ii) together with the fact that μ, ν satisfy (1) we can show successively that the following equalities are identically satisfied in all CA_α's:

$$\mathbf{s}^{\mu_0}_{\nu_0}\mathbf{s}^{\mu_1}_{\nu_1}\ldots\mathbf{s}^{\mu_{\delta-1}}_{\nu_{\delta-1}}(\sigma\cdot - \tau) = \mathbf{0}$$

$$\mathbf{c}_{\mu_0}\mathbf{c}_{\mu_1}\ldots\mathbf{c}_{\mu_{\delta-1}}(\mathbf{d}_{\mu_0\nu_0}\cdot\mathbf{d}_{\mu_1\nu_1}\cdot\cdots\cdot\mathbf{d}_{\mu_{\delta-1}\nu_{\delta-1}}\cdot\sigma\cdot - \tau) = \mathbf{0}$$

$$\mathbf{d}_{\mu_0\nu_0}\cdot\mathbf{d}_{\mu_1\nu_1}\cdot\cdots\cdot\mathbf{d}_{\mu_{\delta-1}\nu_{\delta-1}}\cdot\sigma\cdot - \tau = \mathbf{0}$$

$$\mathbf{d}_{\mu_0\nu_0}\cdot\mathbf{d}_{\mu_1\nu_1}\cdot\cdots\cdot\mathbf{d}_{\mu_{\delta-1}\nu_{\delta-1}}\cdot\sigma \leq \mathbf{d}_{\mu_0\nu_0}\cdot\mathbf{d}_{\mu_1\nu_1}\cdot\cdots\cdot\mathbf{d}_{\mu_{\delta-1}\nu_{\delta-1}}\cdot\tau.$$

Since the equation $\sigma = \tau$ is equivalent to a conjunction of the two inclusions $\sigma \leq \tau$ and $\tau \leq \sigma$, we may conclude that for $\sigma = \tau$ to hold in every RCA_α it is necessary that there exists a finite ordinal δ such that, for every pair of sequences $\mu, \nu \in {}^\delta\alpha$ satisfying (1), the new equation

(2) $$(\mathbf{d}_{\mu_0\nu_0}\cdot\mathbf{d}_{\mu_1\nu_1}\cdot\cdots\cdot\mathbf{d}_{\mu_{\delta-1}\nu_{\delta-1}})\cdot\sigma = (\mathbf{d}_{\mu_0\nu_0}\cdot\mathbf{d}_{\mu_1\nu_1}\cdot\cdots\cdot\mathbf{d}_{\mu_{\delta-1}\nu_{\delta-1}})\cdot\tau$$

holds in every CA_α. On the other hand, it is not hard to see that this condition is also sufficient; actually, for $\sigma = \tau$ to hold in every RCA_α it suffices only that there exist a finite ordinal δ and at least one pair of sequences $\mu, \nu \in {}^\delta\alpha$ satisfying (1) such that the equation (2) holds in every CA_α.[4])

We close this section with one final positive result concerning the $\mathscr{AP}$. It is obvious from Definition $\mathscr{HMT}$ 2.4.61 that, if a CA_α $\mathfrak{A}$ is of *characteristic* κ, then any CA_α which includes a subalgebra isomorphic to $\mathfrak{A}$ is also of characteristic κ. Therefore, since any algebra of positive characteristic is a diagonal algebra by $\mathscr{HMT}$ 2.6.54(ii), we get as an immediate corollary of 2.2.20:

THEOREM 2.2.28. *For any non-zero $\kappa < \omega$ the class of all CA_α's of characteristic κ has the amalgamation property.*

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[4]) These results were obtained independently by Richard Thompson and the author.

REMARK 2.2.29. This theorem is due to Stephen Comer who has shown moreover that it also holds for every finite ordinal α and every positive $\kappa < \omega \cap \alpha$.[5])

As was the case for Theorem 2.2.20, we do not know if 2.2.28 continues to hold when '$\mathscr{AP}$' is replaced by 'strong $\mathscr{AP}$'; on the other hand, as an immediate consequence of 2.2.16 (and the fact that $\mathsf{Di}_\alpha \subseteq \mathsf{RCA}_\alpha$) we get that the theorem does continue to hold when $\mathscr{AP}$ is replaced by 'strong $\mathscr{EP}$'. Moreover, the new theorem thus obtained also holds for all finite $\alpha \geq 2$ (for non-zero $\kappa < \omega \cap \alpha$). This latter result can be proved directly from 2.2.12 in essentially the same way as we proved 2.2.13. Indeed, in order to apply the same method as used in the proof of 2.2.13 we only need the following two facts: any two minimal CA_α's of the same characteristic are isomorphic (cf. $\mathscr{HMT}$ 2.5.30), and every finite-dimensional CA of positive characteristic is representable.[6]) It is also helpful to observe, however, that, if $\mathfrak{A}$ is any CA of characteristic κ and finite (or infinite) dimension $\alpha \geq 2$, and $\mathfrak{B}$ is any CA_β such that $\mathfrak{A}$ is isomorphic to a subalgebra of the α-reduct of $\mathfrak{B}$ (so that $\beta \geq \alpha$), then $\mathfrak{B}$ is also of characteristic κ provided $\kappa > 0$; this last result is an easy consequence of Definition $\mathscr{HMT}$ 2.4.61.

REMARK 2.2.30. In view of $\mathscr{HMT}$ 2.4.65(ii) and the remark immediately following $\mathscr{HMT}$ 1.1.3, a Boolean algebra can be identified in a very natural way with a CA_α of characteristic 1 for each $\alpha \geqslant 2$ (if we disregard the trivial one-element algebras). Thus from Theorem 2.2.28 we infer immediately that the class of Boolean algebras has the amalgamation property. Even more, using the fact that every cylindric algebra of characteristic 1 is locally finite we conclude from 2.2.8 that the class of Boolean algebras has the strong amalgamation property.[7])

However, it should be recalled that the ordinary $\mathscr{AP}$ for the class of Boolean algebras can be obtained much more easily from the general result stated in Theorem 1.2.5; cf. Remark 1.2.6.

2.3 Negative results

The two principal results of this section are Theorems 2.3.4(ii) and 2.3.6. The first is in direct opposition to Theorem 2.2.6, the principal result of the preceding section. Given any class K of CA_α's which includes all representable algebras, 2.3.4(ii) says

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[5]) This result for finite α is announced in Comer [3]; in Comer [2], p. 81, Lemma 4.2 it is proved that the result for finite α implies the result for infinite α (our Theorem 2.2.28).

[6]) For characteristic $\kappa = 2,3$ this second result is due to Comer [2]. Leon Henkin independently showed that the result holds for all positive $\kappa < \omega \cap \alpha$; Henkin's proof has not been published.

[7]) It seems that the result that the class of Boolean algebras has the strong $\mathscr{AP}$ was found at about the same time by several authors. See Dwinger–Yaqub [9] (where the result appears for the first time in print) and in particular the footnote at the end of the article. See also Jónsson [16] and Kiesler [17].

that every free g-algebra over K with more than three free generators fails to have the interpolation property. It follows then that every algebraically closed class of CA_α's which includes all representable algebras fails to have the amalgamation property. This last result, however, can be considerably strengthened, and this is done in 2.3.8.

LEMMA 2.3.1. *Let α be any finite or infinite ordinal ≥ 2, and let $\mathfrak{A}$ be any CA_α. Then for any $x, y, z \in A$ we have*

$$\mathsf{c}_0(x\cdot \mathsf{c}_1 y)\cdot \mathsf{c}_0(x\cdot - \mathsf{c}_1 y) \leq \mathsf{c}_0\mathsf{c}_1(\mathsf{c}_1 z\cdot \mathsf{s}_1^0\mathsf{c}_1 z\cdot - \mathsf{d}_{01}) + \mathsf{c}_0(x\cdot - \mathsf{c}_1 z).$$

PROOF. Set

$$a = x\cdot \mathsf{c}_1 y\cdot - \mathsf{c}_0(x\cdot - \mathsf{c}_1 z),$$
$$b = x\cdot - \mathsf{c}_1 y\cdot - \mathsf{c}_0(x\cdot - \mathsf{c}_1 z).$$

Observe first of all that

$$\begin{aligned} \mathsf{c}_1 a\cdot \mathsf{c}_1 b &\leq \mathsf{c}_1(x\cdot \mathsf{c}_1 y)\cdot \mathsf{c}_1(x\cdot - \mathsf{c}_1 y) && \text{by } \mathscr{HMT}\ 1.2.7 \\ &= \mathsf{c}_1 x\cdot \mathsf{c}_1 y\cdot \mathsf{c}_1 x\cdot - \mathsf{c}_1 y && \text{by } \mathscr{HMT}\ 1.2.11, \end{aligned}$$

and thus

$$\mathsf{c}_1 a\cdot \mathsf{c}_1 b = 0. \tag{1}$$

From the inclusion $x\cdot - \mathsf{c}_1 z \leq \mathsf{c}_0(x\cdot \leq \mathsf{c}_1 z)$ we see at once that

$$x\cdot - \mathsf{c}_0(x\cdot - \mathsf{c}_1 z) \leq \mathsf{c}_1 z.$$

Thus $a, b \leq \mathsf{c}_1 z$ and hence, by $\mathscr{HMT}$ 1.2.9,

$$\mathsf{c}_1 a,\ \mathsf{c}_1 b \leq \mathsf{c}_1 z. \tag{2}$$

We now compute:

$$\begin{aligned} \mathsf{c}_0 a\cdot \mathsf{c}_0 b &\leq \mathsf{c}_0\mathsf{c}_1 a\cdot \mathsf{c}_0\mathsf{c}_1 b && \text{by } \mathscr{HMT}\ 1.2.7 \\ &= \mathsf{c}_0\mathsf{c}_1 a\cdot \mathsf{c}_1\mathsf{s}_1^0\mathsf{c}_1 b && \text{by } \mathscr{HMT}\ 1.5.8\text{(i)}, \\ & && \mathscr{HMT}\ 1.5.9\text{(i)} \\ &= \mathsf{c}_1(\mathsf{c}_0\mathsf{c}_1 a\cdot \mathsf{s}_1^0\mathsf{c}_1 b) \\ &= \mathsf{c}_0\mathsf{c}_1(\mathsf{c}_1 a\cdot \mathsf{s}_1^0\mathsf{c}_1 b) && \text{by } \mathscr{HMT}\ 1.5.9\text{(ii)} \\ &= \mathsf{c}_0\mathsf{c}_1[\mathsf{c}_1 a\cdot \mathsf{s}_1^0\mathsf{c}_1 b\cdot(-\mathsf{d}_{01} + \mathsf{d}_{01})] \\ &= \mathsf{c}_0\mathsf{c}_1[(\mathsf{c}_1 a\cdot \mathsf{s}_1^0\mathsf{c}_1 b\cdot - \mathsf{d}_{01}) + (\mathsf{c}_1 a\cdot \mathsf{s}_1^0\mathsf{c}_1 b\cdot \mathsf{d}_{01})] \\ &= \mathsf{c}_0\mathsf{c}_1[(\mathsf{c}_1 a\cdot \mathsf{s}_1^0\mathsf{c}_1 b\cdot - \mathsf{d}_{01}) + (\mathsf{c}_1 a\cdot \mathsf{c}_1 b\cdot \mathsf{d}_{01})] && \text{by } \mathscr{HMT}\ 1.5.5 \\ &= \mathsf{c}_0\mathsf{c}_1(\mathsf{c}_1 a\cdot \mathsf{s}_1^0\mathsf{c}_1 b\cdot - \mathsf{d}_{01}) && \text{by (1)} \\ &\leq \mathsf{c}_0\mathsf{c}_1(\mathsf{c}_1 z\cdot \mathsf{s}_1^0\mathsf{c}_1 z\cdot - \mathsf{d}_{01}) && \text{by (2)}, \mathscr{HMT}\ 1.2.7. \end{aligned}$$

Therefore, we have proved that

$$\mathsf{c}_0[x\cdot \mathsf{c}_1 y\cdot - \mathsf{c}_0(x\cdot - \mathsf{c}_1 z)]\cdot \mathsf{c}_0[x\cdot - \mathsf{c}_1 y\cdot - \mathsf{c}_0(x\cdot - \mathsf{c}_1 z)] \leq \mathsf{c}_0\mathsf{c}_1(\mathsf{c}_1 z\cdot \mathsf{s}_1^0\mathsf{c}_1 z\cdot - \mathsf{d}_{01}).$$

In view of $\mathscr{HMT}$ 1.2.11 and axiom (C_3) this gives

$$\mathsf{c}_0(x\cdot\mathsf{c}_1y)\cdot\mathsf{c}_0(x\cdot-\mathsf{c}_1y)\cdot-\mathsf{c}_0(x\cdot-\mathsf{c}_1z)\leq\mathsf{c}_0\mathsf{c}_1(\mathsf{c}_1z\cdot\mathsf{s}_1^0\mathsf{c}_1z\cdot-\mathsf{d}_{01}).$$

The conclusion of the lemma now follows immediately.

LEMMA 2.3.2. *Let* K *be any class of* CA_α*'s such that* $\mathsf{K}\supseteq\mathsf{RCA}_\alpha$, *and assume that* $\gamma\geq 3$. *Let* r *and* s *be the elements of* $\mathfrak{Fr}_\gamma\mathsf{K}$ *defined by the formulas*

$$r=\mathsf{c}_0(x\cdot\mathsf{c}_1y)\cdot\mathsf{c}_0(x\cdot-\mathsf{c}_1y),$$
$$s=\mathsf{c}_0\mathsf{c}_1(\mathsf{c}_1z\cdot\mathsf{s}_1^0\mathsf{c}_1z\cdot-\mathsf{d}_{01})+\mathsf{c}_0(x\cdot-\mathsf{c}_1z)$$

where $x=0/Cr_\gamma\mathsf{K}$, $y=1/Cr_\gamma\mathsf{K}$, *and* $z=2/Cr_\gamma\mathsf{K}$. *Furthermore, if* $\gamma\geq 4$, *let*

$$t=\mathsf{c}_0\mathsf{c}_1(\mathsf{c}_1w\cdot\mathsf{s}_1^0\mathsf{c}_1w\cdot-\mathsf{d}_{01})+\mathsf{c}_0(x\cdot-\mathsf{c}_1w)$$

where $w=3/Cr_\gamma\mathsf{K}$. *We then have:*

(i) *there exists no* $u\in Sg\{x\}$ *such that* $r\leq u\leq s$;

(ii) if $\gamma\geq 4$, then there exists no $u\in Sg\{x\}$ such that, for some finite $\Gamma\subseteq\gamma$,

$$r\leq u\leq\mathsf{c}_{(\Gamma)}(s\cdot t).$$

PROOF. Set $\mathfrak{A}=\mathfrak{Fr}_\gamma\mathsf{CA}_\alpha$. We shall first prove part (ii), so assume that $\gamma\geq 4$. We must show that the inclusion

$$r\leq s\cdot t \tag{1}$$

(which holds by 2.3.1) cannot be interpolated by an element of $Sg^{(\mathfrak{A})}\{x\}$, i.e., that the conclusion of (ii) holds. For this purpose we now construct a special set algebra. Let

$$\mathfrak{B}=\langle Sb^{{}^\alpha\alpha},\ \cup,\ \cap,\ \sim,\ 0,\ {}^\alpha\alpha,\ \mathsf{C}_\kappa,\ \mathsf{D}_{\kappa\lambda}\rangle_{\kappa,\lambda<\alpha};$$

that is, $\mathfrak{B}$ is the full cylindric set algebra in the space ${}^\alpha\alpha$. Let E be the set of all equivalence relations on α, and for each $R\in E$ set

$$X_R=\{\varphi:\varphi\in{}^\alpha\alpha\ \text{ and, for all }\ \xi,\eta<\alpha,\ \varphi_\xi=\varphi_\eta\ \text{ iff }\ \xi R\eta\}.$$

Let

$$C=\{\textstyle\bigcup_{R\in L}X_R:L\subseteq E\}.$$

C is clearly closed under the formation of arbitrary unions, and since

$$\sim\textstyle\bigcup_{R\in L}X_L=\bigcup_{R\in E\sim L}X_R$$

for every $L\subseteq E$, we see that C is closed under the formation of complements with respect to ${}^\alpha\alpha$. Thus C is a Boolean subuniverse (indeed, a complete Boolean subuniverse) of $\mathfrak{B}$; moreover, it is obvious that

$$X_R\ \text{ is an atom of }\ \langle C,\ \cup,\ \cap,\ \sim,\ 0,\ {}^\alpha\alpha\rangle\ \text{ for each }\ R\in E. \tag{2}$$

For all $\kappa, \lambda < \alpha$ we have $\mathsf{D}_{\kappa\lambda} = \bigcup\{X_R : \langle\kappa, \lambda\rangle \in R \in E\}$ and hence $\mathsf{D}_{\kappa\lambda} \in B$. Also,

$$\mathsf{C}_\kappa X_R = \bigcup\{X_S : S \in E,\ {}^2(\alpha \sim \{\kappa\}) \cap S = {}^2(\alpha \sim \{\kappa\}) \cap R\}$$

for any $\kappa < \alpha$ and $R \in E$. Thus, because C_κ is completely additive (cf. $\mathscr{HMT}$ 1.2.6(i) and the remark preceding it), we see that C is closed under the operation C_κ for every $\kappa < \alpha$. Therefore, we have succeeded in showing that

(3) $$C \text{ is a subuniverse of } \mathfrak{B}.$$

To prove that (1) cannot be interpolated by an element of $Sg^{(\mathfrak{A})}\{x\}$ it suffices to show that there is a subset Y of ${}^\alpha\alpha$ such that

(4) $X_{\alpha 1 Id} \cap Fr \neq 0$ for every $F \in Hom(\mathfrak{A}, \mathfrak{B})$ such that $Fx = X_{\alpha 1 Id}$ and $Fy = Y$,

and also that for every finite $\Gamma \subseteq \alpha$ there are subsets Z, W of ${}^\alpha\alpha$ such that

(5) $$X_{\alpha 1 Id} \sim \mathsf{C}_{(\Gamma)}\, G(s \cdot t) \neq 0 \text{ for every } G \in Hom(\mathfrak{A}, \mathfrak{B}) \text{ such that } Gx = X_{\alpha 1 Id},\ Gz = Z, \text{ and } Gw = W.$$

For suppose, on the contrary, that these conditions are not sufficient. Then there exists a finite $\Gamma \subseteq \alpha$ and an element $u \in Sg^{(\mathfrak{A})}\{x\}$ such that the double inclusion of (ii) holds, and there also exist Y, Z, $W \subseteq {}^\alpha\alpha$ such that (4) and (5) hold. Take any $H \in Hom(\mathfrak{A}, \mathfrak{B})$ such that $Hx = X_{\alpha 1 Id}$, $Hy = Y$, $Hz = Z$, and $Hw = W$ (that at least one such H exists follows from 1.1.4 and the premise $\mathsf{RCA}_\alpha \subseteq \mathsf{K}$). Then using the fact that $X_{\alpha 1 Id} \cap Hr$ is non-empty by (4) we get

$$X_{\alpha 1 Id} \cap Hu = H(x \cdot u) \supseteq H(x \cdot r) \neq 0,$$

and using the fact that $X_{\alpha 1 Id} \sim \mathsf{C}_{(\Gamma)} H(s \cdot t)$ is non-empty by (5) we get

$$X_{\alpha 1 Id} \sim Hu = H(x \cdot - u) \supseteq H(x \cdot - \mathsf{c}_{(\Gamma)}(s \cdot t)) \neq 0.$$

However, in view of (2), it is impossible for $X_{\alpha 1 Id}$ to intersect both Hu and its complement since $Hu \in C$; to see that Hu is indeed contained in C recall that $u \in Sg^{(\mathfrak{A})}\{x\}$, and then observe that because of (3) and the fact that $X_{\alpha 1 Id} \in C$ we must have

(6) $$H^* Sg^{(\mathfrak{A})}\{x\} \subseteq C.$$

Therefore, (4) and (5) are sufficient conditions for (1) not to be interpolated by an element of $Sg^{(\mathfrak{A})}\{x\}$.

Let σ be the point in ${}^\alpha\alpha$ such that $\sigma_0 = 0$, and $\sigma_\kappa = \kappa + 1$ for every non-zero $\kappa < \omega$; if $\alpha > \omega$, let $\sigma_\xi = \xi$ for each ξ such that $\omega \leq \xi < \alpha$. Let $\tau = (\alpha \sim \{0\}) \upharpoonright \sigma \cup \{\langle 0, 1\rangle\}$. Then $\sigma, \tau \in X_{\alpha 1 Id}$. Take

$$Y = \{\sigma\}.$$

Then

$$\sigma \in X_{\alpha 1 Id} \cap \mathsf{C}_1 Y \quad \text{and} \quad \tau \in X_{\alpha 1 Id} \sim \mathsf{C}_1 Y,$$

and hence

$$\sigma \in \mathsf{C}_0 (X_{\alpha 1 Id} \cap \mathsf{C}_1 Y) \cap \mathsf{C}_0 (X_{\alpha 1 Id} \sim \mathsf{C}_1 Y). \tag{7}$$

Therefore, we have $\sigma \in Fr$ for every $F \in Hom(\mathfrak{A}, \mathfrak{B})$ such that $Fx = X_{\alpha 1 Id}$ and $Fy = Y$, and thus (4) holds.

We now want to show that for any given finite $\Gamma \subseteq \alpha$ there exist sets Z, $W \subseteq {}^{\alpha}\alpha$ such that (5) holds; it is clear that no generality is lost if we assume that $0, 1 \in \Gamma$, so we make this assumption. Take

$$Z = \{\varphi : \varphi \in X_{\alpha 1 Id}, \varphi_0 < \varphi_1\} \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\}$$

and

$$W = \{\varphi : \varphi \in X_{\alpha 1 Id}, \varphi_1 > \varphi_0\} \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\}.$$

We want to show that

$$\alpha \upharpoonright Id \in X_{\alpha 1 Id} \sim \mathsf{C}_{(\Gamma)} G(s \cdot t) \tag{8}$$

for any $G \in Hom(\mathfrak{A}, \mathfrak{B})$ such that $Gx = X_{\alpha 1 Id}$, $Gz = Z$, and $Gw = W$; to do this we simply compute the value of $\mathsf{C}_{(\Gamma)} G(s \cdot t)$. For the purpose of this computation we make use of the following property of ordinals: if Δ is any set of ordinals, then $\bigcap \Delta$ is the smallest ordinal in Δ, and if, in addition, Δ is finite, then $\bigcup \Delta$ is the largest element in Δ. Also, in this computation we shall assume that φ always represents an arbitrary sequence in ${}^{\alpha}\alpha$. Then, setting

$$\Delta\varphi = \Gamma \sim \varphi^*(\Gamma \sim \{0, 1\})$$

for every φ, we successively compute:

$$\mathsf{C}_1 Z = \{\varphi : |\Delta\varphi| = 2, \varphi_0 = \bigcap \Delta\varphi\} \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\},$$

$$(X_{\alpha 1 Id} \sim \mathsf{C}_1 Z) \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\} = \{\varphi : |\Delta\varphi| = 2, \varphi_0 = \bigcup \Delta\varphi, \varphi_1 = \bigcap \Delta\varphi\} \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\},$$

and, finally,

$$\mathsf{C}_0 (X_{\alpha 1 Id} \sim \mathsf{C}_1 Z) \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\} = \{\varphi : |\Delta\varphi| = 2, \varphi_1 = \bigcap \Delta\varphi\} \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\}. \tag{9}$$

Similarly, we obtain

$$\mathsf{C}_0 (X_{\alpha 1 Id} \sim \mathsf{C}_1 W) \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\} = \{\varphi : |\Delta\varphi| = 2, \varphi_1 = \bigcup \Delta\varphi\} \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\}.$$

The last two formulas together give

$$\mathsf{C}_0 (X_{\alpha 1 Id} \sim \mathsf{C}_1 Z) \cap \mathsf{C}_0 (X_{\alpha 1 Id} \sim \mathsf{C}_1 W) \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\} = 0. \tag{10}$$

Continuing the computation we successively obtain:

$$\mathsf{C}_1 Z \cap \mathsf{D}_{01} = \{\varphi : |\Delta\varphi| = 2,\ \varphi_0 = \varphi_1 = \bigcap \Delta\varphi\} \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\},$$
$$\mathsf{S}_1^0 \mathsf{C}_1 Z = \{\varphi : |\Delta\varphi| = 2,\ \varphi_1 = \bigcap \Delta\varphi\} \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\},$$
$$\mathsf{C}_1 Z \cap \mathsf{S}_1^0 \mathsf{C}_1 Z = \{\varphi : |\Delta\varphi| = 2,\ \varphi_0 = \varphi_1 = \bigcap \Delta\varphi\} \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\};$$

hence we finally get

$$\mathsf{C}_0 \mathsf{C}_1 (\mathsf{C}_1 Z \cap \mathsf{S}_1^0 \mathsf{C}_1 Z \cap \sim \mathsf{D}_{01}) = \mathsf{C}_0 \mathsf{C}_1 0 = 0, \tag{11}$$

and similarly we get

$$\mathsf{C}_0 \mathsf{C}_1 (\mathsf{C}_1 W \cap \mathsf{S}_1^0 \mathsf{C}_1 W \cap \sim \mathsf{D}_{01}) = 0. \tag{12}$$

Now take G to be any homomorphism from $\mathfrak{A}$ into $\mathfrak{B}$ such that $Gx = X_{\alpha \upharpoonright Id}$, $Gz = Z$ and $Gw = W$. Then from (10)–(12) we have

$$G(s \cdot t) \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\} = 0.$$

Then applying $\mathsf{C}_{(\Gamma)}$ to both sides of this equation we get

$$\mathsf{C}_{(\Gamma)} G(s \cdot t) \cap \mathsf{C}_{(\Gamma)} \{\alpha \upharpoonright Id\} = 0.$$

Thus (8) holds and we have shown that for any finite $\Gamma \subseteq \alpha$ there are Z, $W \subseteq {}^{\alpha}\alpha$ such that (5) holds. This completes the proof of part (ii).

Part (i) is proved analogously. We must show that (under the assumption $\gamma \geq 3$) the inclusion

$$r \leq s \tag{13}$$

cannot be strongly interpolated by an element of $Sg^{(\mathfrak{A})}\{x\}$, i.e., that the conclusion of (i) holds. For this purpose take $\mathfrak{B}$, C, $X_{\alpha \upharpoonright Id}$, Y, and Z as in the first part of the proof, and consider any $H \in Hom(\mathfrak{A}, \mathfrak{B})$ such that $Hx = X_{\alpha \upharpoonright Id}$, $Hy = Y$, and $Hz = Z$. Then $X_{\alpha \upharpoonright Id} \cap Hr \neq 0$ by (4) while, at the same time, $X_{\alpha \upharpoonright Id} \sim Hs \neq 0$ since $\alpha \upharpoonright Id \in X_{\alpha \upharpoonright Id} \sim Hs$ by (9) and (11). Therefore, in view of (2), (3), and (6) we see that the inclusion (13) cannot be strongly interpolated by an element of $Sg^{(\mathfrak{A})}\{x\}$. This completes the proof of the lemma.

Remark 2.3.3. The question naturally arises whether the inclusion (13) in the proof of 2.3.2 can be interpolated in the ordinary sense by an element of $Sg^{(\mathfrak{A})}\{x\}$, i.e., whether there exists a member u of the subuniverse of $\mathfrak{A} = \mathfrak{F}\mathrm{r}_\gamma \mathsf{K}$ generated by the single element $x = 0/Cr_\gamma \mathsf{K}$, and a finite subset Γ of α such that

$$r \leq u \leq \mathsf{c}_{(\Gamma)} s.$$

As it turns out, (13) can be so interpolated. In fact, by an argument very similar to the

one used in the proof of 2.3.1 one can show that for every $\mathfrak{A} \in \mathsf{CA}_\alpha$ the inclusions

$$\mathsf{c}_0(x \cdot \mathsf{c}_1 y) \cdot \mathsf{c}_0(x \cdot - \mathsf{c}_1 y) \leq \mathsf{c}_0 \mathsf{c}_1(\mathsf{c}_1 x \cdot \mathsf{s}_1^0 \mathsf{c}_1 x \cdot - \mathsf{d}_{01}) \leq$$
$$\leq \mathsf{c}_1[\mathsf{c}_0 \mathsf{c}_1(\mathsf{c}_1 z \cdot \mathsf{s}_1^0 \mathsf{c}_1 z \cdot - \mathsf{d}_{01}) + \mathsf{c}_0(x \cdot - \mathsf{c}_1 z)]$$

hold for all $x, y, z \in A$.

THEOREM 2.3.4. *Let* K *be any class of* CA_α*'s such that* $\mathsf{K} \supseteq \mathsf{RCA}_\alpha$.
(i) *If* $\gamma \geq 3$, *then* $\mathfrak{Fr}_\gamma \mathsf{K}$ *fails to have the strong* $\mathscr{IP}$.
(ii) *If* $\gamma \geq 4$, *then* $\mathfrak{Fr}_\gamma \mathsf{K}$ *fails to have the* $\mathscr{IP}$.
PROOF: by 2.3.1 and 2.3.2.

REMARK 2.3.5. Let K be a fixed but arbitrary class of CA_α's such that $\mathsf{K} \supseteq \mathsf{RCA}_\alpha$. We do not know if the conclusion of 2.3.4(ii) continues to hold for $\gamma = 3$ (cf. 2.3.3); on the other hand, we know from 2.2.17 that it no longer holds for $\gamma = 2$ if $\mathsf{K} = \mathsf{RCA}_\alpha$. We also do not know if the conclusion of 2.3.4(i) continues to hold for $\gamma = 2$; cf. Remark 2.2.18.

As an immediate consequence of 1.1.38, 1.2.8(i), 2.1.11, and 2.3.4(ii) we get that K fails to have the amalgamation property whenever K is algebraically closed; thus, in particular, RCA_α fails to have the $\mathscr{AP}$ since it is algebraically closed by $\mathscr{HMT}$ 2.6.32. We shall obtain a much stronger result however as a corollary of our next theorem; see Corollary 2.3.8.

THEOREM 2.3.6. RCA_α *fails to have the* $\mathscr{AP}$ *with respect to* CA_α.
PROOF. Assume that RCA_α does have the $\mathscr{AP}$ with respect to CA_α. We shall derive a contradiction.

Set

$$\mathfrak{A} = \mathfrak{Fr}_4 \mathsf{CA}_\alpha.$$

Set $x = 0/Cr_4\mathsf{CA}_\alpha$, $y = 1/Cr_4\mathsf{CA}_\alpha$, $z = 2/Cr_4\mathsf{CA}_\alpha$, and $w = 3/Cr_4\mathsf{CA}_\alpha$, and let r, s, and t be as in the statement of Lemma 2.3.2. Set $I = \{0, 1\}$ and $J = \{0, 2, 3\}$ so that $A^{(I)} = Sg^{(\mathfrak{A})}\{x, y\}$, $A^{(J)} = Sg^{(\mathfrak{A})}\{x, z, w\}$, and

$$A^{(I \cap J)} = Sg^{(\mathfrak{A})}\{x\}; \tag{1}$$

hence

$$r \in A^{(I)} \quad \text{and} \quad s, t \in A^{(J)}. \tag{2}$$

Finally, let R be the ideal of $\mathfrak{A}$ such that

$$\mathfrak{A}/R \cong \mathfrak{Fr}_4 \mathsf{RCA}_\alpha \tag{3}$$

(R is unique by 1.1.28(i)).

By (2) and 2.3.1 we have

$$r \in Ig^{(\mathfrak{A})}\{s \cdot t\} \cap A^{(I)}. \tag{4}$$

Let

$$M = Ig^{(\mathfrak{A}^{(J)})}[\{s \cdot t\} \cup (R \cap A^{(J)})], \tag{5}$$

$$N = Ig^{(\mathfrak{A}^{(I)})}[(M \cap A^{(I \cap J)}) \cup (R \cap A^{(I)})]; \tag{6}$$

clearly then

$$M \supseteq R \cap A^{(J)} \quad \text{and} \quad N \supseteq R \cap A^{(I)}. \tag{7}$$

From the first of these inclusions we see that

$$M \cap A^{(I \cap J)} \supseteq (R \cap A^{(J)}) \cap A^{(I \cap J)} = (R \cap A^{(I)}) \cap A^{(I \cap J)};$$

hence, by (6) and 2.1.9.10, $N \cap A^{(I \cap J)} = M \cap A^{(I \cap J)}$. From this last equality we get, using 1.1.30,

$$(\mathfrak{A}^{(J)}/M)^{(I \cap J)} \cong \mathfrak{A}^{(I \cap J)}/(M \cap A^{(I \cap J)}) \cong \mathfrak{A}^{(I \cap J)}/(N \cap A^{(I \cap J)}) \cong (\mathfrak{A}^{(I)}/N)^{(I \cap J)} \tag{8}$$

From (3), (7), 1.1.28(ii), and 1.1.30 we have

$$(\mathfrak{A}^{(J)}/M) \preccurlyeq \mathfrak{A}^{(J)}/(R \cap A^{(J)}) \cong (\mathfrak{A}/R)^{(J)} \cong (\mathfrak{F}\mathfrak{r}_4 \mathsf{RCA}_\alpha)^{(J)},$$

so that $\mathfrak{A}^{(J)}/M \in \mathbf{RCA}_\alpha$ (since $\mathbf{RCA}_\alpha$ is closed under the formation of homomorphic images by $\mathscr{HMT}$ 2.6.32); by a similar argument, $\mathfrak{A}^{(I)}/N \in \mathbf{RCA}_\alpha$. Thus, in view of the assumption that RCA_α has the $\mathscr{AP}$ with respect to CA_α, we conclude from (8) that there exists a $\mathfrak{B} \in \mathbf{CA}_\alpha$ such that $In\,\mathfrak{B} = 4$ and

$$\mathfrak{B}^{(J)} \cong \mathfrak{A}^{(J)}/M \quad , \quad \mathfrak{B}^{(I)} \cong \mathfrak{A}^{(I)}/N.$$

Therefore, if P is the (unique) ideal of $\mathfrak{A}$ such that $\mathfrak{A}/P \cong \mathfrak{B}$, we see by 1.1.30 that

$$\mathfrak{A}^{(J)}/(P \cap A^{(J)}) \cong (\mathfrak{A}/P)^{(J)} \cong \mathfrak{B}^{(J)} \cong \mathfrak{A}^{(J)}/M$$

and hence by 1.1.28(i) that

$$P \cap A^{(J)} = M; \tag{9}$$

in an analogous manner we get

$$P \cap A^{(I)} = N. \tag{10}$$

In view of (2) and (5), (9) implies that $s \cdot t \in P$, and hence $r \in P$ by (4). Consequently,

from (2) and (10) we get $r \in N$, and thus by (6) and 2.1.9(iii) there exist elements

$$u \in M \cap A^{(I \cap J)} \tag{11}$$

and $b \in R$ such that

$$r \leq u + b\,. \tag{12}$$

Since $u \in M$ we have by (5), $\mathscr{HMT}$ 2.3.10(i) and 2.1.9(iii) that there is a finite $\Gamma \subseteq \alpha$ and a $c \in R$ such that

$$u \leq \mathsf{c}_{(\Gamma)}\,(s \cdot t) + c\,. \tag{13}$$

Let h be the unique homomorphism from $\mathfrak{A}$ onto $\mathfrak{Fr}_4\mathsf{RCA}_\alpha$ such that $h(\kappa/Cr_4\mathsf{CA}_\alpha) = \kappa/Cr_4\mathsf{RCA}_\alpha$ for $\kappa = 0, 1, 2, 3$; then it is clear from (3) that $hb = hc = 0$. Hence (12) and (13) imply that

$$hr \leq hu \leq \mathsf{c}_{(\Gamma)}\,(hs \cdot ht)\,;$$

but $hu \in Sg^{(\mathfrak{Fr}_4\mathsf{RCA}_\alpha)}\{hx\}$ by (1) and (11), and thus this double inclusion contradicts 2.3.2(ii) since it is clear that hr, hs, and ht respectively equal the r, s, and t of 2.3.2 when we take $x = 0/Cr_4\mathsf{RCA}_\alpha$, $y = 1/Cr_4\mathsf{RCA}_\alpha$, $z = 2/Cr_4\mathsf{RCA}_\alpha$, and $w = 3/Cr_4\mathsf{RCA}_\alpha$.

Therefore, under the assumption that RCA_α has the $\mathscr{AP}$ with respect to CA_α, we have obtained a contradiction. Thus the theorem is proved.

REMARK 2.3.7. The proof just given is easily seen to be a relativization of portions of the proofs of 1.2.8(i) and 2.1.11. Indeed, although we shall not do it here, it is easily seen how to formulate and prove a relativized version of 1.2.8(i) in which the $\mathscr{AP}$ is replaced by the $\mathscr{AP}$ with respect to an arbitrary class $\mathbf{L}$ of g-algebras, and the congruence-extension property is given an appropriately relativized form. The reader is warned the relativized version of the $\mathscr{CP}$ required here is slightly different from the one defined in 1.2.7(i).)

COROLLARY 2.3.8. K *fails to have the amalgamation property for every class* K *such that* $\mathsf{RCA}_\alpha \subseteq \mathsf{K} \subseteq \mathsf{CA}_\alpha$.

REMARK 2.3.9. Comer [4] has shown that 2.3.8 holds also for all finite $\alpha \geq 2$. More generally, he shows that, for any finite $\alpha \geq 2$ and any K such that $\mathsf{RCA}_\alpha \subseteq \mathsf{K} \subseteq \mathsf{CA}_\alpha$, K fails to have even the embedding property. Comer's result taken together with 1.2.8(ii) and 2.1.11 implies that, for every finite $\alpha \geq 2$ and each class K of CA_α's which contains every representable algebra, the free g-algebra $\mathfrak{Fr}_\gamma\mathsf{K}$ fails to have the restricted interpolation property for at least one non-zero cardinal γ. However, Leon Henkin and the author have independently shown this to be the case by a more direct argument. We now state formally, but in metamathematical terms, their exact result; we do not however give a proof.

THEOREM 2.3.10. *If α is any finite ordinal ≥ 2, and we take*

$$\sigma = \mathsf{c}_0\mathsf{c}_{(\alpha\sim\{0\})}\mathsf{v}_0 \cdot \mathsf{c}_0 - \mathsf{c}_{(\alpha\sim\{0\})}\mathsf{v}_0$$

and

$$\tau = \mathsf{c}_0\mathsf{c}_{(\alpha\sim\{0\})}\mathsf{v}_1 + \mathsf{c}_0 - \mathsf{c}_{(\alpha\sim\{0\})}\mathsf{v}_1,$$

then the inclusion $\sigma \leq \tau$ holds in all CA_α's but cannot even be weakly interpolated relative to the class RCA_α; i.e, there is no term ρ in which no individual variables occur such that the two inclusions

$$\mathsf{c}_{(\alpha)}{}^{\partial}\sigma \leq \rho \leq \mathsf{c}_{(\alpha)}\tau$$

hold in all representable CA_α's.

2.4 Summary. Open problems

In this section we summarize in tabular form the main results of Chapter 2. It will be seen that the positive results of Section 2.2 in combination with the negative ones of the preceding section seem to answer most of the natural questions one would ask about amalgamation and interpolation properties in the present state of the development of the theory of cylindric algebras.

The final portion of the section is taken up by a brief discussion of some open problems.

In the top row of Table 2.4.1 we find a list of eight different amalgamation (and embedding) properties and in the left column a comprehensive list of classes of cylindric algebras of both finite and infinite dimension. At each matrix point we have indicated, by putting yes or no, that the corresponding class has or does not have the amalgamation property in question, and, by putting ?, that the problem is open. (A word of explanation in connection with the fifth and eighth columns of the table: for a class K to have the $\mathscr{AP}$, or $\mathscr{EP}$, *for simple algebras* means that the subclass L of simple members of K has this property.) Practically all of the more important results which were formally proved, or mentioned in the informal remarks, are included in the table. In addition, we have included for completeness a few results which were mentioned nowhere in the preceding discussion. One of these results, that the class CA_1 has the $\mathscr{AP}$, is known from the literature; cf. Comer [4]. In all other cases the result is either an easy consequence of those which have been discussed, or it can be obtained by an argument very similar to one used previously to obtain some closely related result. For example, that every class K such that $\mathsf{RCA}_\alpha \subseteq \mathsf{K} \subseteq \mathsf{CA}_\alpha$ has the $\mathscr{AP}$ for simple algebras is an easy consequence of 2.2.23 and the fact that every simple CA_α ($\alpha \geq \omega$) is representable. Also, the result that, for finite $\alpha \geq 2$, every class K such that $\mathsf{RCA}_\alpha \subseteq \mathsf{K} \subseteq \mathsf{CA}_\alpha$ fails to have the $\mathscr{EP}$ for simple algebras follows from the weaker result that K does not have the $\mathscr{EP}$, together with the fact that every finite-dimensional CA is semisimple; cf. $\mathscr{HMT}$ 2.4.53. On the other hand, by observing that one can

TABLE 2.4.1

		strong AP	strong AP wrt* RCA_α	AP	AP wrt* RCA_α	AP for simple algebras	strong EP	EP	EP for simple algebras
$\alpha \geq \omega$	Lf_α	yes[1]	yes	yes	yes	yes	yes	yes	yes
	Dc_α	no	yes[2]	no	yes	no	no	no	no
	semisimple CA_α's	?	?	yes[3]	yes	yes	?	yes	yes
	diagonal CA_α's	?	?	yes[4]	yes	yes	?	yes	yes
	RCA_α	no	no	no[5]	no	yes[6]	yes	yes[7]	yes
	$RCA_\alpha \subset K \subseteq CA_\alpha$	no	no	no[5]	no	yes[6]	?	?	yes
	CA_α's of characteristic $\varkappa > 0$	?	?	yes[8]	yes	yes	yes	yes	yes
$2 \leq \alpha < \omega$	$RCA_\alpha \subseteq K \subseteq CA_\alpha$	no	no	no[9]	no	no	no	no[9]	no
	CA_α's of characteristic $\varkappa > 0$	?	?	yes[8]	yes	yes	yes	yes	yes
	CA_1	?	?	yes[10]	yes	yes	?	yes	yes
	CA_0	yes[11]	yes	yes	yes	yes	yes	yes	yes

1) Daigneault [7] (see also 2.2.8)
2) 2.2.14
3) 2.2.24
4) 2.2.20
5) 2.3.8
6) 2.2.23
7) 2.2.15
8) Comer [2], [3] (cf. 2.2.29)
9) Comer [4] (cf. 2.3.9)
10) Comer [4]
11) Dwinger–Yaqub [9]; see 2.2.30

* with respect to

choose the algebras $\mathfrak{A}$ and $\mathfrak{B}$ of 2.2.9 to be simple, we immediately get that the class Dc_α fails to have the $\mathscr{AP}$ for simple algebras.

For the more significant of the results in Table 2.4.1 we have indicated their source in the literature, or, if they appear for the first time in this paper, we indicate where in the text they may be found.

Our second table, 2.4.2, contains a summary of results involving the interpolation properties. In order to keep things simple we have restricted our attention to free algebras with ω generators. Most of the results listed in 2.4.2 are almost automatic consequences of corresponding results from Table 2.4.1 and Theorems 1.2.8, 1.2.10, 2.1.11, 2.1.13, and 2.1.20 where the general relationships holding between the various amalgamation and interpolation properties are stated (cf. also 2.1.22); the remaining

TABLE 2.4.2

		strong IP	IP	weak IP	strong restricted IP	restricted IP	weak restricted IP
$\omega \leq \alpha$	$Fr_\omega^{(\Delta)}\,CA_\alpha$ which is in Lf_α	yes[1)]	yes	yes	yes	yes	yes
	$Fr_\omega^{(\Delta)}\,CA_\alpha$ which is in Dc_α	yes[2)]	yes	yes	yes	yes	yes
	$Fr_\omega\,RCA_\alpha$	no	no[3)]	yes	?	yes	yes
	$Fr_\omega\,K, RCA_\alpha \subset K \subseteq CA_\alpha$	no	no[3)]	yes	?	?	yes
	$Fr_\omega\,K, K$ class of CA_α's of characteristic $\chi, 0 < \chi < \omega$	?	yes	yes	?	yes	yes
$2 \leq \alpha < \omega$	$Fr_\omega\,K, RCA_\alpha \subseteq K \subseteq CA_\alpha$	no	no[4)]	no	no	no[4)]	no
	$Fr_\omega\,K, K$ class of CA_α's of characteristic $\chi, 0 < \chi < \omega \cap \alpha$	?	yes	yes	?	yes	yes
	$Fr_\omega\,CA_1$	?	yes	yes	?	yes	yes
	$Fr_\omega\,CA_0$	yes	yes	yes	yes	yes	yes

1) Craig [5] and Daigneault [8]
2) 2.2.6
3) 2.3.4(ii)
4) 2.3.9 and 2.3.10

results are in turn trivial consequences of ones obtained in this way. However, in those cases where a result in 2.4.2 either originated outside the present work or played a fundamental role in our discussion we have indicated its source.

REMARKS 2.4.3. Several open problems are indicated in Tables 2.4.1 and 2.4.2. It seems to us that the most interesting of these is whether CA_α $(\alpha > \omega)$ has the embedding property, or, equivalently, whether $\mathfrak{Fr}_\omega CA_\alpha$ has the restricted interpolation property. It would also be interesting to know if $\mathfrak{Fr}_\omega RCA_\alpha$ has the strong restricted $\mathscr{IP}$.

It is obvious from the definitions involved that the strong version of each of the $\mathscr{AP}$, $\mathscr{EP}$, $\mathscr{IP}$, and restricted $\mathscr{IP}$ implies the corresponding ordinary version of the same property, which in turn implies the property for simple algebras in the case of the $\mathscr{AP}$ or $\mathscr{EP}$, and the weak property in the case of the $\mathscr{IP}$'s. It is also obvious that each property implies its restricted version. The question whether any of the corresponding implications in the opposite direction hold still involves some open problems. The only conclusions that can be drawn from the discussion of the present chapter is that neither the $\mathscr{AP}$ for simple algebras nor the $\mathscr{EP}$ implies either

the $\mathscr{AP}$ or strong $\mathscr{AP}$. All these results are obtained by considering the class RCA_α. Consequently, by considering $\mathfrak{Fr}_\omega\mathsf{CA}_\alpha$ we see that neither the weak $\mathscr{IP}$, nor the restricted $\mathscr{IP}$, need imply either the $\mathscr{IP}$ or strong $\mathscr{IP}$. If one restricts oneself to arbitrary algebraically closed classes of cylindric algebras and to the free algebras over these classes, the questions about which of the remaining implications hold seem to all be open.

On the other hand if one is allowed to consider classes K of CA_α's which are required only to be closed under the formation of subalgebras and homomorphic images (and not necessarily under the formation of direct products), and to consider K-independently generated algebras in K, then the situation changes radically. For instance, let $\mathfrak{A}$ be any simple, minimal CA_α and let K be the class of all isomorphic images of $\mathfrak{A}$. Then, clearly, K has the $\mathscr{AP}$ but not the strong $\mathscr{AP}$. Also, if a is any system of elements of $\mathfrak{A}$ whose range is the whole universe of $\mathfrak{A}$, then the K-independently generated g-algebra $\langle\mathfrak{A}, a\rangle$ has the $\mathscr{IP}$ but not the strong $\mathscr{IP}$.

We have not discussed polyadic algebras in this paper, but the question as to which classes of polyadic algebras the amalgamation property applies has been intensively examined in the literature; for completeness we would like to give a brief outline of the major results in this area.

For each set I let PA_I denote the class of polyadic algebras which have I as their set of variables, and let PEA_I denote the class of PA_I's with equality, that is, the class of polyadic equality algebras with I as their variable set. As we have previously mentioned, Daigneault shows in [7] that for infinite I the class of locally finite PEA_I's has the strong $\mathscr{AP}$; he shows there that this is true also for the class of locally finite PA_I's. Johnson [14] extends Daigneault's result to the classes PEA_I and PA_I; see Johnson [13]. On the other hand, Comer in [4] shows that for any finite set I with $|I| \geq 2$ and any class K of PEA_I's containing all representable algebras, K fails to have even the embedding property; Comer's results apply also to polyadic algebras without equality as well as to diagonal-free cylindric algebras and relation algebras.

In conclusion we want just to mention the possibility of extending the entire discussion of this paper to algebras with infinitary operations, and to languages with expressions of infinite length. In particular we have in mind γ-complete (γ any uncountable cardinal) CA_α's or PEA_I's treated as algebras where the sum $\sum_\gamma$ of γ-sequences of elements is taken to be a fundamental operation. We refer to Preller [22] for some work in this direction; Preller does not however deal with γ-complete cylindric algebras or polyadic algebras, but with a certain related algebra with infinitary operations.

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